PECTRACK: A Web-Based Order Management System with Predictive Analytics and Inventory System for Pecto's Bakery in Lucban, Quezon

A Capstone Project Proposal
Presented to the Faculty of the
Information and Communications Technology Program
STI College Lucena

In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Information Technology

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ENDORSEMENT FORM FOR PROPOSAL DEFENSE

TITLE OF RESEARCH: PECTRACK: A Web-Based Order Management System with Predictive Analytics and Inventory System for Pecto's Bakery in Lucban, Quezon

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In Partial Fulfilment of the Requirements
for the degree Bachelor of Science in Information Technology
has been examined and is recommended for Proposal Defense

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APPROVAL SHEET

This capstone project proposal titled PECTRACK: A Web-Based Order Management System with Predictive Analytics and Inventory System for Pecto's Bakery in Lucban Quezon, prepared and submitted by Christian V. Luza, Christian C. Valencia, Yancy L. Ymata, and Carla Mhaey G. Zoleta in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, has been examined and is recommended for acceptance and approval.

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INTRODUCTION

Project Context

Order Management Systems (OMS) have become an important part of modern business operations because they help businesses organize transactions, coordinate order fulfillment, and monitor product availability. As businesses increasingly accept orders through different channels, managing transactions through separate platforms and manual records can make it difficult to maintain consistent and accurate order information. Jimenez et al. (2024) examined the design and evaluation of order management software for small and medium-sized enterprises (SMEs) in Lima, Peru. Their study involved 308 employees and reported positive evaluations of the system's usability and perceived operational efficiency. Similarly, Sinaga and Johan (2024) designed a web-based Order Management System to help a company consolidate transactions received through multiple online marketplaces. These studies demonstrate how centralized order management can help organize order information and simplify the work involved in handling transactions.

Order management also depends on effective inventory control and coordination throughout the order fulfillment process. Labastida et al. (2024) developed a web-based ordering system for a start-up business that integrated online ordering, inventory management, and demand forecasting. Their work demonstrates the potential of combining these functions within a single platform to support business operations. In addition, Panigrahi et al. (2024) reviewed research on inventory management and its relationship with SME performance. Their findings highlight the importance of inventory-related knowledge, technology, and appropriate management practices in supporting business operations. For small businesses, integrating order records with inventory information can help personnel monitor product availability, keep stock records organized, and make better-informed decisions about order fulfillment.

In food-related businesses, accurate order handling and reliable fulfillment are particularly important because customers often expect their orders to be prepared and

released within a specified period. Bonfanti et al. (2023) examined expectations regarding online food ordering and delivery services and identified accuracy, punctuality, and the ability to meet expected service requirements as important considerations. Wu et al. (2024) also examined food delivery service quality and found that reliability, system operation, and order traceability were among the factors associated with customer satisfaction and the intention to reuse a service. These findings support the importance of maintaining accurate order details, providing clear status information, and coordinating the preparation and delivery of orders. A web-based Order Management System can help address these operational needs by providing a centralized record of transactions and making order progress easier to monitor.

Furthermore, predictive analytics can complement order and inventory management by helping businesses estimate future product demand. Hübner et al. (2024) investigated the use of machine-learning-based demand forecasting in a bakery setting, focusing on its potential to reduce food waste through improved sales forecasting. Their study illustrates the relevance of demand estimation to bakeries, where unsold products and inaccurate production planning can create operational challenges. When combined with order records and inventory monitoring, predictive analytics can provide additional information for planning product quantities and identifying expected demand patterns. However, the usefulness of forecasting depends on the quality and availability of historical data, and predictions should be evaluated according to the data on which they are based.

Taken together, the reviewed studies demonstrate the importance of integrating order management, inventory monitoring, and demand forecasting into business operations. Research on web-based order management highlights the value of centralized transaction records, while studies on inventory management and food ordering emphasize accurate stock information, reliable order fulfillment, and effective monitoring. These findings provide a basis for developing PECTRACK, a web-based Order Management System with Predictive Analytics and Inventory System for Pecto’s Bakery in Lucban, Quezon. The proposed system aims to consolidate online, walk-in, and advanced orders, record

payment information, monitor order status, and maintain organized product inventory records. It will also demonstrate demand forecasting using assumed historical sales data, as actual historical records are unavailable for the study. Therefore, its predictive results will demonstrate the application of forecasting methods rather than represent verified predictions of the bakery's actual sales.

Background of the Problem

Pecto's Bakery, established in 1947 and located in Lucban, Quezon, is one of the long-standing local bakeries that has continuously served its community and nearby areas. Over the years, the bakery has expanded its reach by offering delivery services within Lucban and across different parts of Quezon Province. With decades of operation and continuous business growth, the bakery has established a strong customer base and remains one of the recognized local bakeries in the area.

Based on the interview conducted with the management, the bakery accommodates approximately 500–700 customers daily, particularly during regular operating days, with higher customer volume during weekends, holidays, and bulk-order seasons. Despite its long history and growing customer demand, the bakery continues to rely heavily on traditional and manual procedures for managing customer orders, customer records, and transactions.

At present, customers place their orders either by sending messages through the bakery's official Facebook page or by directly visiting the physical store. While these methods provide customers with different ways to place orders, they do not provide a single, centralized platform for organizing transactions. Customer information is recorded manually in logbooks or on paper, while confirmed orders are written down and forwarded to the personnel responsible for preparation. This workflow involves several steps before an order is completed. Based on the information gathered during the interview, the bakery follows these procedures in managing customer orders:

1. Receiving customer inquiries.

2. Checking product availability.
3. Manually listing orders.
4. Confirming order details.
5. Forwarding orders to the assigned personnel.
6. Preparing the products.
7. Monitoring inventory manually.
8. Confirming payments.
9. Preparing orders for pickup or delivery.
10. Manually updating customers regarding their order status.

To better understand these operational challenges, an interview was conducted with the General Manager of Pecto's Bakery, Mrs. Katherine R. Abillar, who has been managing the bakery for many years as part of their family-run business. The following are the significant problems identified:

- **Decentralized Order Management:**

  The interview revealed that Pecto's Bakery receives customer orders through its official Facebook page and direct verbal communication with cashiers for walk-in transactions. The admin or manager handles online orders, while the sales lady manages walk-in customers. This decentralized approach may lead to inconsistent order records, miscommunication, and discrepancies in transaction handling and inventory tracking. To address this problem, PECTRACK will provide a centralized web-based platform for managing customer orders and related transactions. This will allow authorized personnel to access organized order information and coordinate order handling more efficiently.
- **Manual Order Recording and Processing:**

  Mrs. Abillar explained that confirmed orders are written on paper and forwarded to the personnel responsible for preparation. Order details are sometimes verified through customer messages and pictures before being recorded. These manual procedures may result in misplaced order slips, incomplete order details,

transcription errors, and communication delays, particularly during peak operating hours and when handling large-volume transactions. PECTRACK aims to reduce these issues by digitally recording and organizing customer orders, making order information more accessible to authorized personnel and improving the monitoring of order preparation and fulfillment.

- **Lack of Centralized Customer Information and Order History:**
  The interview revealed that Pecto's Bakery does not have a centralized database for managing customer information and previous transactions. As a result, retrieving customer records and reviewing order history may become time-consuming, especially when serving an estimated 500–700 customers daily. This limitation can make it difficult to verify repeat orders and maintain organized customer information. PECTRACK aims to address this concern by providing centralized customer records and order history, allowing authorized users to retrieve relevant information more efficiently and maintain organized transaction records.
- **Limited Visibility into Order Status and Scheduling:**
  Mrs. Abillar explained that delays may occur when production schedules do not align with customers' expected pickup times. For example, an order scheduled for 6:00 may not be ready until 7:00 because the baked products still require time to finish baking and cool before release. When such delays occur, customers must contact the bakery to ask about their order status. To address this problem, PECTRACK will provide order status monitoring so that authorized personnel can track order progress and customers can check available updates through the system. This feature aims to improve communication, provide better visibility into order fulfillment, and reduce the need for repeated inquiries about order status.
- **Manual Payment Verification and Order Release:**
  According to Mrs. Abillar, orders must be paid for before they are released, while delivery orders require full payment in advance to minimize losses from refused deliveries. However, coordinating payment information with individual orders

may become challenging when transaction records are handled separately. PECTRACK aims to address this concern by linking every payment to its order. A customer who pays online scans the bakery's QR Ph code and enters the payment's reference number, and the order is held until a staff member confirms that payment. The system refuses a reference number that was already used, records who confirmed each payment and when, and releases an order only once it is paid. Staff no longer need to match payments with messages and paper records, although the check against the bakery's bank or e-wallet app remains a manual step, as stated in the Scope and Limitations.

- **Manual Inventory Monitoring and Production Planning** :

  According to Mrs. Abillar, orders must be paid for before they are released, while delivery orders require full payment in advance to minimize losses from refused deliveries. However, coordinating payment information with individual orders may become challenging when transaction records are handled separately. PECTRACK aims to address this concern by providing a centralized means of recording payment details and associating payment status with corresponding orders. This feature will support personnel in verifying payment completion and coordinating order release in accordance with the bakery’s payment policies.

- **No Sales History for Planning :**

  Because orders and sales are recorded on paper, the bakery has no organized record of its past sales by product and by day. Production quantities therefore rely on experience, and the bakery cannot readily review which products sell more on particular days, weekends, or events. For the same reason, actual historical sales data are not available for this study. To address this problem, PECTRACK will record every sale by product and date, so that a sales history builds up from the first day of use. The system will report sales, best-selling products, and peak ordering periods, and will demonstrate demand forecasting on this history. Until enough actual sales are recorded, the forecast will be demonstrated on assumed

historical data, as approved by the adviser, and it will switch to actual sales after 56 days.

The findings from the interview demonstrate that Pecto's Bakery encounters challenges related to decentralized order management, manual record-keeping, customer information retrieval, order status monitoring, payment verification, inventory monitoring and production planning, and the lack of a sales history. These concerns may affect the efficiency of daily operations, particularly when the bakery handles hundreds of customers and large-volume orders. Without a centralized system, personnel must coordinate information through separate communication channels and manually maintained records, making order management more difficult during busy operating periods.

In response to these identified problems, the proposed study aims to develop PECTRACK: A Web-Based Order Management System with Predictive Analytics and Inventory System for Pecto's Bakery in Lucban, Quezon. The system is designed to centralize order management, customer information, transaction records, and inventory monitoring within a single web-based platform. It will support the digital recording of orders, the retrieval of customer order history, the monitoring of order status, and the recording of payment details. Through these functions, the system aims to reduce repetitive manual tasks, minimize errors associated with paper-based records, improve coordination among authorized personnel, and provide customers with better access to their order information.

Furthermore, the integration of predictive analytics and inventory management is intended to support the bakery in monitoring product availability and demonstrating how demand trends can be used to assist inventory planning. Since actual historical sales data are not available for the predictive analytics component, the study will use assumed historical data for demonstration and evaluation purposes, subject to the approval

provided by the adviser. Therefore, the predictive results will illustrate the potential application of analytics rather than represent verified forecasts of Pecto's Bakery's actual sales.

Ultimately, PECTRACK aims to provide Pecto's Bakery with a more organized and accessible approach to managing customer orders, transaction records, and inventory information. By addressing the identified operational challenges through a centralized web-based system, the proposed study seeks to improve workflow efficiency, support more systematic inventory planning, strengthen order monitoring, and enhance the overall service experience for both bakery personnel and customers.

Purpose and Description

Pectrack is a web-based Order Management System with predictive analytics and inventory system made to improve Pecto's Bakery In Lucban Quezon in its daily transactions. It helps make ordering, product tracking, and inventory management faster and more organized. Instead of using paper and manual processes, the system stores everything digitally, which reduces errors and saves time.

The name Pectrack was coined from Pecto's Bakery and the word Track, reflecting the system's capability of monitoring and tracking the final baked products keeping track of how many items are produced, currently available, and sold. This ensures that the bakery always has an up-to-date view of its product inventory.

The system caters to different user roles including Admin, Cashier and Customer each with specific access and functions suited to their responsibilities. All data is stored digitally, which reduces errors and saves time, providing a more reliable and efficient experience for both the business and its customers.

Below are the following concepts that can incorporate in the proposed study:

- **Login Module**

  The Login Module secures the system by verifying the identity of each user before granting access. The sign-in form opens as a pop-up window over the page the visitor is currently viewing and provides two entrances: Customer and Admin. The Admin entrance is used by the administrator and cashiers, each entrance refuses accounts that belong to the other. To sign in, users enter their username or email address together with their password, after which the system validates the credentials and determines the level of access based on the user's assigned role. To protect accounts from unauthorized access, the system locks an account for 15 minutes after five failed sign-in attempts. Users who forget their password may request a password reset link through email. When the "Keep me logged in" option is selected, the user remains signed in on that device for up to seven days; otherwise, the session ends once the browser is closed. All users may sign out through the My Profile page.

  New customers register by providing their first and last name, username, number and is verified through a six-digit code sent by SMS before the account is created. The customer is then required to confirm the email address through a link sent by email before placing an order.

  The administrator account is subject to an additional layer of security. After passing credential validation, the administrator must enter a six-digit One-Time Password (OTP) sent by email, which remains valid for five minutes. The administrator may mark the device as trusted for 30 days, after which signing in on that device requires only the password; changing the password removes all trusted devices. Once verified, the administrator is directed to the Dashboard and is given full access to all modules.

The other users are likewise directed to the page appropriate to their role. Cashiers are directed to the Dashboard and are given access to the operational features of the system, namely orders, walk-in and advanced orders, inventory and production, payments and the payment log, and deliveries, while the User Management and Reporting and Analytics modules remain exclusive to the administrator. Customers are directed to the online Menu, where they browse products, place and pay for orders, and track the progress of their orders.

- **Dashboard Module**

  The Dashboard Module provides the administrator and the cashier with an at-a-glance overview of the matters that require attention. At the top, a "Today" row presents the orders paid for the day, the day's sales excluding delivery fees, and the cash orders that remain unpaid. Below this row, color-coded status tiles, in which green indicates Good, yellow indicates Okay, and red indicates Needs Attention, summarize the soft breads about to be sold out, the hard breads and ingredients low on stock, the orders for the day, and the deliveries awaiting a rider. Selecting a tile opens the corresponding module already filtered to the concern. The Dashboard also displays a Running Out list of products with low stock, a Needed By list of paid orders still to be picked up or delivered, each marked according to how near its requested time is, and a list of the most recent orders.

- **User Management Module**

  The User Management Module maintains the records of the individuals who use the system. It is accessible only to the administrator and is organized into two tabs: Customers and Staff. Under the Customers tab, the administrator can search the customer list, correct a customer's details, and deactivate or reactivate an account. Customers create their own accounts through registration, which refuses any username, email address, or contact number already in use, and they manage their own profile, saved delivery addresses, and password through their account

page. Under the Staff tab, the administrator creates and manages the accounts of cashiers, edits their details, resets their passwords, and deactivates or reactivates their accounts.

- **Order Management Module**

  The Order Management Module oversees each order from placement to completion. In line with the policy of the bakery, every order is paid before it is released to the customer. An order progresses from Ready for Pickup or Ready for Delivery, to Out for Delivery in the case of deliveries, and finally to Completed. An online order paid through GCash is held for 15 minutes while the customer completes the payment; if the payment is not made within this period, the system releases the order and returns its stock to inventory. Alternatively, a pickup order may be paid in cash at the counter. Such an order is immediately placed as Ready for Pickup and is marked Not Paid until the cashier records the cash payment, with the order details indicating the payment method. A customer may have only one unpaid cash order at a time, and any cash order not collected by the end of its scheduled day is automatically cancelled. Delivery orders, on the other hand, are payable through GCash only and carry a delivery fee of ₱50. Since the bakery does not grant refunds, paid orders cannot be cancelled; however, customers may cancel their own unpaid cash orders.

  Staff members view the orders for the current day, the upcoming orders, or the orders for any chosen date, which may be filtered by status and searched by order number or customer name. Each order displays its items, requested time, payment, and progress, and a delivery order awaiting a rider may be assigned to one directly from its details.

  Counter orders are handled by the cashier through the Walk-in tab and are classified into walk-in orders and advanced orders. A walk-in order is placed by a customer who is present at the counter, requires no customer details, and is paid in

full through cash or GCash; a printable receipt showing the cash received and the change serves as its proof of order. An advanced order, by contrast, is picked up on a later date and requires the customer's name, contact number, which is verified through an SMS code, and the preferred pickup date and time. It is likewise paid in full through cash or GCash, and the order number is sent to the customer by SMS as proof of order, which remains accessible even without an internet connection. To add products to either type of order, the cashier types the first letters of a product's name and selects it from the list of matching products, and quantities may be entered directly.

For online orders, customers browse the Menu, add items to the cart, and choose between pickup and delivery. Customers may leave the time blank to receive the order as soon as it is ready, or they may select a date and one of the available time slots, which are offered every 30 minutes from 5:00 AM to 10:00 PM for up to 30 days in advance. A pickup order may be paid in cash at the counter or immediately through GCash, whereas a delivery order is paid through GCash with the ₱20 delivery fee added to the total. Saved delivery addresses are limited to Lucban, Quezon, with the customer providing the street, barangay, and other necessary details.

- **Inventory Management Module**

  The Inventory Management Module keeps the stock of every product up to date. Stock is automatically deducted when an order is placed, with the stock of an online GCash order set aside while the payment is being made, and is automatically returned when an unpaid order is released or cancelled. The module follows the two methods of stocking practiced by the bakery, one for soft breads and another for hard breads.

  Each soft bread is assigned a daily standard stock. Every day at 5:00 AM, the system automatically tops up each soft bread to its standard, wherein the leftovers

from the previous day are sold first and only the difference is baked. Soft breads are not restocked during the day. A soft bread is labeled About to Be Sold Out when its stock falls below one-fifth of its standard and Sold Out Today when its stock reaches zero. A soft bread ordered in advance is baked on its pickup day in addition to the standard, so that it does not reduce the stock available to walk-in customers for that day. Each hard bread, on the other hand, is assigned a minimum level, which signals when to produce, and a stop level, which determines how much to produce. A hard bread is labeled Low Stock when its stock is at or below its minimum level.

Stock may be adjusted in two ways. When stock is added, the pieces received are added to the quantity on hand, and the system asks the user to confirm the new total before saving. When a count is corrected, the administrator sets the actual quantity after a physical inspection, such as when bread is discarded. Every change is recorded together with the user who made it and the time it was made.

The Production tab lists the quantity of each product to be baked for a chosen day. For a soft bread, the quantity is computed as the standard less the stock on the shelf, plus the advanced orders for that day. For a hard bread at its minimum level, the quantity is the stop level less the stock on hand. Hard breads are automatically built up to twice the stop level beginning Thursday in preparation for the weekend, and to the multiplier of an event, which is four times the stop level by default, beginning one week before the event. The administrator maintains an Events calendar covering Holy Week, the Pahiyas Festival, Christmas, New Year, and long weekends, and the Production list indicates which event or weekend raised the quantity. Beside each product, a forecast check displays the expected sales for that day. When the forecast expects more sales than the planned stock, the list shows the shortfall, for example "+20 more," and a count of such products appears at the top of the list. While the stocking rules continue to determine the

quantities to be baked, the forecast check serves as a warning before a product sells out. The Production list may also be printed.

The module also monitors the stock of ingredients. Staff members record ingredient deliveries, the administrator corrects the counts, and ingredients that reach their minimum level are flagged. Through the Edit Products feature, the administrator maintains the product catalog, including each product's category, bread type (soft or hard), price, size or pack, description, photo, and availability for sale.

- **Payment and Billing Module**

  The Payment and Billing Module processes and records every payment made in the system. At the counter, the cashier records cash payments, for which the system computes the change from the amount received, or GCash payments together with their reference numbers. Online orders for cash pickup are likewise paid at the counter upon collection. Online payments are processed through QR Ph using GCash, and the system confirms each payment automatically. Delivery orders are paid in full in advance, inclusive of the delivery fee, as cash on delivery is not offered**.**

  Each paid order is issued a digital receipt showing the items purchased, the delivery fee if applicable, the total amount, the amount received, and the change. Staff members can view the orders ready for pickup or delivery as well as the completed orders, each with its payment method and payment status. The Payment Log lists every payment received, may be filtered by date and payment method, and may be exported as a PDF file. Customers, for their part, can view their own receipts and payment records.

- **Notifications**

  The system keeps its users informed through several channels. Within the application, a notification bell alerts staff members and riders of new orders, payments, deliveries, and orders due within the hour. When a product falls to its minimum level, the administrator and the cashier receive a red low-stock warning accompanied by a distinct sound, which opens the Inventory module when selected. Through email, customers are notified when their order is ready, out for delivery, and delivered. Through SMS, the order number of an advanced order is sent to the customer.

- **Reporting and Analytics Module**

  The Reporting and Analytics Module transforms the recorded data of the bakery into reports and forecasts. It is accessible only to the administrator. Every report may be printed or saved as a PDF file, and the sales report and the forecast may also be exported in CSV format. The module is organized into five tabs. The Sales tab presents the sales, defined as the money received excluding delivery fees, together with the number of orders, and the delivery fees for any date range, grouped by day, week, or month. It also presents a chart of sales and orders, the sources of sales by payment method (cash or GCash) and by order type (walk-in, online pickup, or delivery), the unpaid cash orders, and the peak ordering periods by hour and by day of the week. The Products tab ranks the best-selling products by the number of pieces sold, for all breads or for soft or hard breads only, together with their revenue. The Inventory tab shows, for the selected dates, the number of pieces of each product baked or added, sold, and lost through rejects and downward count corrections. The Forecast tab presents the expected sales of each product for the next seven days and identifies the busiest day ahead. Finally, the Export History tab records every report downloaded or printed, together with the user who did so and the time it was done.

The payment log, as previously discussed, is located in the Payment and Billing Module. The reports reflect the actual transactions of the bakery; however, a Demo Data switch adds the practice orders used to demonstrate the reports, and the screen indicates when this switch is turned on.

The forecasting feature of the module employs three methods: a seasonal naïve baseline, Holt-Winters exponential smoothing, and multiple linear regression. The multiple linear regression is a supervised machine learning model that learns from the daily sales history the extent to which each day of the week, payday, holiday, and bakery event raises or lowers sales. The three methods are compared for each product through a rolling four-week backtest, and the most accurate method is selected. The forecast runs automatically every night after the sales of the day are complete. It learns from the assumed sales history until the bakery accumulates 56 days of actual sales, after which it learns from the actual sales. The administrator may also run the forecast manually at any time. The forecast guides the setting of the daily standards and the hard bread levels, and the forecast check in the Production list warns when a product is expected to sell more than the planned quantity.

These modules work together to form an integrated system that supports the operational needs of the business. The system is designed with simplicity and ease of use as its primary considerations, enabling staff and customers to use its features without extensive technical knowledge. By integrating these modules into a single web-based platform, the system provides a structured, reliable, and efficient approach to order and business management.

Objectives

General Objective

The general objective of this study is to design, develop, and implement PECTRACK: A Web-Based Order Management System with Predictive Analytics and Inventory System for Pecto's Bakery in Lucban, Quezon. The system centralizes the bakery's online, walk-in, and advanced orders, customer records, order status, payments, and product inventory in a single web-based platform, and demonstrates demand forecasting to support the bakery's inventory planning.

Specific Objectives

1. **To centralize the bakery’s order management in a single web-based platform.**

   The system receives online orders through the bakery's web storefront, and walk-in and advanced orders through the cashier's counter screen, and lists all of them in one order list that every authorized staff member sees. This replaces the separate handling of Facebook messages by the admin and walk-in orders by the sales lady, and reduces inconsistent records and miscommunication.

2. **To record and process customer orders digitally.**

   Every order is saved with its items, quantities, prices, requested pickup or delivery time, and a daily order number, and each paid order has a printable receipt. Recording an order also deducts its stock automatically. This removes handwritten order slips and the misplaced slips, missing details, and copying errors that come with them.

3. **To maintain centralized and secure customer records and order history**

   Customers with accounts keep their contact details, saved delivery addresses, and past orders with their receipts, so the bakery and the customer can retrieve them at once. Access is limited to authorized users: customers verify their mobile number with an SMS code, the administrator signs in with a one-time password sent by email, accounts are locked for 15 minutes after five failed sign-in attempts, and each role sees only its own functions.

4. **To provide order status monitoring and scheduling.**

   Customers choose a pickup or delivery time in 30-minute slots from 6:00 AM, an hour after the 5:00 AM bake, so the breads have cooled before release, and staff see which orders are needed next. Each order moves through Ready for pickup or Ready for delivery, Out for delivery, and Completed; staff update the status in the system, and customers see it on their orders and receive email updates. Advanced orders reserve their bread when they are placed: hard breads are taken from the stock, and soft breads are added to the bake list for the pickup day. Customers see when their order is ready instead of messaging the bakery to ask.

5. **To record payments and link them to order release.**

   The system records cash and QR Ph payments and shows each order's payment status. For a QR Ph payment, the customer scans the bakery's QR Ph code, pays in their own bank or e-wallet app, and enters the reference number; a staff member confirms each payment before the order is released. An order is released only once paid, and delivery orders are paid in full beforehand, following the bakery's policy. A payment log lists every payment and can be exported as a PDF.

6. **To automate the monitoring of the bakery’s product inventory**

Stock is deducted as orders are placed and returned when an unpaid order is cancelled, and the counts update on screen in real time. Soft breads are topped up to their daily standard every morning, hard breads are flagged for production at their minimum level, and a warning appears when a product runs low. Staff records added stock and deductions, each logged with who made it.

7. **To provide reports and demonstrate demand forecasting for inventory planning.**

   Every sale is recorded by product and date, building a sales history the bakery did not have.the system reports sales, best-selling products, peak ordering periods, and inventory movements. It forecasts each product's sales for the next seven days, comparing a machine-learning regression with two simpler methods and using the most accurate one. The forecast is demonstrated on assumed historical sales, as approved by the adviser, so its results illustrate the method rather than verified predictions of the bakery's sales.

Scopes and Limitations

The general purpose of this study is to design, develop, and implement PECTRACK, a web-based Order Management System with Predictive Analytics and Inventory System for Pecto's Bakery in Lucban, Quezon. The system operates as an e-commerce platform that enables customers to browse products, place and pay for orders, and track their orders online, while providing the bakery with a centralized system for managing its daily operations.

The target users and beneficiaries of this study are:

The Customers, who benefit from browsing products and ordering online without a physical visit to the store;

The Administrator, who gains full system control, including products, inventory, production, users, and reports

The Cashiers, who are provided with a structured way to take walk-in and advanced orders, record payments, manage stock, and assign deliveries

The proposed system includes the following features organized by module:

LOGIN MODULE

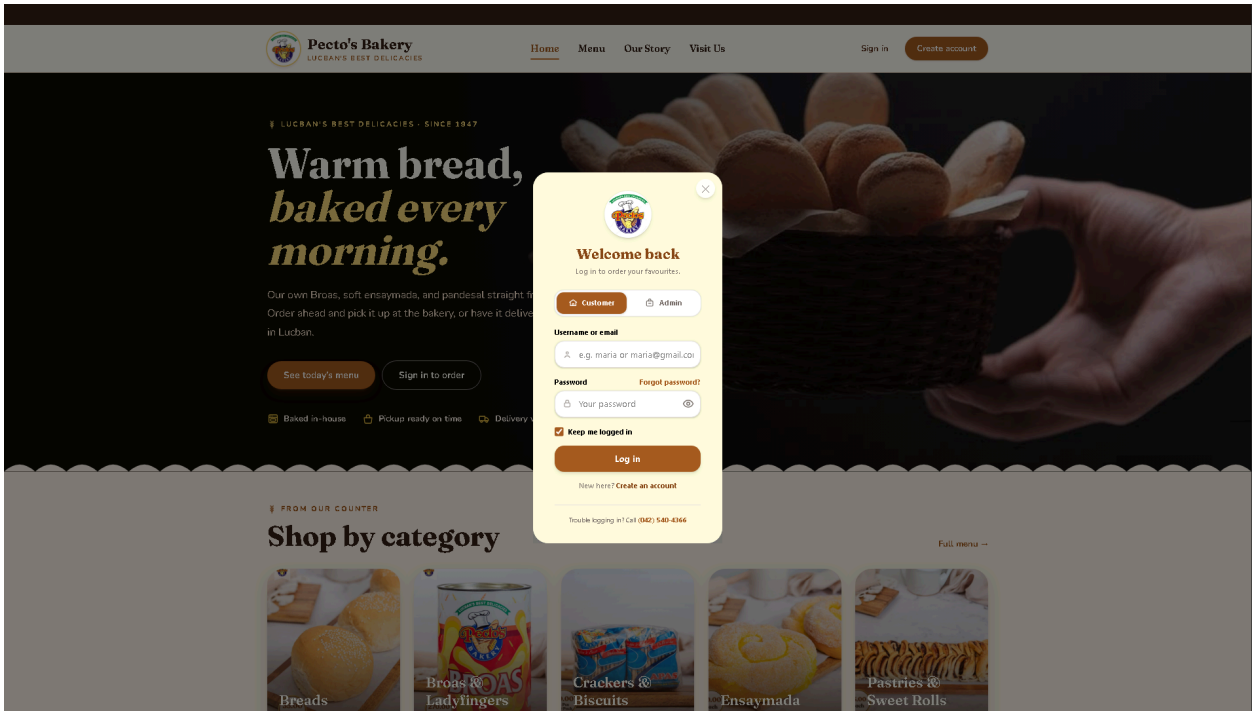


Figure 1. *Proposed login page design*

The log-in pop-up displays the bakery logo, a short welcome message, the Customer and Admin switch, the fields for the username or email address and the password, the "Keep me logged in" option, the "Forgot password?" link, and a single Log In button. Customers are also shown a "Create an account" link. For the administrator, a second step asks for the code sent by email, with the option to trust the device for 30 days.

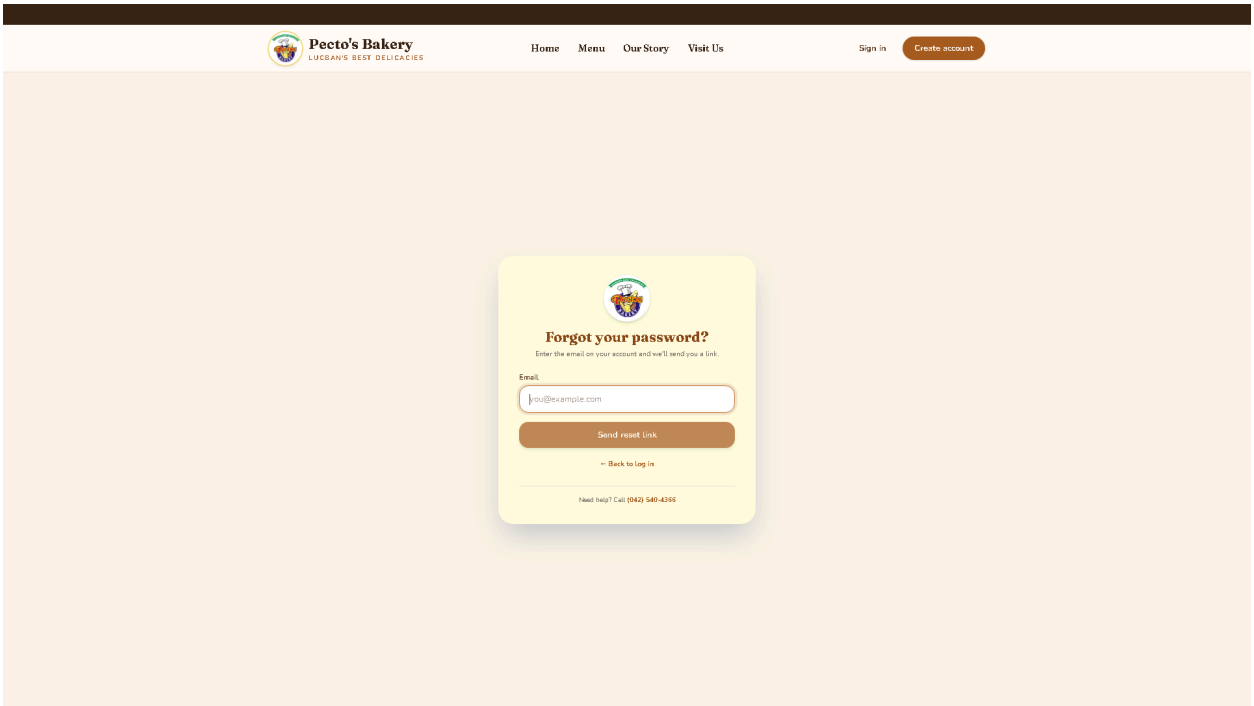


Figure 1.1 *Figure 1.1 Forgot password*

Selecting "Forgot password?" opens a page designed in the same style as the log-in, where the user enters an email address to receive a password reset link. The link then opens a page where the user sets a new password.

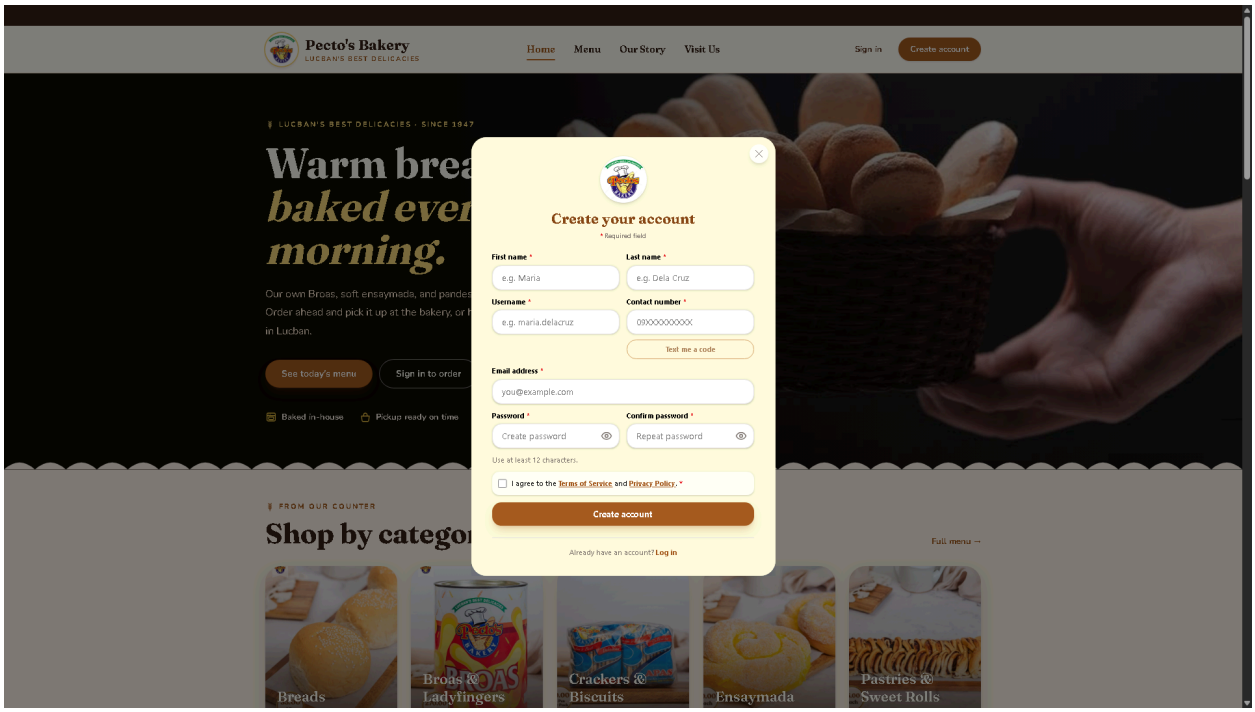


Figure 1.2 *Create account*

The registration form asks for the customer's first and last name, username, contact number, email address, password, and password confirmation, and requires agreement to the Terms of Service and Privacy Policy. The password must be at least 12 characters long and must contain uppercase and lowercase letters, a number, and a symbol; common passwords are refused. The contact number accepts digits only and is verified through a code sent by SMS. A username, email address, or contact number that is already in use is refused.

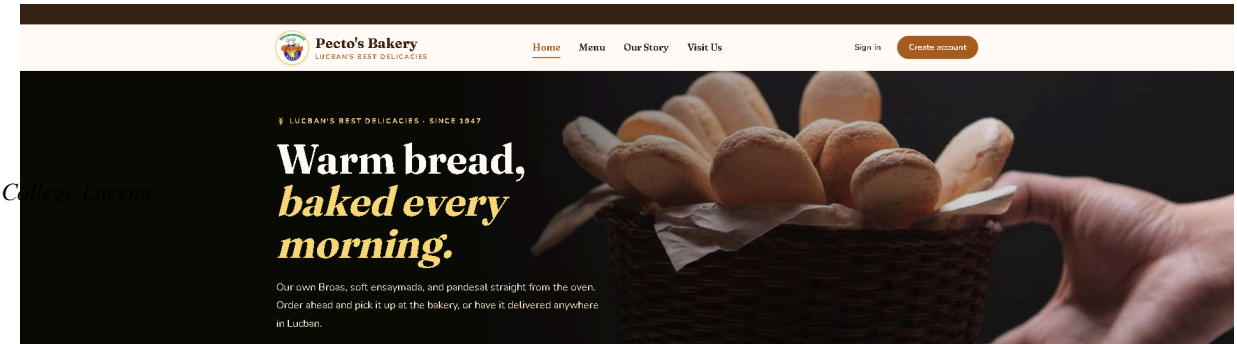


Figure 1.3 *Home page*

The home page presents the bakery in its own colors and photographs, with links to Home, Menu, Our Story, and Visit Us, as well as the history of the bakery since 1947.

DASHBOARD MODULE

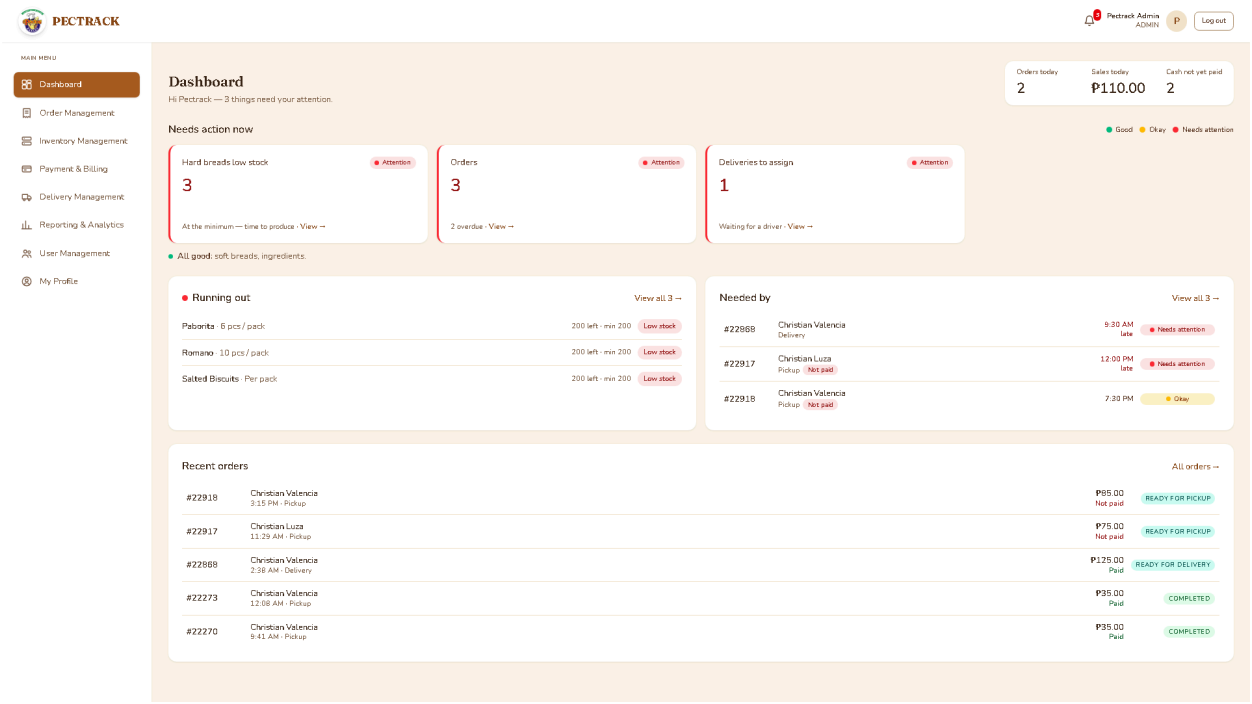


***Figure 2**. Admin dashboard*

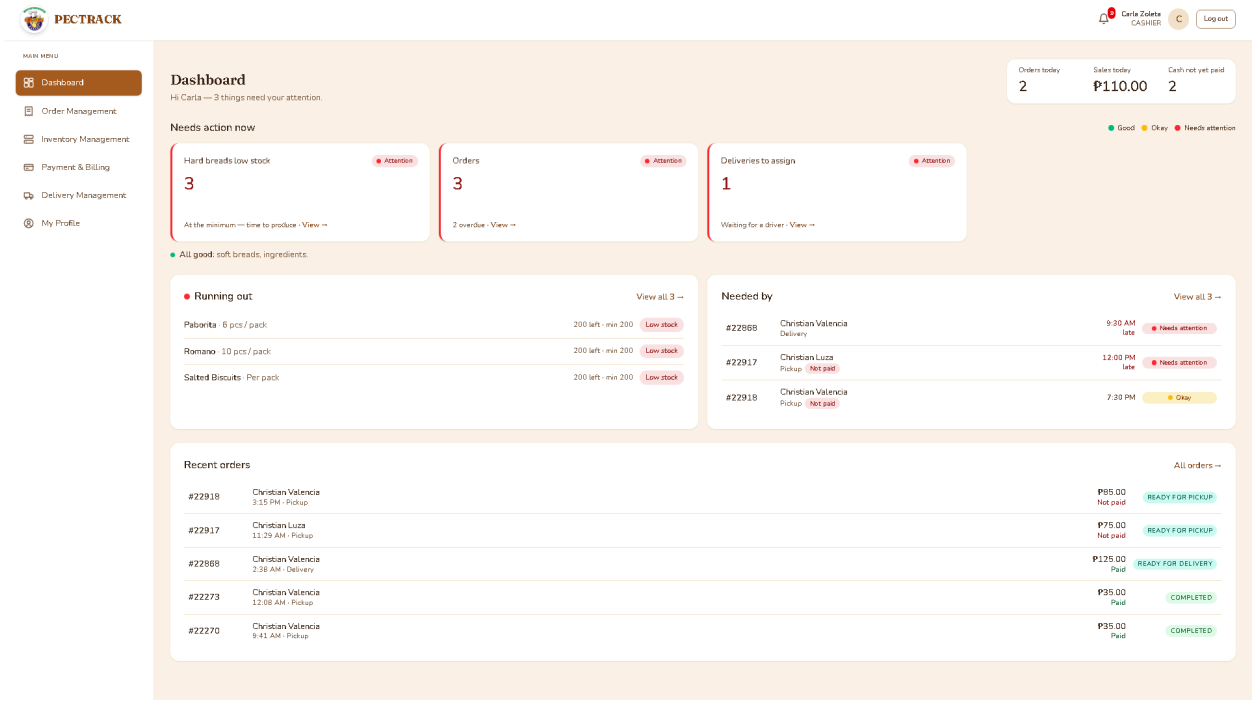


Figure 2.1 *Cashier dashboard*

The dashboard opens with a greeting that states how many matters need attention. It is followed by the Today row, which shows the orders and sales for the day and the cash not yet paid, the status tiles, and the Running Out, Needed By, and Recent Orders lists.

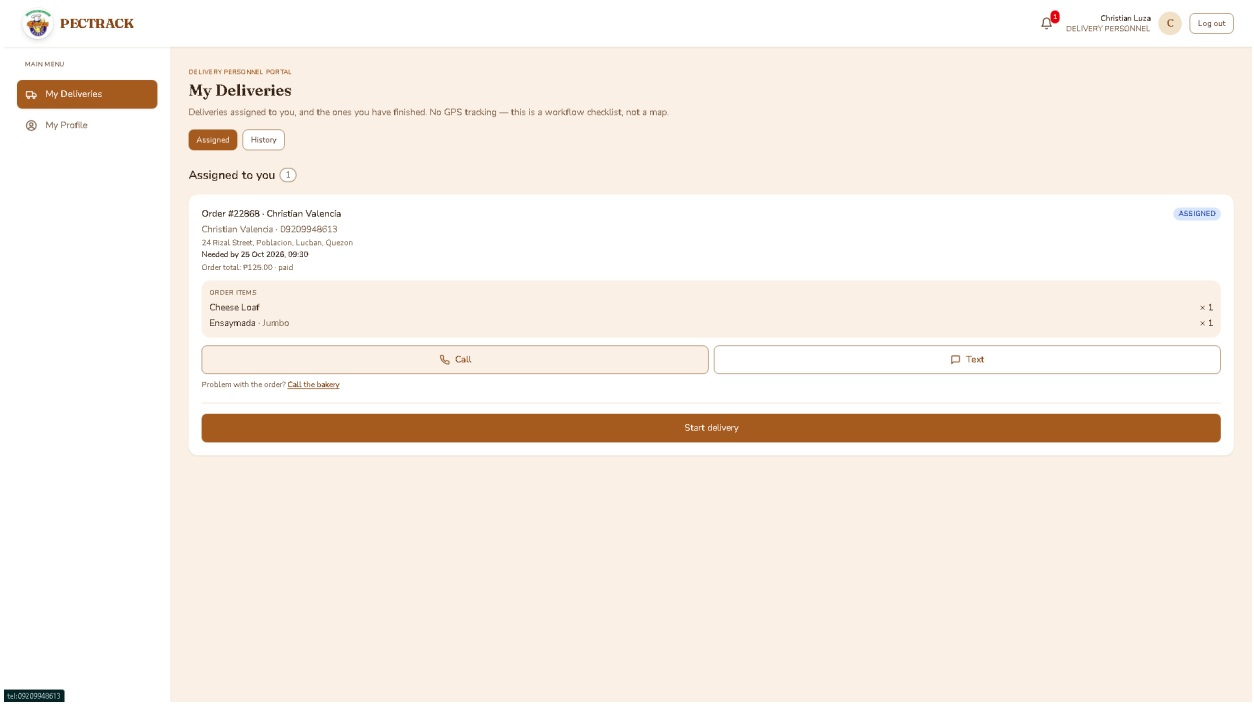


***Figure 2.2** My Deliveries (delivery personnel)*

Delivery personnel are directed to My Deliveries, which is designed for use on a mobile phone. Each delivery card shows the customer's name with a tap-to-call contact number, the delivery address, the order total marked as paid, and a Start Delivery button. The system does not provide a map or GPS tracking; instead, each delivery is managed as a status checklist.

USER MANAGEMENT MODULE

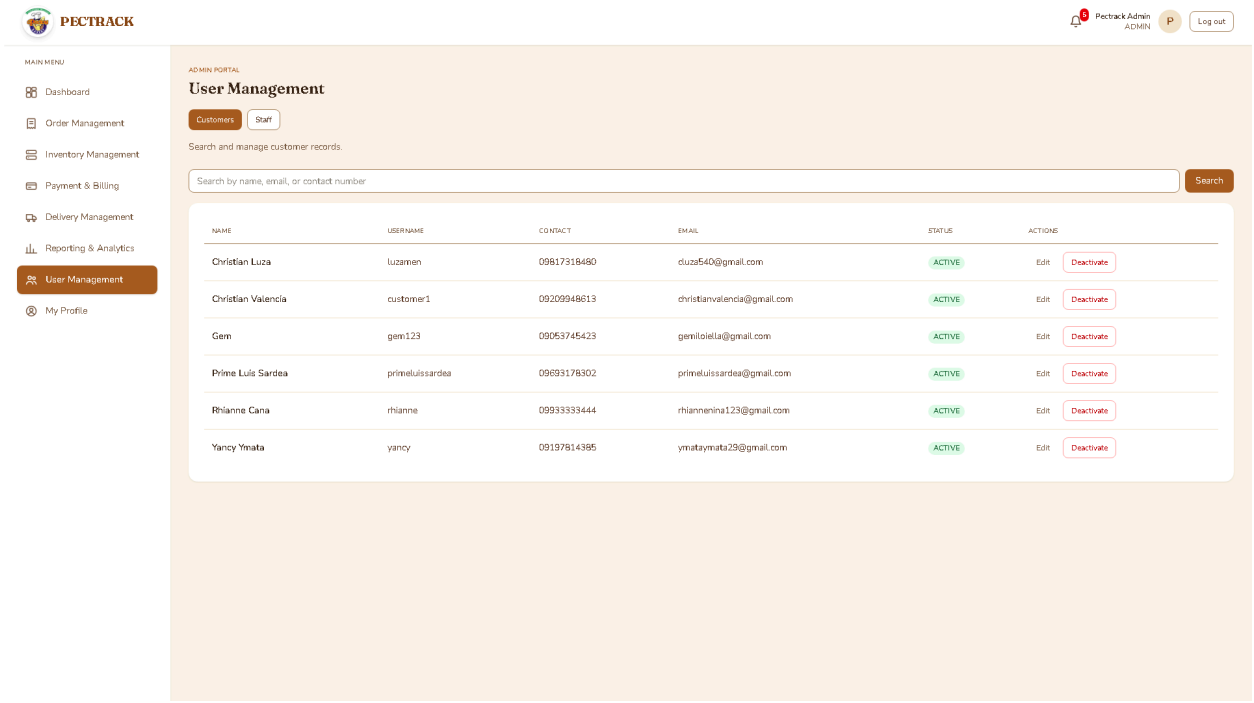


Figure 3. *User Management —Customers*

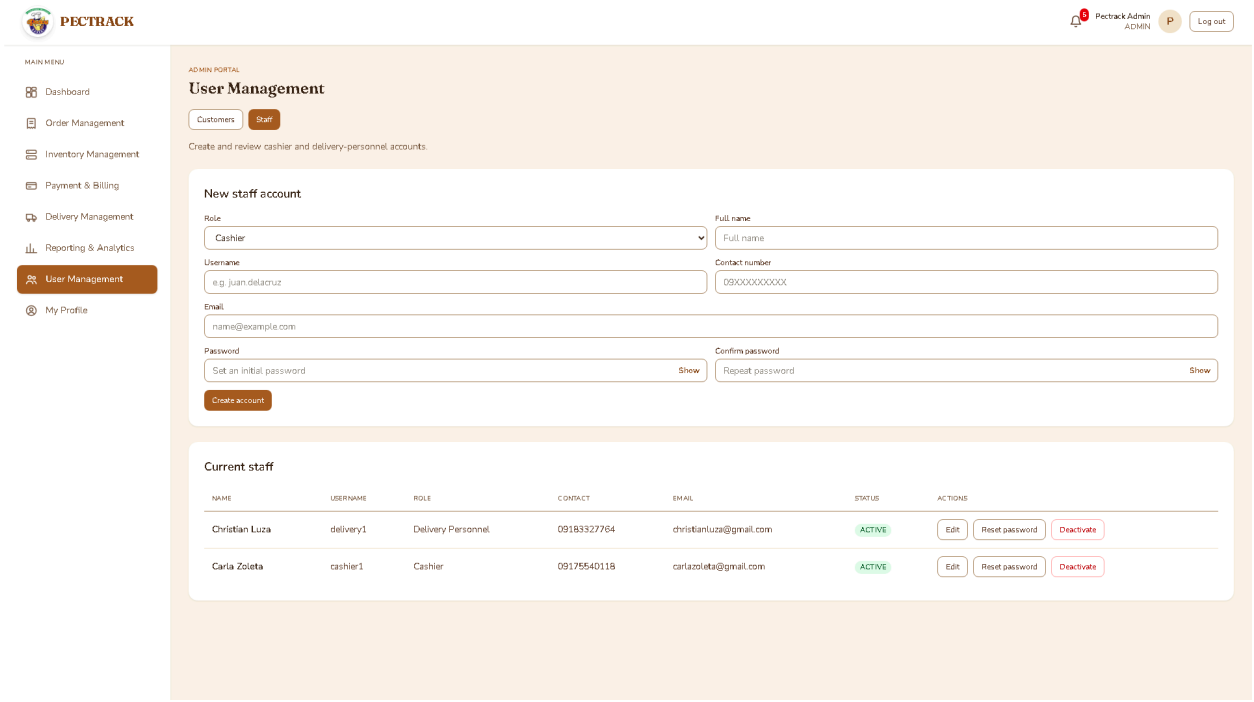


Figure 4. *User Management — Staff*

The Customers tab lists every registered customer with the name, username, contact number, email address, and account status, together with a search box, and allows the administrator to edit or deactivate an account. The Staff tab is used to create and manage the accounts of cashiers . Both tabs are accessible to the administrator only.

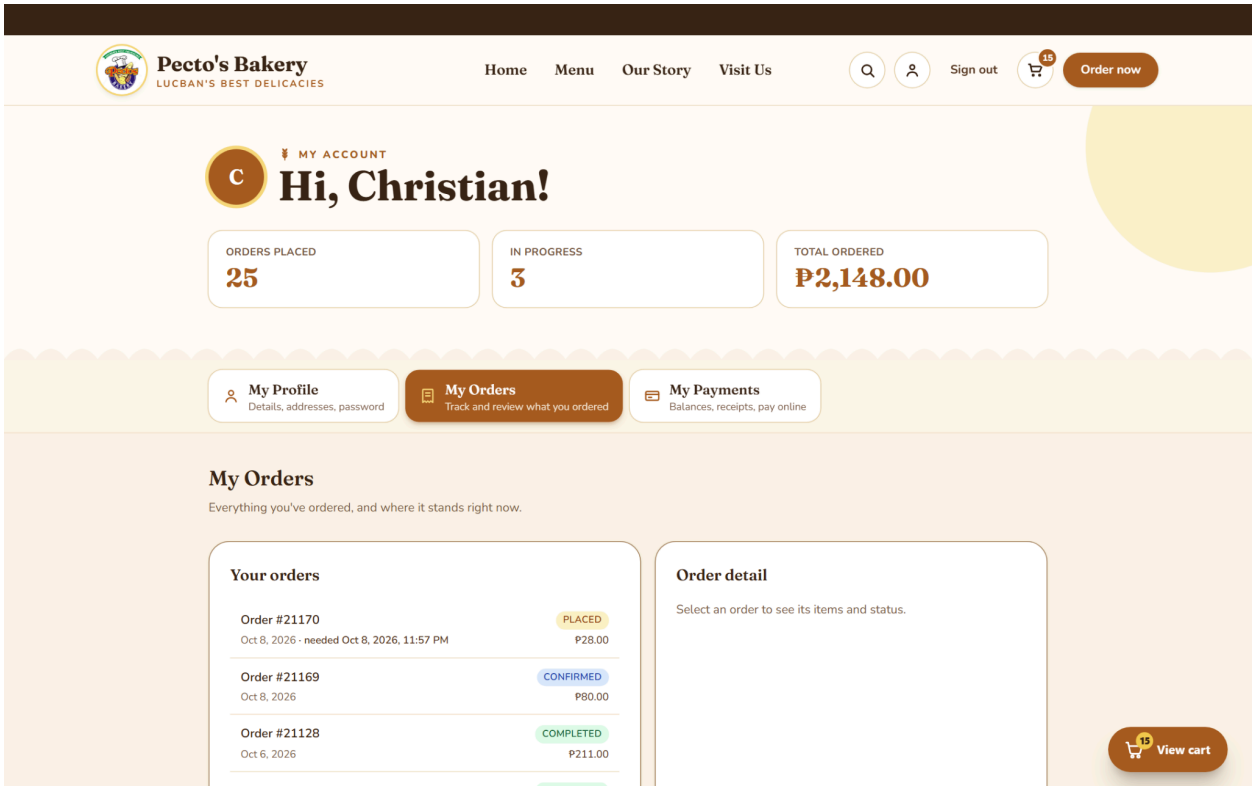


Figure 5. *Customer account — My Orders*

The customer's account page greets them by name and shows three figures from their own orders: orders placed, orders in progress, and the total amount ordered. Three tabs follow: My Profile, My Orders, and My Payments. My Orders lists the customer's orders, newest first, with their status; selecting one shows its items, requested time, delivery progress, the amount paid and balance due with a "Pay now" button, and, while the order is still Placed, a Cancel button that asks for a reason.

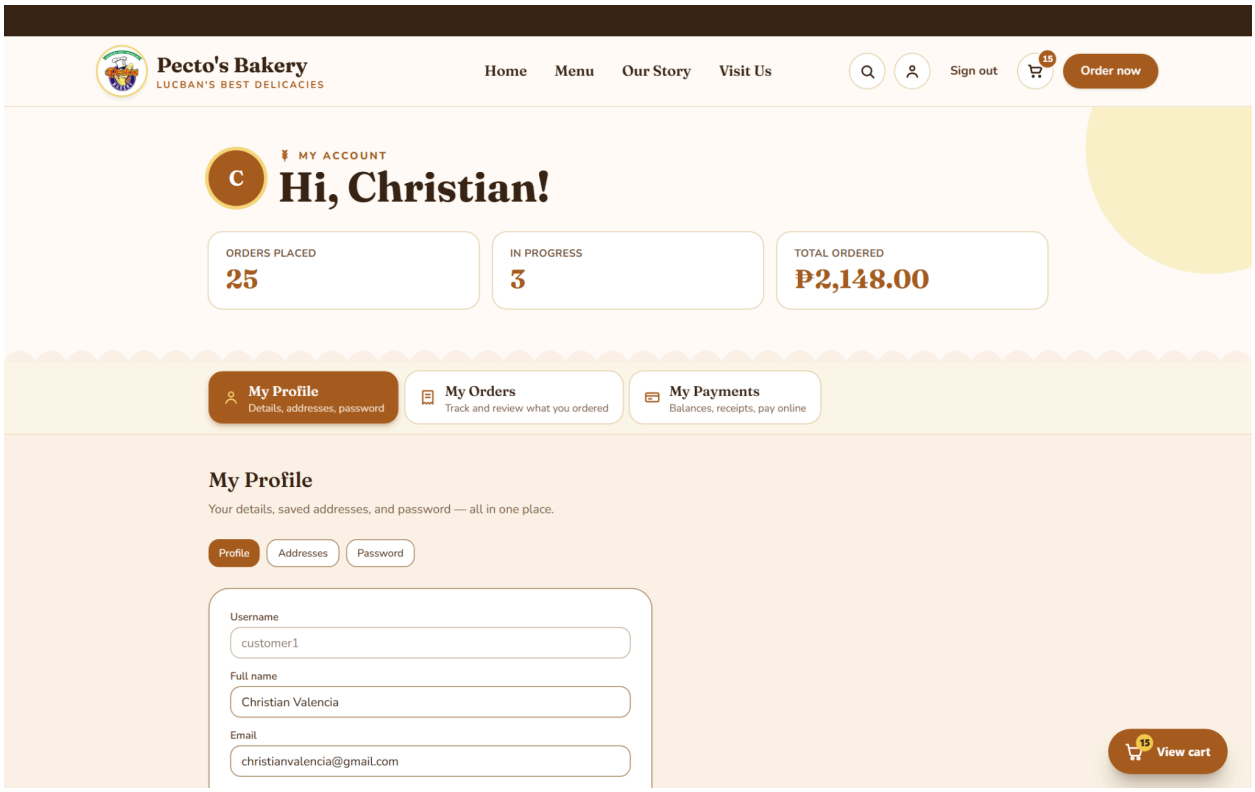


Figure 5.1 *My Profile*

My Profile has three sections: Profile, Addresses, and Password. The Profile section shows the username and lets the customer update their full name and contact number. Changing the email address requires the current password, and a notice is sent to the old address.

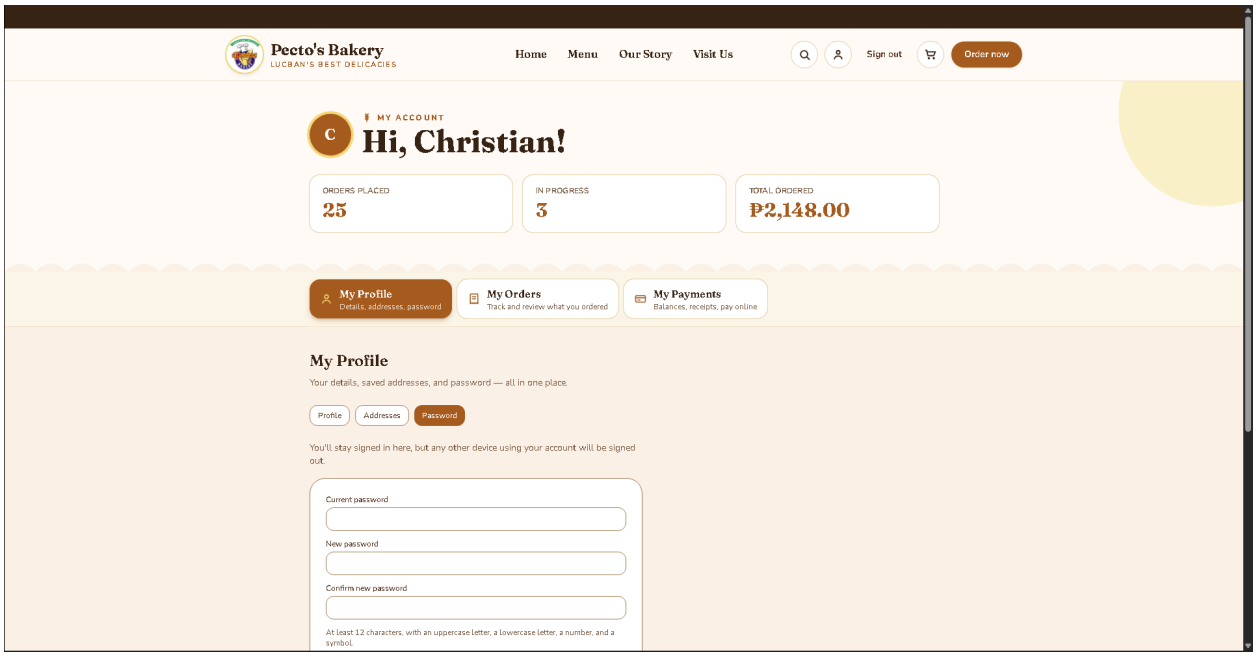


Figure 5.2 *Customer account — change password*

The Password section changes the account password. The current password is required, and the new password must meet the same rules as at registration. In the final system, the customer's account page has three tabs — My Profile, My Orders, and My Payments — and the profile holds the name, username, email address, contact number, and saved addresses.

The customer account page has three tabs. The My Profile tab contains the customer's profile, saved addresses, and password; the My Orders tab lists the customer's orders together with their status and receipts; and the My Payments tab lists the customer's payments.

ORDER MANAGEMENT MODULE

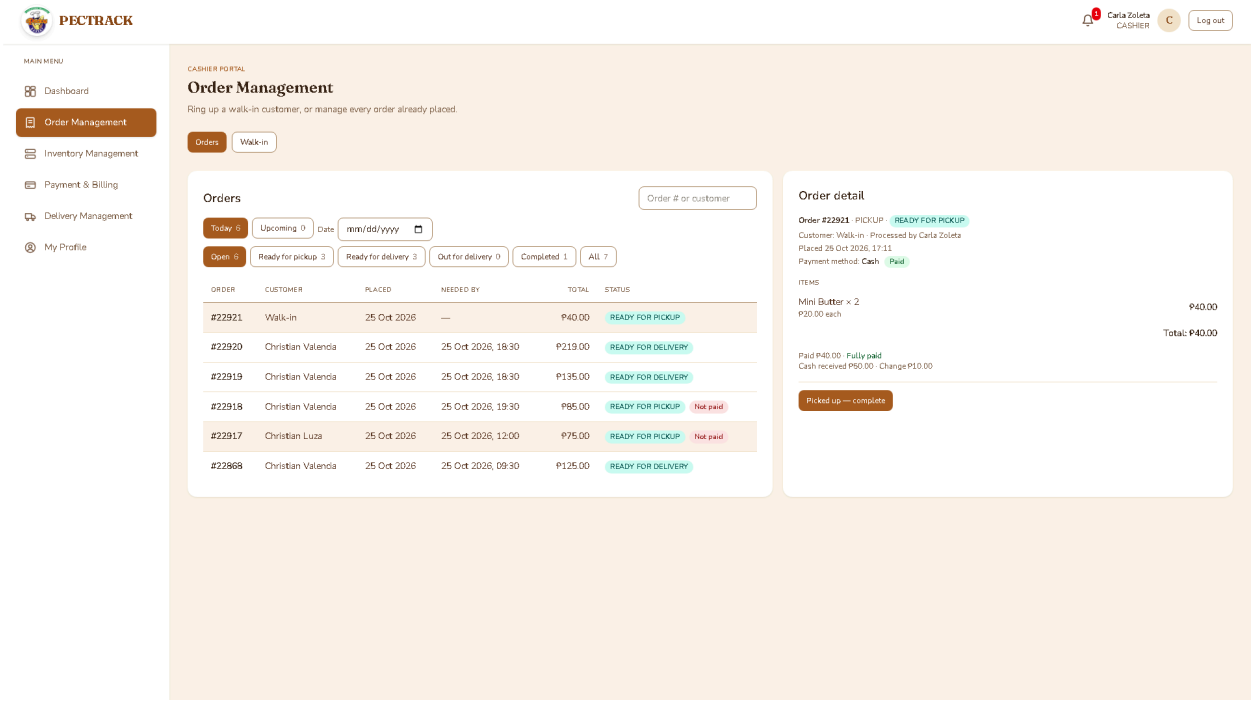


Figure 6. *Order Management - Orders (admin)*

The admin's Order Management page lists the orders with their customer, date placed, requested time, total, and status, and can be searched and filtered by status. Selecting an order opens its details: the customer, order type, requested time, delivery progress, items and prices, the amount paid and balance due, and the status history with who changed it and when. A status can only move forward; cancelling requires a reason.

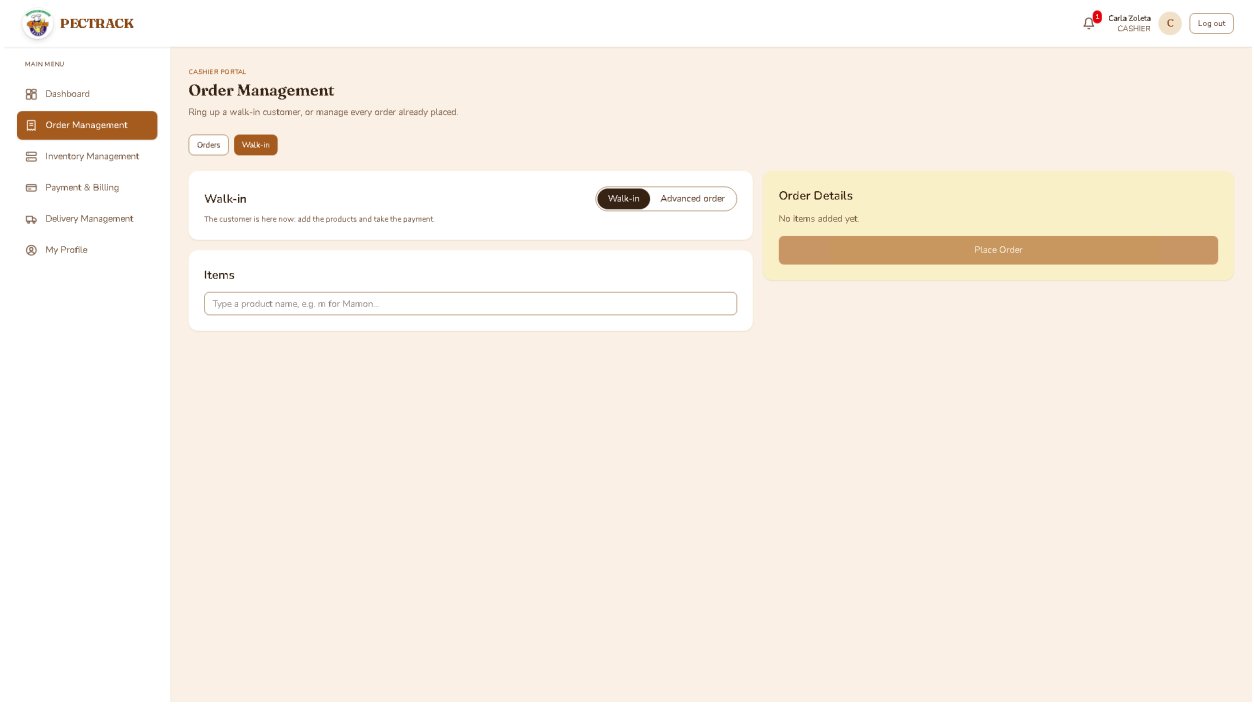


Figure 7. *Walk-in / Advanced order (cashier)*

The cashier's Order Management page has two tabs: Orders, the same searchable order list and detail as the admin's, and New order, for orders taken at the counter.

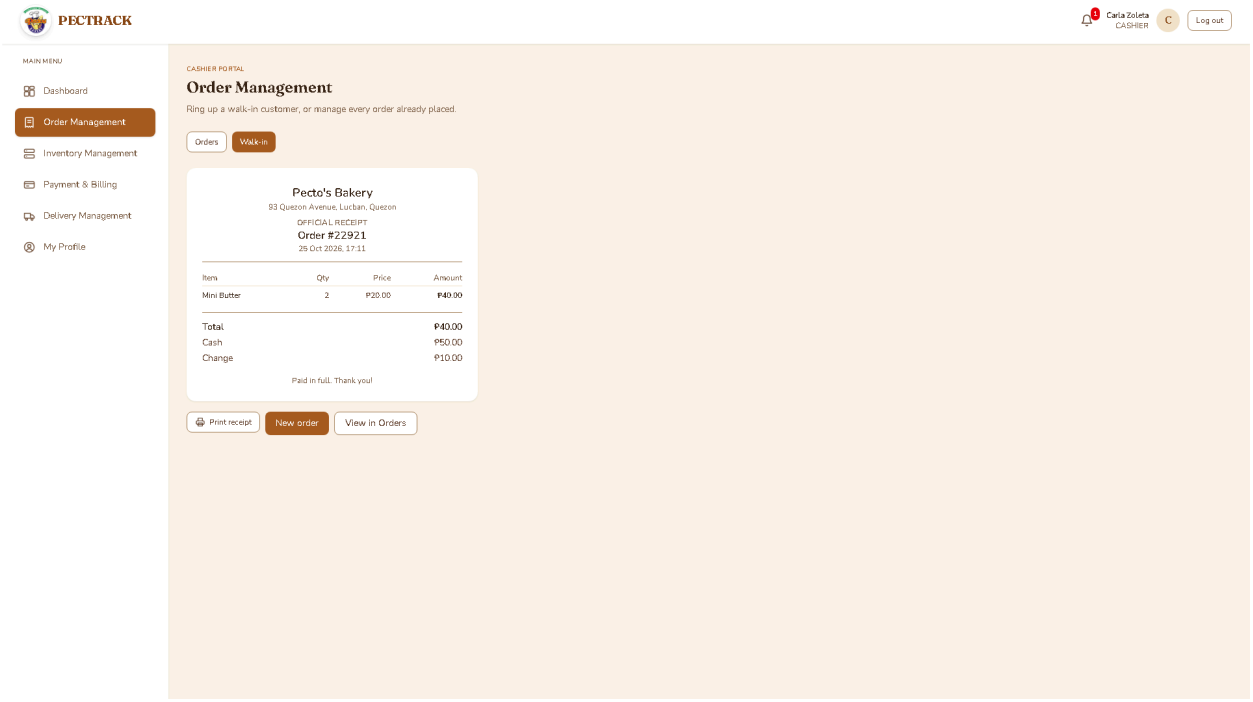


Figure 7.1 *Walk-in receipt (cashier)*

The Orders tab shows the orders for the current day, the upcoming orders, or the orders for a chosen date, each with its status, the customer, and a Not Paid tag for cash orders still awaiting payment. The order details show the items, the requested time, and the payment method and payment status, including the cash received and the change. For a delivery awaiting a rider, the order details also provide an Assign Rider control. The Walk-in tab switches between walk-in and advanced orders, as described in the Purpose and Description.

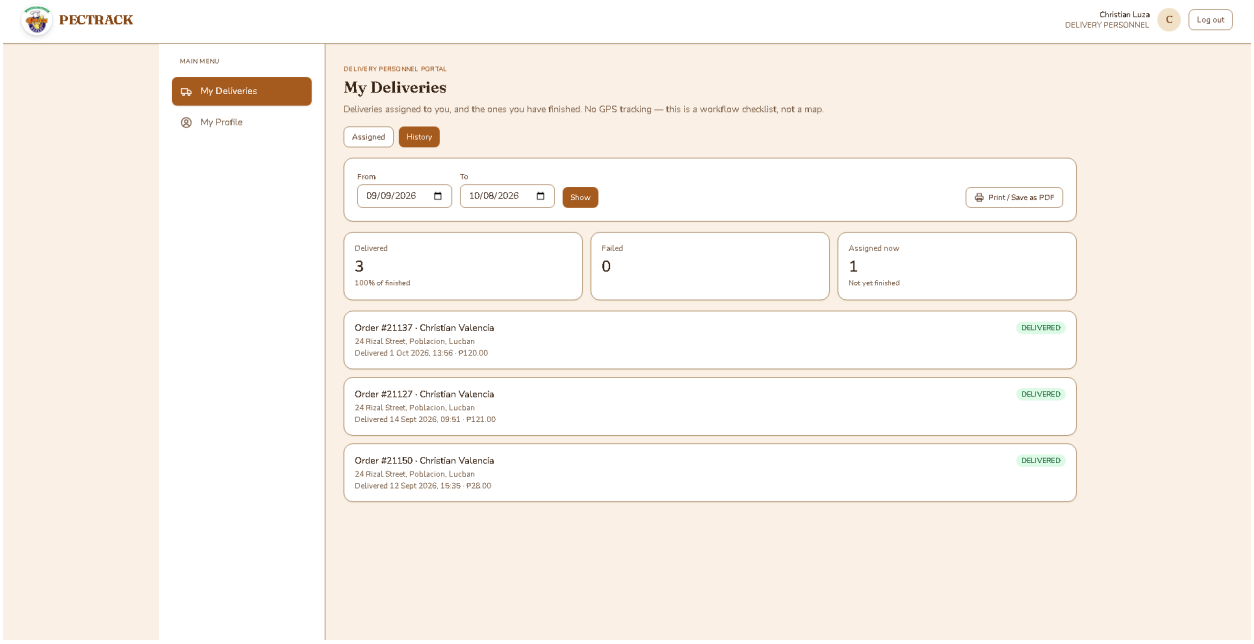


Figure 8. *Delivery in progress (delivery personnel)*

After the rider selects Start Delivery, the card shows the proof-of-delivery photo upload, which accepts up to five JPEG or PNG photos, the Mark Delivered button, and the Couldn't Deliver option with a field for the reason.

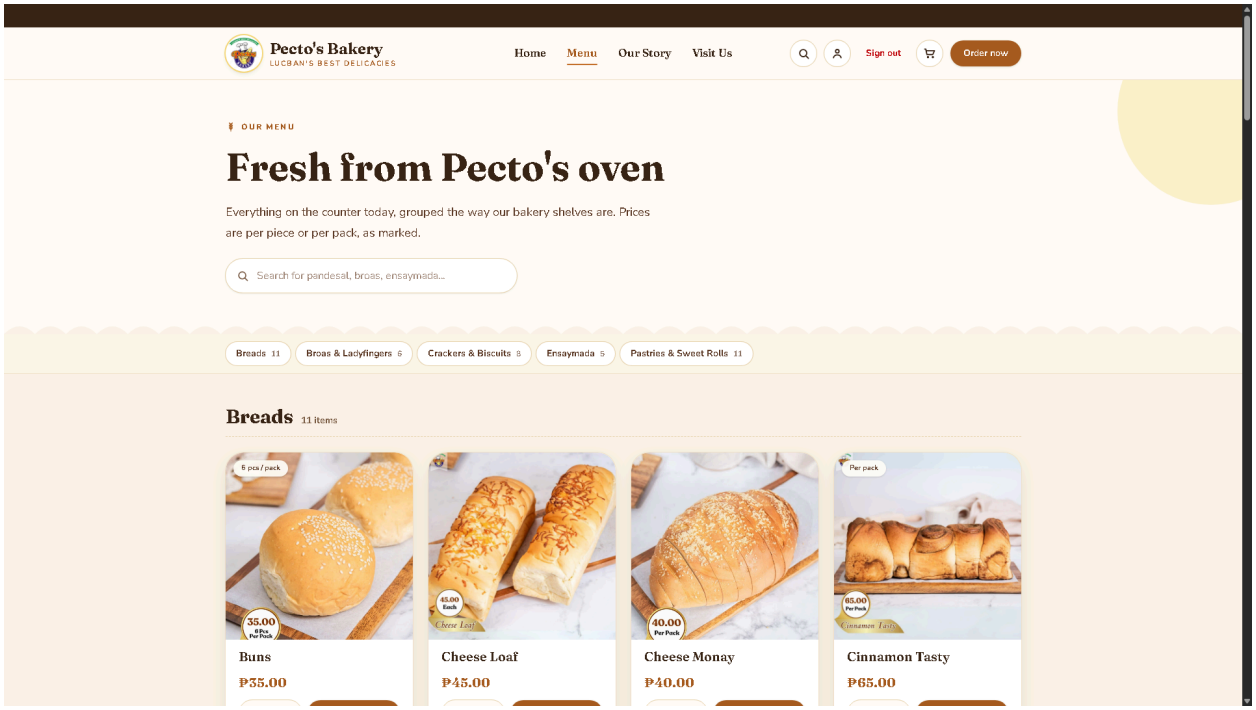


Figure 9. *Menu (customer)*

The Menu displays the products of the bakery grouped by category, with a search box. Each product shows its photo, name, size or pack, and price, together with an Add to Cart button. Sold-out products are marked as such and cannot be ordered.

INVENTORY MANAGEMENT MODULE

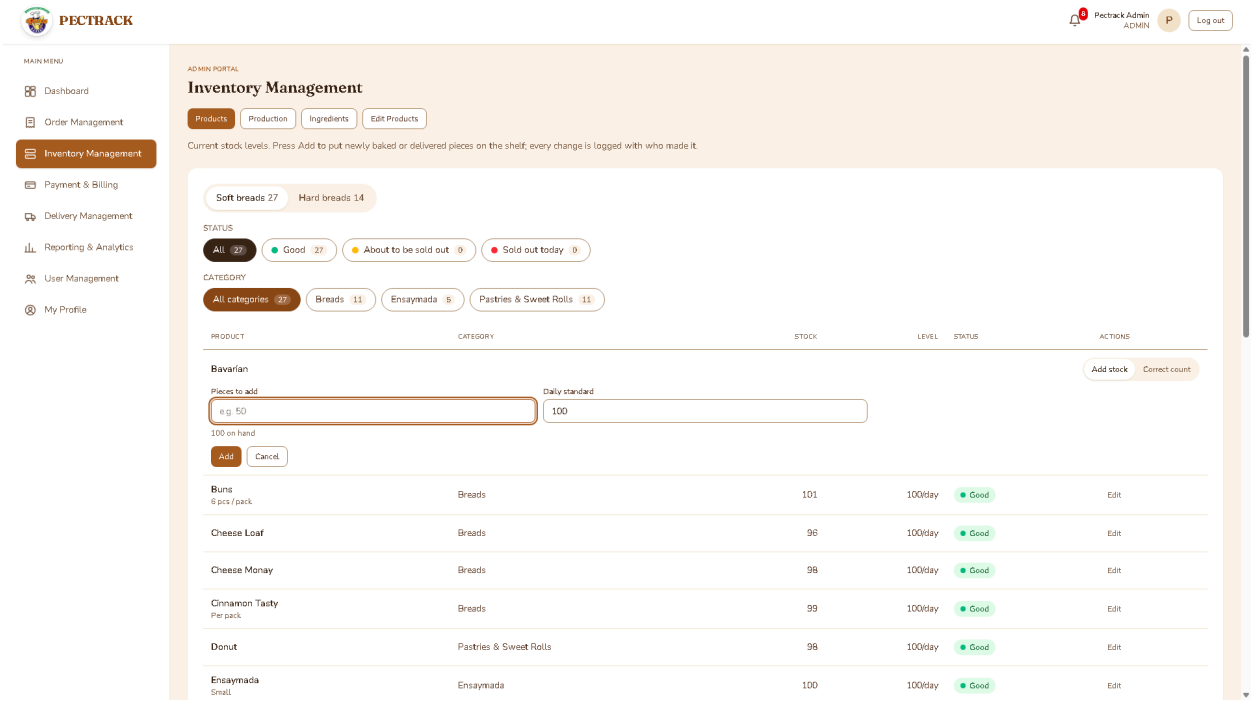


Figure 10. *Inventory — Products (soft and hard breads)*

The Pectrack Product & Inventory admin portal features a navigation menu highlighting "Product & Inventory," a main dashboard table with an "Add Product" button that lists product IDs, names, prices, and availability statuses like "In Stock" or "Out of Stock," and a "Product Details" section below for a selected item that includes a product image, an inventory summary, and action buttons to edit or delete the product.

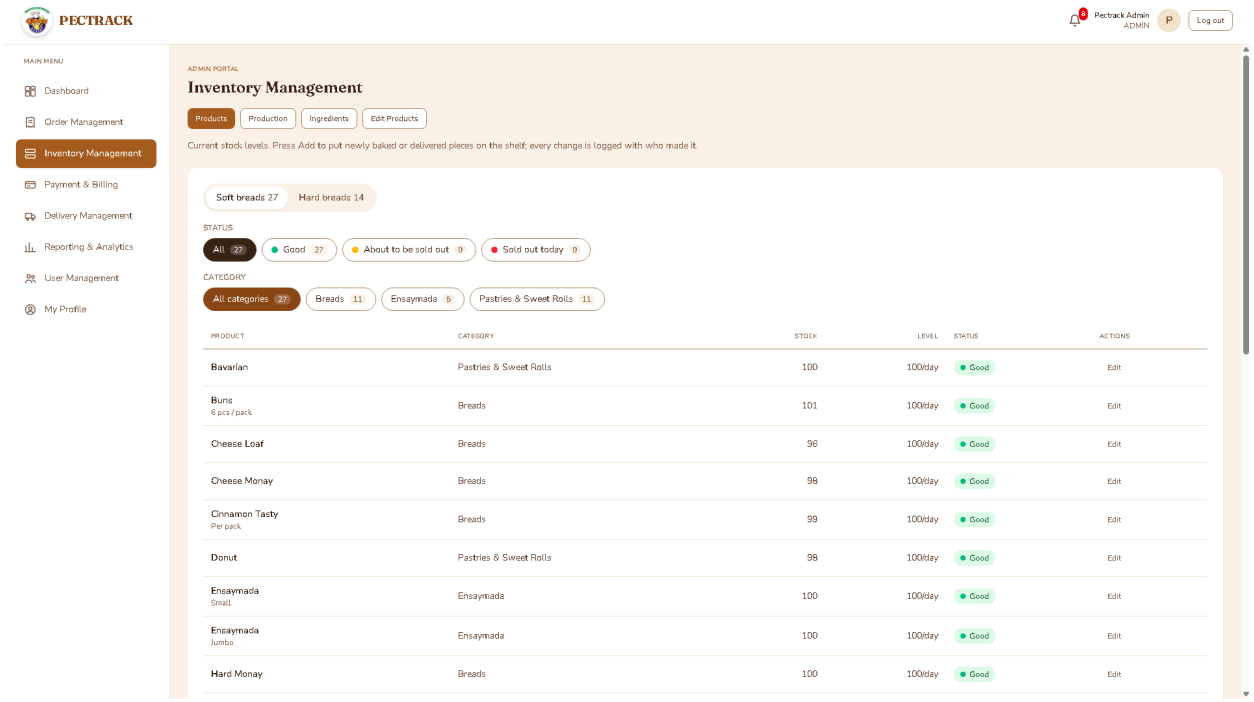


Figure 10.1 *Edit (Add stock / Correct count)*

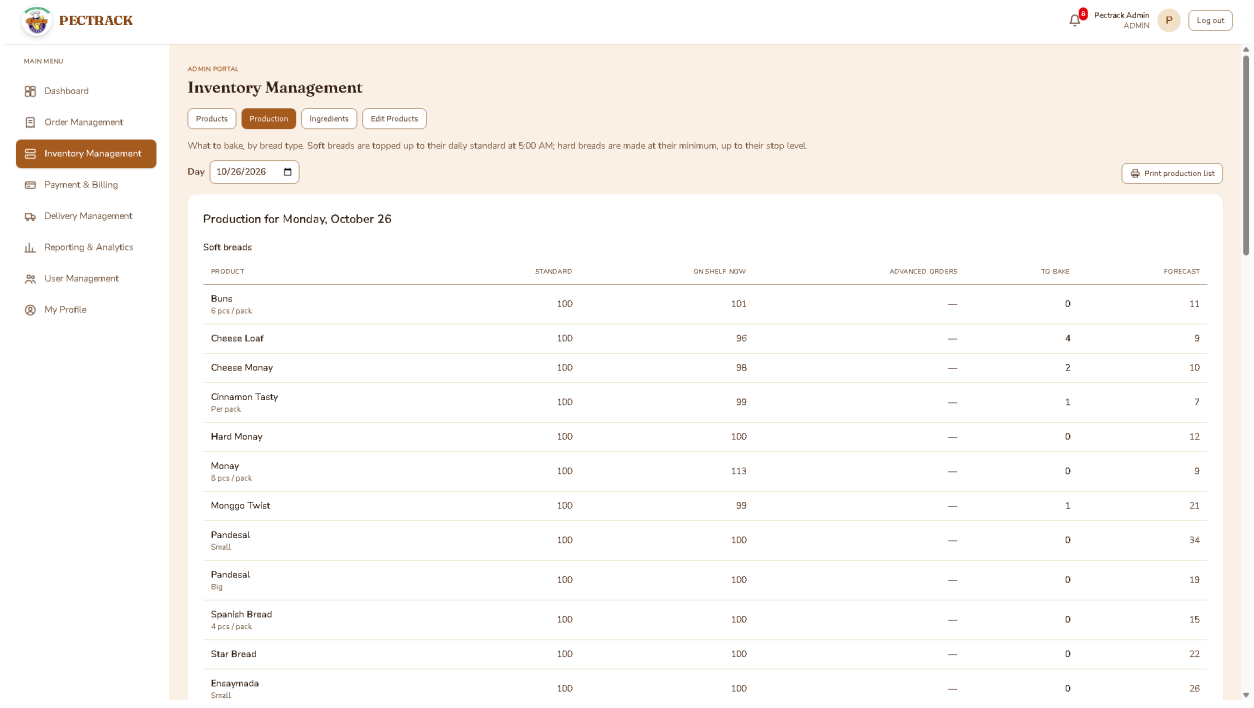


Figure 10.2 *Production and Events, with the forecast check*

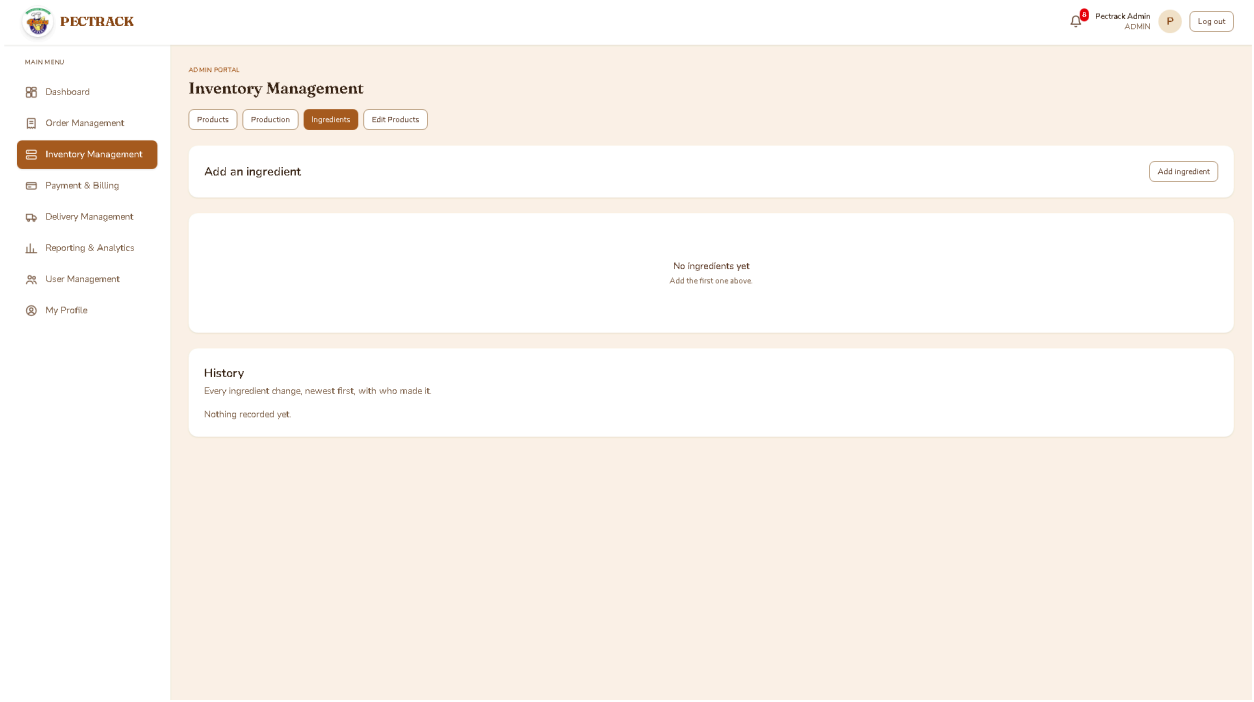


Figure 11. *Ingredients*

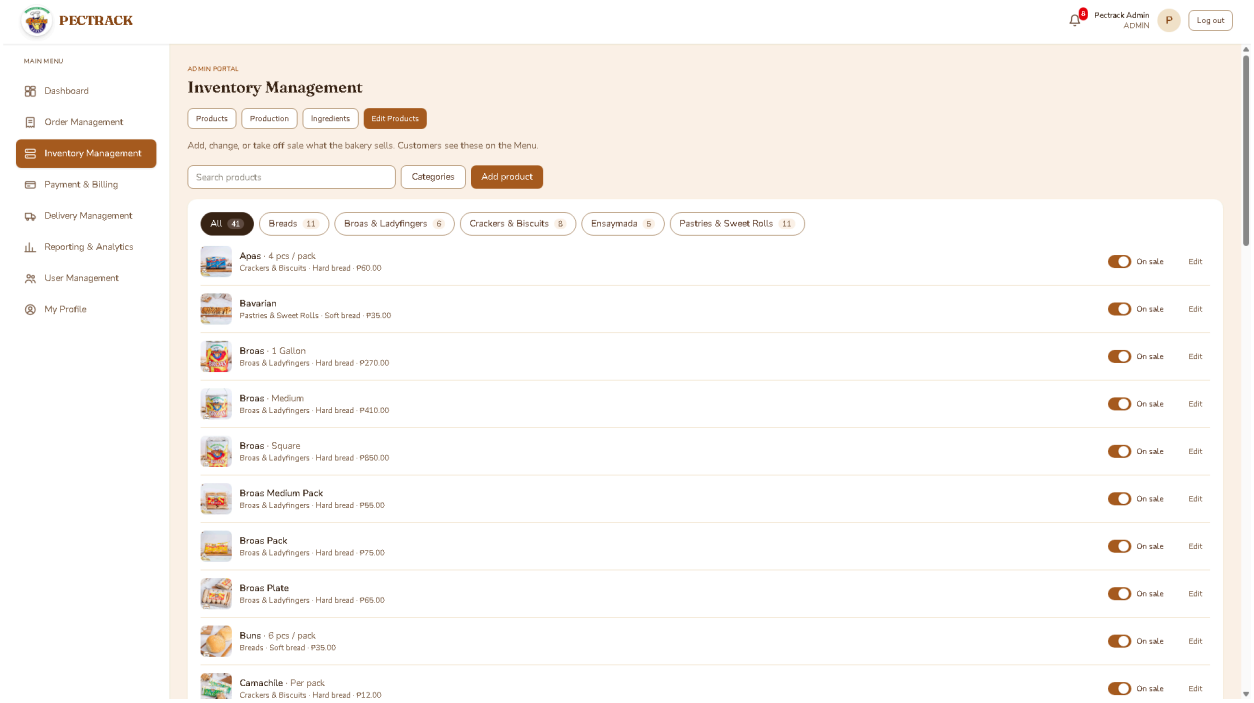


Figure 11.1 *Edit Products*

The Products tab lists soft breads and hard breads separately, each with its own status and category filters. The Edit button opens the Add Stock option and, for the administrator, the Correct Count option, each of which requires confirmation. The Production tab lists the quantities to be baked for a chosen day, with the forecast for each product and the Events calendar below the list. The Ingredients tab monitors the stock of ingredients, while Edit Products maintains the product catalog.

PAYMENT AND BILLING MODULE

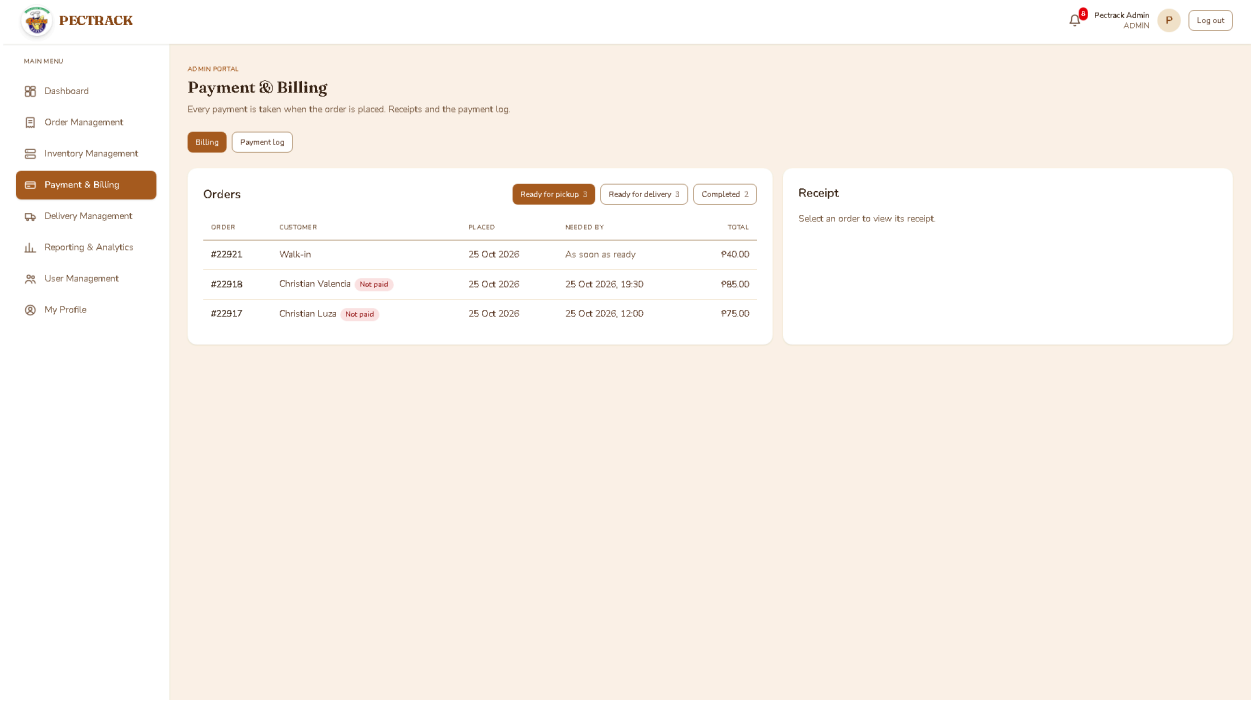


Figure 12. *Payment & Billing — Billing*

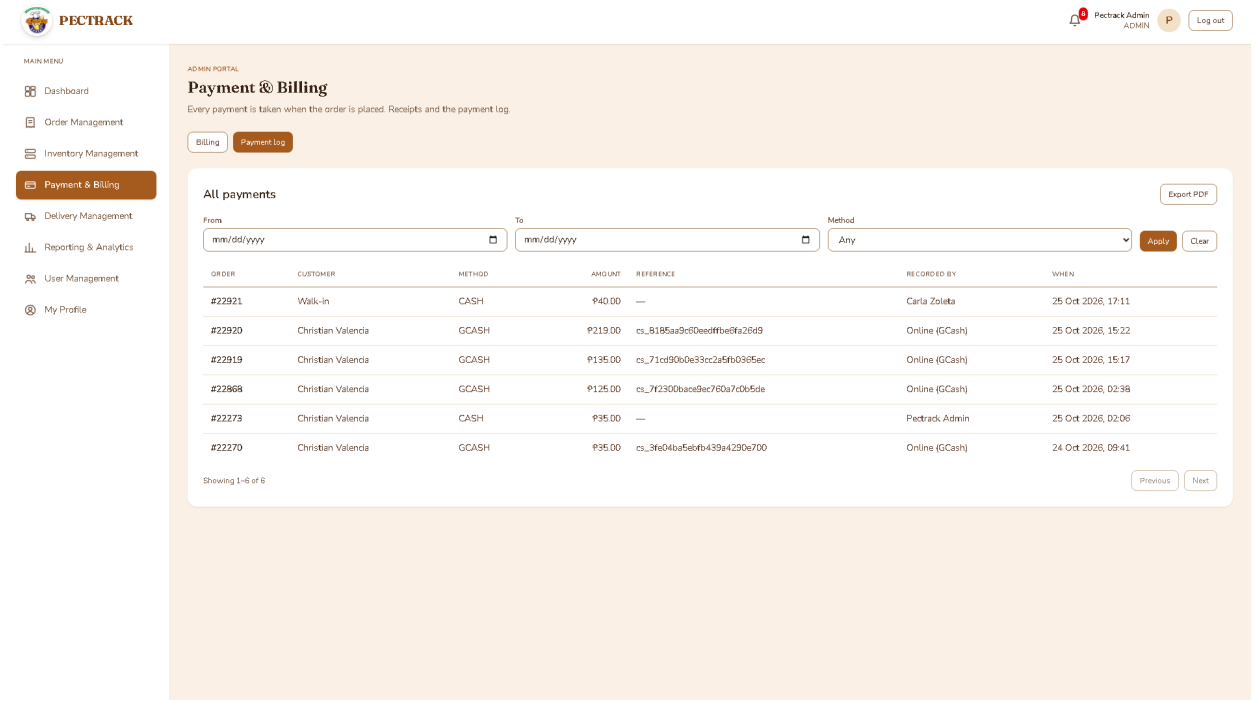


Figure 13. *Payment log and its PDF*

The Billing tab lists the orders that are ready for pickup, ready for delivery, and completed, each with its receipt, payment method, and a Not Paid tag for cash orders still awaiting payment. The Payment Log lists every payment received and can be exported as a PDF file that shows the total.

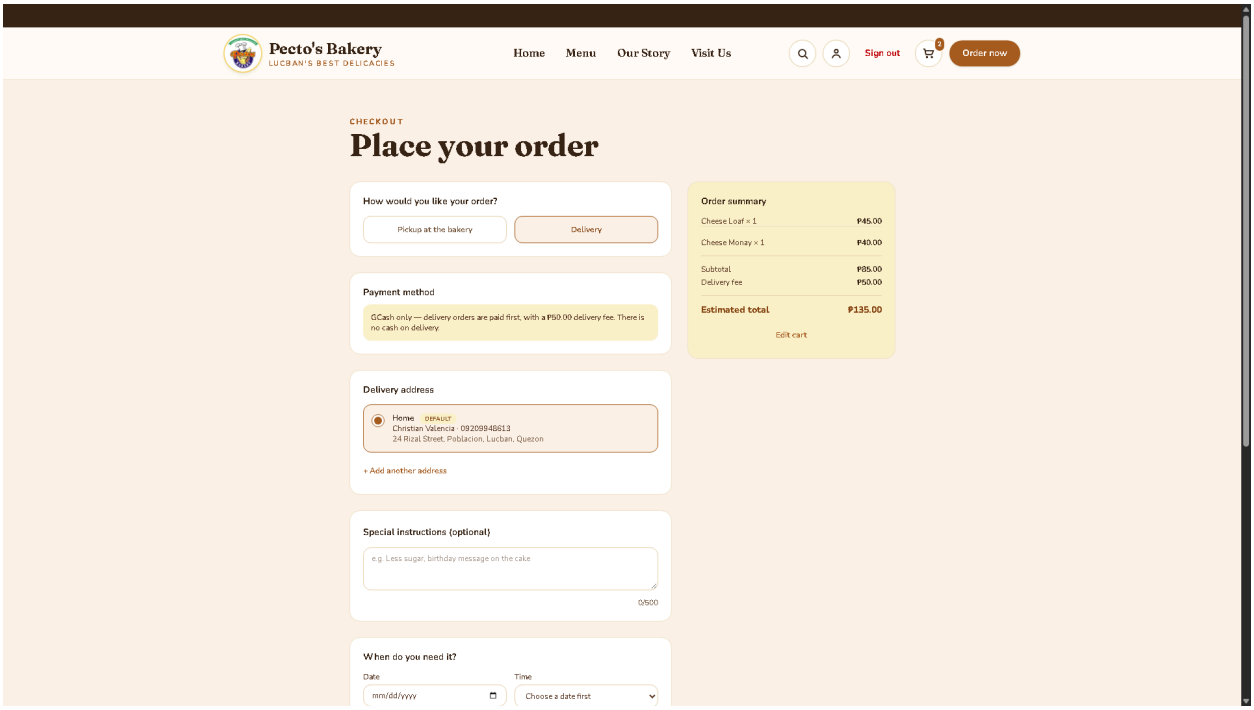


Figure 14. *Checkout — delivery order (customer)*

At checkout, the customer chooses between pickup and delivery and selects the date and an available time, as well as a saved address in Lucban in the case of delivery. A pickup order is paid in cash at the counter or through GCash, whereas a delivery order is paid through GCash, with the order summary showing the ₱50 delivery fee included in the total.

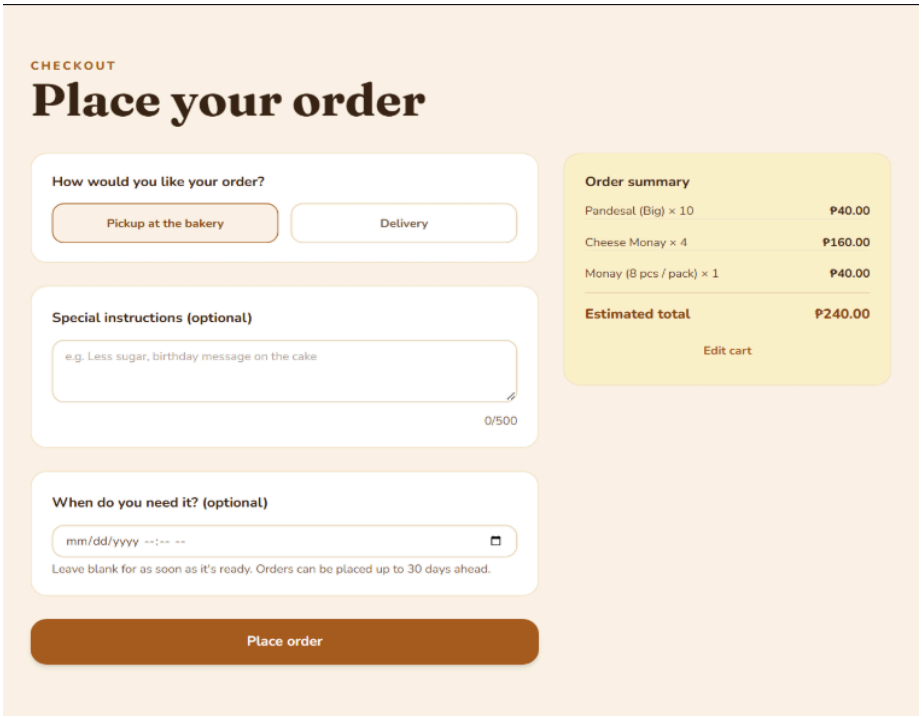


Figure 14.1 *Checkout-delivery order(customer)*

A pickup order needs no address. The customer pays online beforehand from My Payments, or at the counter when collecting the order.

DELIVERY MANAGEMENT MODULE

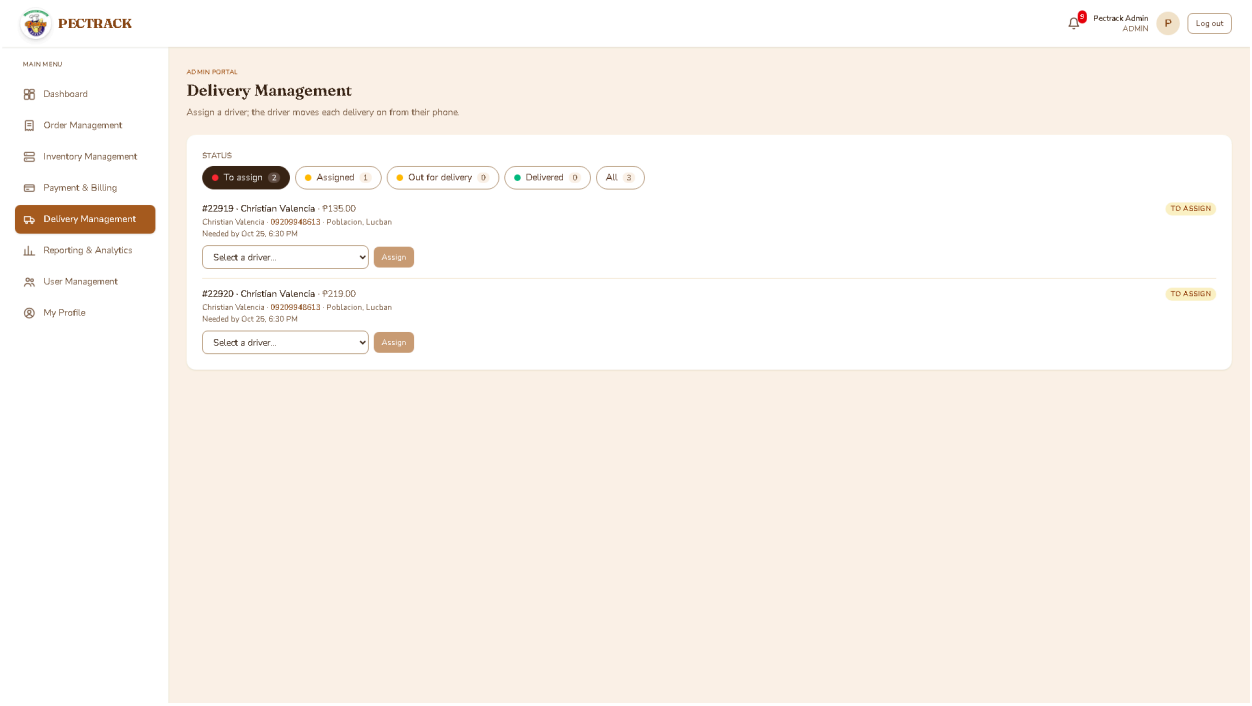


Figure 15. *Delivery Management*

Staff members view the deliveries that are awaiting assignment, assigned, out for delivery, and delivered, and assign each waiting delivery to a rider.

REPORTING AND ANALYTICS MODULE

The Reporting and Analytics Module has six tabs:

- Sales: daily, weekly, or monthly totals over any date range, with a sales trend chart and the peak ordering periods by hour of day and day of week.
- Products: the best sellers.
- Inventory: stock movements, and products currently low on stock.
- Forecast: expected sales for the next seven days and how much of each product to bake.
- Transactions: the payment log, filterable by date, payment method, and status.
- Export History.

The cashier sees the same reports as the admin, except Export History. Every report can be printed or saved as PDF.

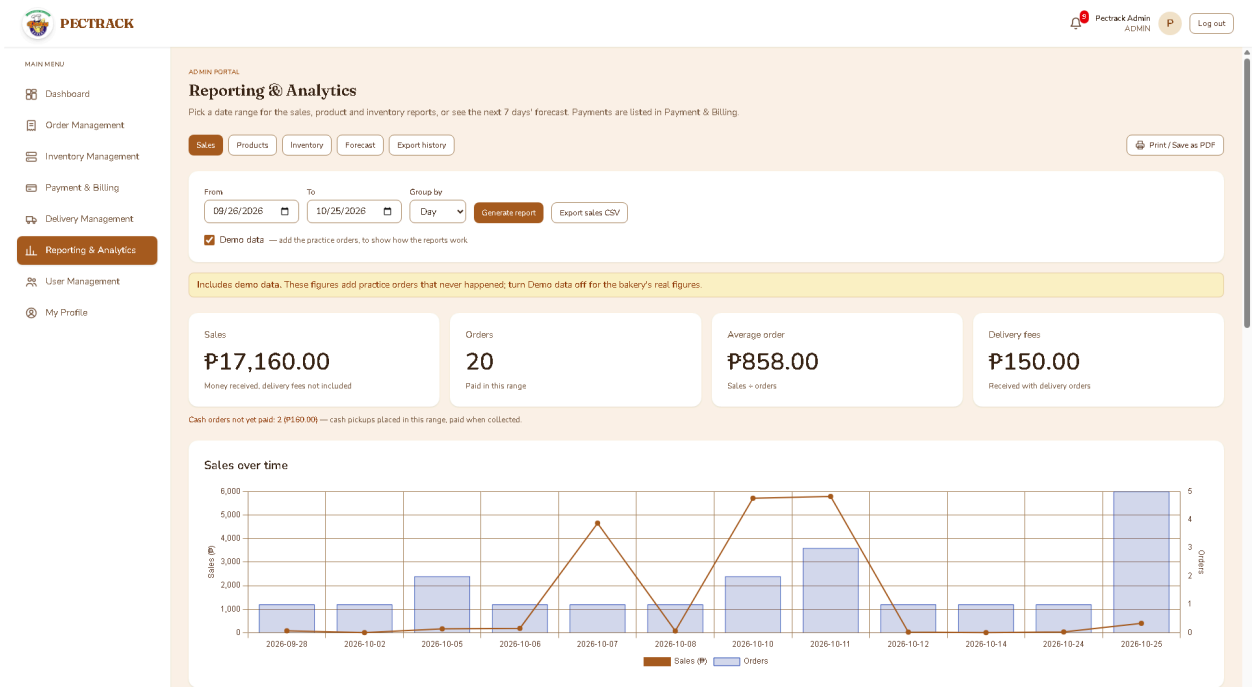


***Figure 16**. Sales(admin)*

The Sales tab: the date range and grouping at the top, then the orders placed, the amount ordered, the amount collected, and refunds for the period, followed by the sales chart.

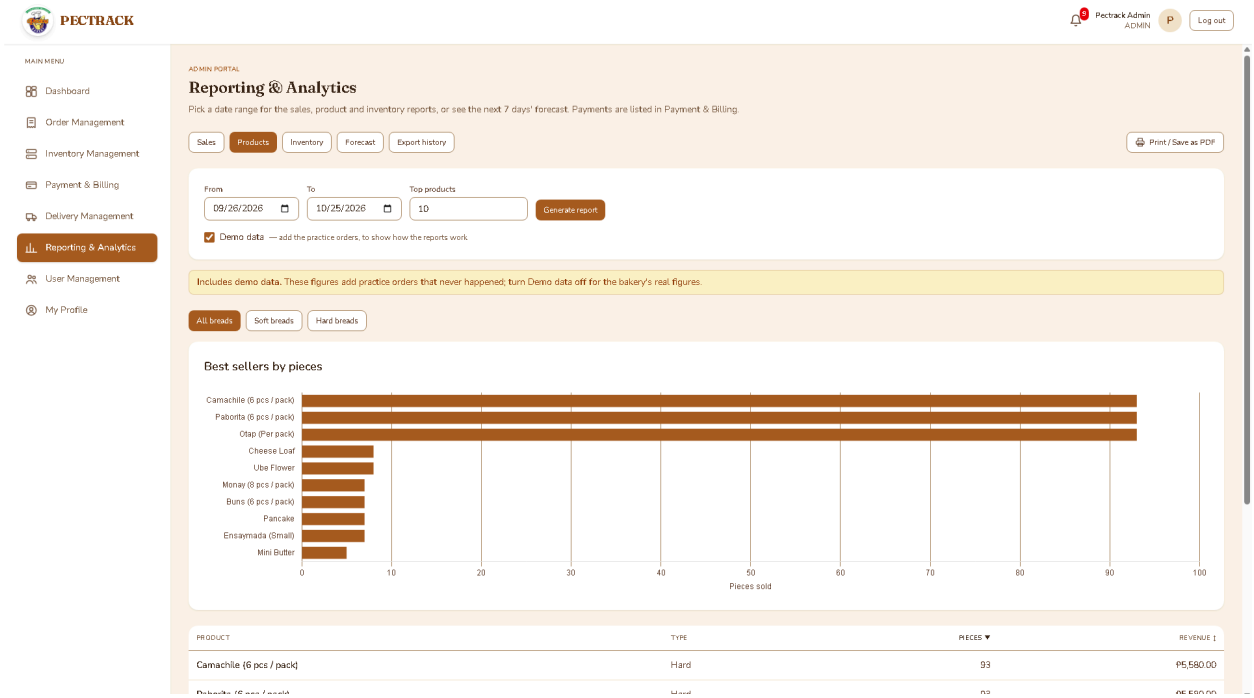


***Figure 16.1** Products*

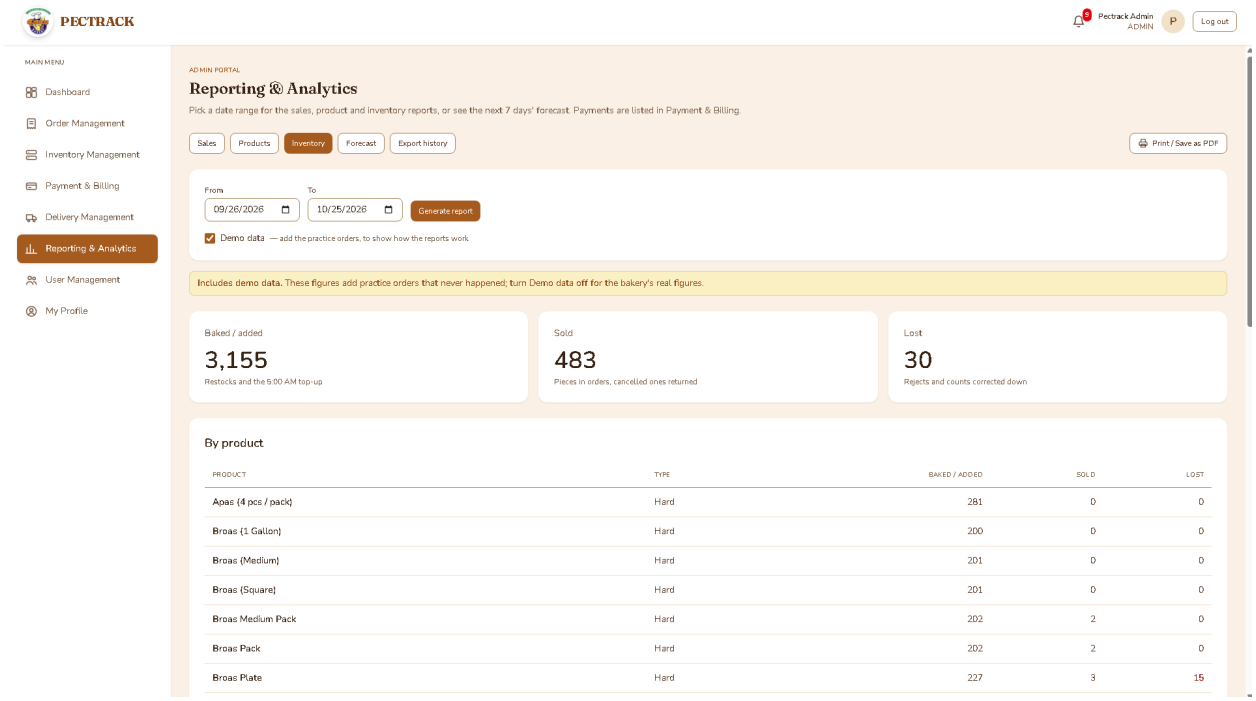


***Figure 16.2** Inventory*

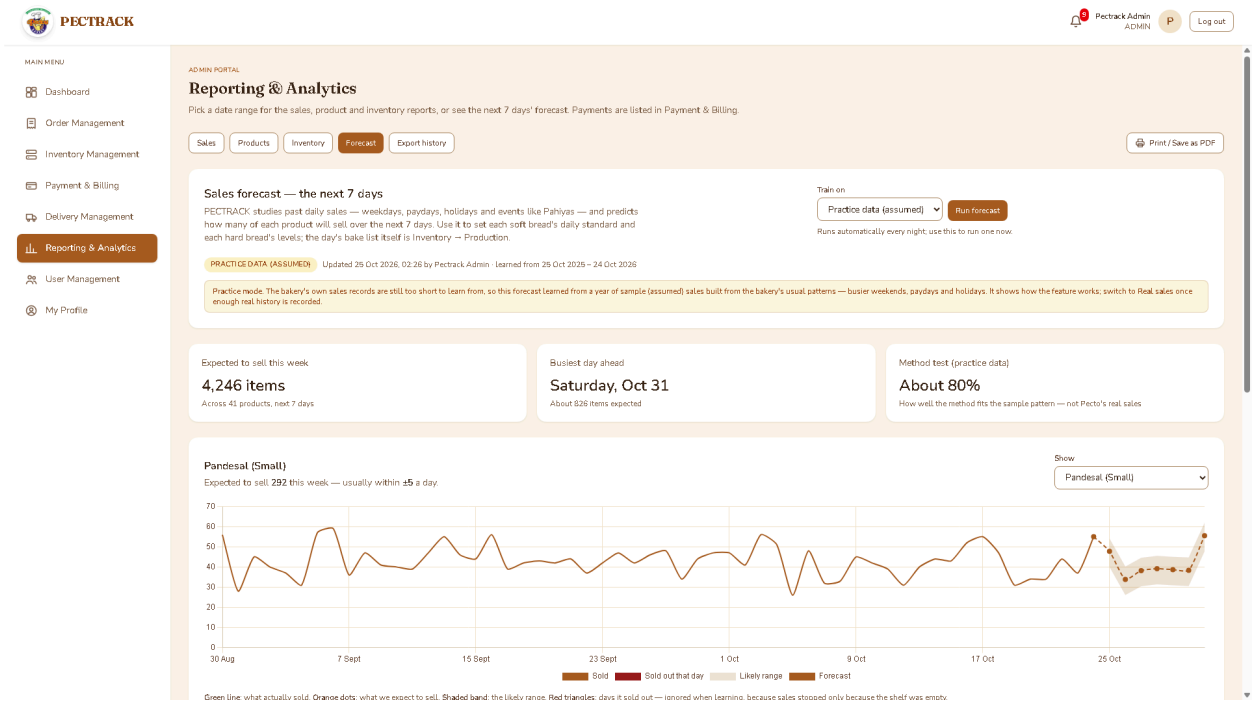


Figure 16.3 *Forecast*

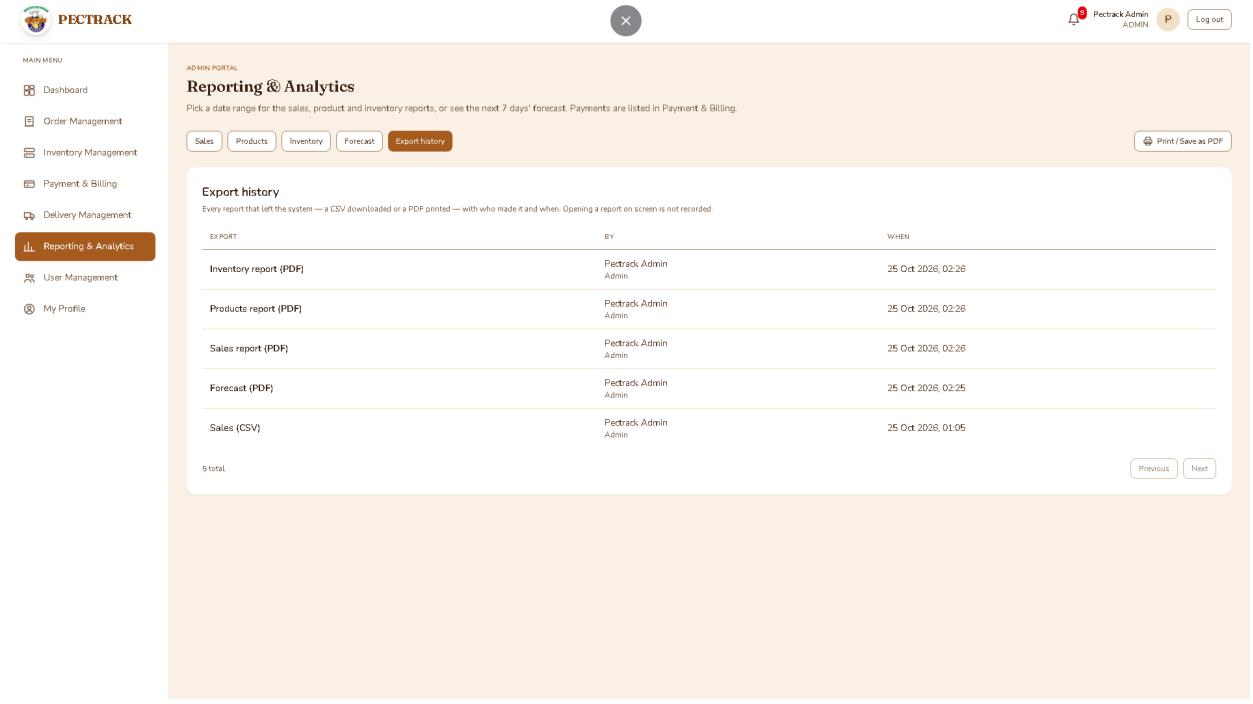
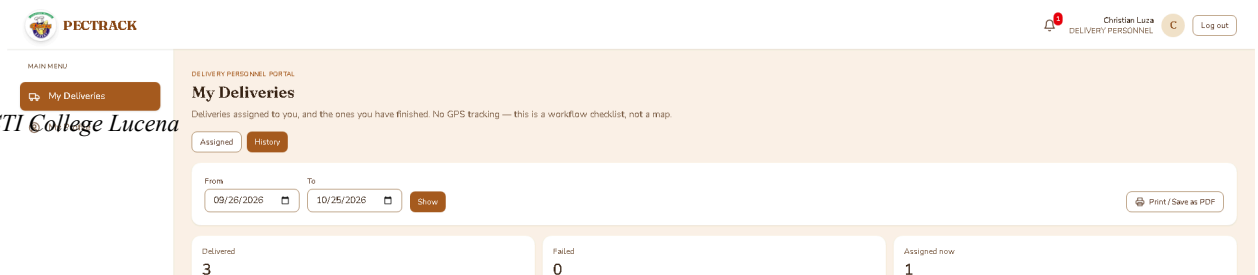


Figure 17. *Reporting & Analytics — Transactions (cashier)*



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Figure 18. *Delivery history (delivery personnel)*

The module contains the Sales, Products, Inventory, Forecast, and Export History tabs described in the Purpose and Description, and is accessible to the administrator only. Delivery personnel, on the other hand, can view their own delivery history over a chosen date range.

Limitations

1. Prototype and Data Exclusivity for Pecto's only.

The PECTRACK system is developed for, and tailored to, the operational needs of Pecto's Bakery in Lucban, Quezon, and all of its modules, workflows, and database structures follow the existing business processes of the bakery. All data stored, processed, and generated within the system, including product information, customer records, transaction histories, inventory data, and reports, belong to Pecto's Bakery and are used solely for the purpose of this capstone project. Delivery is limited to the municipality of Lucban, Quezon, for a flat delivery fee of ₱50. Delivery orders must be paid in full beforehand through

GCash, as the system does not offer cash on delivery. Cash is accepted only at the counter, including for online pickup orders paid upon collection, and paid orders are not refunded.

2. Network Dependency.

As a web-based platform, the system requires a stable internet connection. Any network interruption on the part of the cashier or customers may disrupt order placement, status updates, and payment processing.

3. Ingredient Monitoring Only.

Ingredient stock is monitored but not automated. Since the system does not store recipes, ingredients are not deducted when bread is baked; instead, ingredient counts are kept current through recorded deliveries and the administrator's stock checks.

4. Assumed Sales History for the Forecast.

The bakery has not kept sales records that the system can import. As approved by the adviser, the forecast is therefore demonstrated on an assumed one-year sales history built from the patterns described by the bakery, namely higher sales on weekends, sales of about four times the usual during events, mainly in hard breads, and payday effects. The assumed data are labeled as such and are kept out of the sales reports. The forecasting method is validated on this assumed history; hence, the accuracy figure on screen, labeled "Method test (practice data)," shows how well the method fits the assumed pattern rather than the actual demand of the bakery. Its accuracy on the actual sales of Pecto's Bakery will be measured once 56 days of actual sales exist, at which point the system switches to them automatically. To demonstrate the reports, practice orders may be added; these are marked as practice data and appear only when the Demo Data switch is turned on. Furthermore, on days when a product sells out, the true demand is unknown, so

those days are excluded from the forecast's learning.

Review of Related Literature/Studies/Systems

Related Literature

Local Literature

Predictive Analytics

Predictive Analytics for Sales Forecasting and Inventory Management

Divina, Olan, and their colleagues (2025) developed a data-driven point-of-sale and inventory system for Pastil sa Tabi that combines sales forecasting with predictive

analytics. The study aimed to improve transaction processing, inventory monitoring, and demand prediction by integrating these functions into one system. Its approach demonstrates how historical sales records can help a food business estimate future demand and make better inventory decisions. However, forecasting accuracy depends on the quality of the available records and the sales patterns of the business. This study is relevant to PECTRACK because it provides a local example of integrating predictive analytics with sales and inventory management for a food-related enterprise.

Predictive Analytics and Linear Regression for Sales Forecasting

Lo, Tolentino, and their colleagues (2025) developed RELTRACK, a web-based system for inventory, appointments, and sales forecasting for REL Glass and Aluminum Works. The system uses linear regression to analyze sales data and estimate future sales performance. By combining forecasting with inventory monitoring, the system supports decisions based on recorded business transactions rather than relying entirely on manual estimates. However, its business setting differs from a bakery, where products may be perishable and daily demand may change considerably. This study is relevant to PECTRACK because it demonstrates how linear regression can be integrated into a web-based system to support sales forecasting and inventory planning.

Sales Forecasting for Inventory Control in Small Businesses

Orquesta (2026) examined the development of a computerized point-of-sale and inventory system with sales forecasting for micro, small, and medium enterprises in Pagadian City. The system incorporated several forecasting methods, including moving averages, linear regression, weighted moving averages, quadratic regression, and Holt-Winters seasonal smoothing. These methods can help businesses examine historical sales patterns and estimate future demand. However, the most suitable method depends on the availability of historical records and the characteristics of the products being sold.

This study is relevant to PECTRACK because it provides potential forecasting methods that can be evaluated using Pecto's Bakery sales data.

Predictive Analytics in E-Commerce and Market Demand Estimation

Culili (2026) developed an e-commerce platform for the Department of Agrarian Reform Region 02's Agrarian Reform Beneficiary Organizations, incorporating predictive analytics for sales forecasting and market demand estimation. The study addressed challenges involving sales tracking, inventory management, and market reach. Its findings highlight how integrating online transactions with predictive analytics can support planning and data-driven decision-making. However, agricultural enterprises have different operational needs from a bakery that produces and sells fresh products daily. This study is relevant to PECTRACK because it demonstrates how predictive analytics can work alongside online ordering and inventory records to help a business anticipate demand.

Predictive Sales Analytics and Inventory Decision-Making

Custodio, Lucero, and their colleagues (2026) presented a predictive sales and inventory system that uses linear regression for sales forecasting and K-means clustering for customer segmentation. The system combines sales transactions, inventory monitoring, and customer information to support business decisions. However, its customer segmentation feature may not be necessary for every small business, and the usefulness of its predictions depends on the quality of the data collected. This study is relevant to PECTRACK because its sales forecasting component could help Pecto's Bakery estimate demand for individual products and make more informed inventory decisions.

Foreign Literature

Predictive Analytics

Predictive Analytics for Retail Inventory and Demand Forecasting

Krishnamurthy, Nadukuru, and their colleagues (2024) discussed how predictive analytics can improve inventory management and demand forecasting in retail businesses. Their work examined approaches such as time-series analysis, machine learning, and causal forecasting to estimate future demand and support inventory decisions. The study also discussed challenges related to data quality and system integration. However, some analytical approaches may require technical resources that smaller businesses do not have. This literature is relevant to PECTRACK because it explains how historical sales data can support inventory planning and help reduce the risk of product shortages or excess stock.

Integrating Demand Forecasting with Inventory Control

Van der Haar and Wellens (2024), together with their co-authors, investigated how supervised learning can connect forecasting with inventory control. Rather than treating demand prediction and inventory decisions as completely separate activities, their approach directly estimates inventory-related decisions using available data. The study also considered situations involving perishable goods and lost sales. However, its methods may require sufficient historical data and careful adjustment to a business's operating conditions. This study is relevant to PECTRACK because bakery products have limited shelf lives, making it important to consider expected demand when planning stock and production quantities.

Machine Learning for Retail Sales Forecasting

Pattnaik, Padhi, and their colleagues (2025) examined artificial intelligence and machine learning approaches for forecasting sales using data from Big Bazaar. Their research compared different predictive models and grouped products into categories, including fresh and perishable foods, packaged foods, and household items. The study highlights the importance of evaluating different models to determine which provides more accurate forecasts for particular product categories. However, a large retail chain may have more extensive transaction records and resources than a small bakery. This study is relevant to

PECTRACK because it supports the use of product-level sales data to estimate future demand and improve production planning.

Automated Machine Learning for Retail Forecasting

The researchers of Comparative Study of AutoML and Machine Learning in Retail Forecasting (2025) compared traditional machine-learning models with automated machine-learning approaches using transaction data from an Indonesian traditional store. The study evaluated forecasting performance using several error measurements and found that model performance varied depending on the method and characteristics of the data. However, automated tools may still require data preparation and model evaluation to produce reliable results. This study is relevant to PECTRACK because it emphasizes the importance of selecting a forecasting method that balances accuracy with practical implementation requirements.

Predictive Analytics for Sales Forecasting and Inventory Optimization

Mansur, Sattar, and their colleagues (2025) investigated a hybrid neural-network approach for retail sales forecasting. Their model combined convolutional neural networks and long short-term memory networks while considering factors such as holidays, salary days, protests, and weather conditions. The study reported improved forecasting performance by considering both historical sales and relevant external factors. However, a complex neural-network model may not be necessary for a small bakery if simpler methods can produce sufficiently accurate predictions. This research is relevant to PECTRACK because it highlights how historical sales patterns and relevant calendar factors can help estimate future demand during weekends, holidays, and other periods when bakery sales may change.

Related Studies

Local Studies

A Web-Based Sales and Inventory System for Small Business Enterprises

Cruz, Mendoza, and Dela Cruz (2022) developed a web-based sales and inventory system for small business enterprises. The study aimed to improve transaction processing and record management by replacing manual operations with a digital platform. The results indicated that the system improved transaction accuracy and reduced processing time. It also helped businesses keep organized and accessible records, thus enhancing operational efficiency. However, the study mainly targeted general retail environments and did not include features specific to food-related businesses, such as order customization, product perishability, and ingredient inventory management.

An Online Ordering System with Integrated Inventory Management for Local Food Businesses

Lopez, Garcia, and Reyes (2023) created an online ordering system integrated with inventory management specifically for local food businesses. The study aimed to enhance both customer experience and internal operations by combining ordering and stock monitoring in one system. The findings revealed that the system boosted customer satisfaction and reduced order errors. Furthermore, the integration ensured consistency between sales and inventory levels, lowering discrepancies in stock records. Despite these advantages, the study lacked advanced tracking and reporting features, which are essential for monitoring business performance and making informed decisions.

Web-Based Inventory Management System for Small and Medium Enterprises

According to Michael B. Santos, and Liza P. Villanueva (2023), web-based inventory systems significantly improve stock monitoring, product tracking, and transaction recording in small businesses. Their study emphasizes that integrating inventory monitoring into a web-based platform minimizes stock discrepancies and improves the

accessibility of product records. The researchers explained that automated stock deduction and inventory updates improve operational efficiency and reduce human error. However, their study mainly focused on inventory processes and lacked customer management and order tracking features. This study is relevant to the proposed system because PECTRACK integrates inventory management together with customer records and order processing in one centralized platform.

Motormania's Web-Based Streamline Point of Sales System

Taqueban, and Villamor (2024) developed a web-based point-of-sale system to streamline sales transactions and improve record management. Their study found that digital transaction recording improved sales accuracy, reduced processing time, and simplified report generation. The study also highlighted the importance of centralized sales records for better business analysis. However, the system focused more on sales transactions and lacked integrated customer management and delivery monitoring. This study supports the proposed system because PECTRACK also centralizes sales recording while expanding its functions to customer management, delivery tracking, and bakery inventory management.

Integrated Information System for Retail Transaction Management

Jerome V. Salazar (2024) emphasized that integrated business systems improve transaction flow, inventory visibility, and customer servicing by connecting different business modules into one platform. Their study found that businesses using integrated systems improved efficiency and reduced operational delays. However, the study focused on retail businesses in general and did not specifically address food production workflows. This study is related to PECTRACK because the proposed system also follows an integrated system approach, connecting customer management, order processing, inventory, billing, and reporting into one platform.

Foreign Studies

Order Management Systems and Transaction Efficiency

Order Management Systems (OMS) play a significant role in improving the efficiency, accuracy, and coordination of business transactions. According to Alzoubi and Snider (2022), order management systems help businesses automate order entry, monitor transaction progress, and improve coordination between inventory, sales, and customer service operations. Their study explains that integrating OMS with centralized database systems allows businesses to manage customer orders systematically while reducing delays, data inconsistencies, and human error.

Furthermore, the study highlights that digital order management systems improve operational visibility by allowing businesses to monitor order status, inventory availability, and transaction records in real time. Businesses that implement OMS benefit from faster transaction processing, improved customer communication, and more accurate order handling. However, the study also notes that many existing order management systems are designed for medium and large-scale enterprises, making them costly and complex for small businesses with limited technical resources.

This study is relevant to the proposed system because it supports the need for a structured and automated Order Management System that improves transaction efficiency, customer order monitoring, and inventory coordination. It also reinforces the importance of developing a system specifically tailored to the operational needs of small businesses such as Pecto's Bakery by focusing on usability, accessibility, and integration of essential functions including order management, inventory tracking, customer records, and payment monitoring.

Related System

Local Systems

Foodpanda(2024)

In the Philippines, digital platforms such as Foodpanda are widely used to enhance business operations through automation and online services. Foodpanda operates as an online food delivery platform that connects customers, partner merchants, and delivery riders through a centralized system. It utilizes a platform-based architecture supported by client-server architecture and cloud infrastructure, allowing users to access services via mobile applications or web interfaces. The system also applies concepts from e-commerce systems and service-oriented architecture (SOA), where services such as ordering, payment processing, and delivery tracking function as interconnected components. While Foodpanda provides convenience and faster transactions, it acts as a third-party intermediary, limiting business control over pricing, operations, and customer data.

GrabFood(2024)

GrabFood, another widely used platform in the Philippines, functions similarly as an online food delivery system. It follows a platform ecosystem model where multiple stakeholders—customers, merchants, and delivery personnel—interact within a unified system. GrabFood employs cloud-based client-server architecture and integrates real-time processing to manage orders, payments, and delivery tracking efficiently. It also applies principles of distributed systems, ensuring scalability and reliability across high-demand usage. However, like Foodpanda, GrabFood operates as a third-party platform, which restricts direct control of businesses over their operations, customer relationships, and

data management.

Jollibee Delivery System (2024)

Jollibee Foods Corporation uses a web-based and mobile-based order processing system that follows client-server architecture and e-commerce system architecture. It allows customers to place orders, process payments, and track deliveries in real-time. The system uses centralized order management and transaction processing principles. However, it is designed for large-scale fast-food operations. PECTRACK adopts similar ordering concepts but is customized for bakery operations.

Red Ribbon Online Ordering System (2024)

Red Ribbon Bakeshop implements an online ordering platform that uses cloud-based architecture and integrated transaction systems. It allows customers to browse bakery products, place orders, and process payments online. This system is related to PECTRACK because both focus on improving bakery ordering transactions and making order processing more convenient and organized. Similarly, PECTRACK aims to streamline order management, monitor product availability, and help ensure accurate and efficient order fulfillment at Pecto's Bakery.

Goldilocks Online Delivery System (2024)

Goldilocks Bakeshop operates a digital ordering platform supported by cloud computing and service-oriented architecture (SOA). It integrates order processing, billing, and customer order history in one system. However, it focuses mainly on large-scale bakery operations. PECTRACK adopts similar concepts while focusing specifically on small-scale bakery operations.

Foreign Systems

Shopify(2024)

Shopify is a widely used e-commerce platform that enables businesses to create and manage online stores. It operates using cloud computing and follows e-commerce architecture, integrating content management system (CMS) principles to manage products, orders, and customer data. Shopify also uses scalable infrastructure and database systems to support high volumes of transactions. Although it enhances online presence and accessibility, it is not specifically designed for food service operations and lacks specialized features for managing bakery production workflows.

Square for Restaurants(2023)

Square for Restaurants is another foreign system designed for food service businesses. It utilizes cloud computing and SaaS architecture to deliver services through web and mobile interfaces. The system follows a service-oriented architecture (SOA), where functionalities such as billing, order tracking, and reporting are handled by separate but integrated services. It also incorporates secure payment processing technologies and database systems to manage financial transactions and customer data. While it provides a comprehensive solution, its features and infrastructure may be too extensive for small businesses with simpler operational needs.

StoreHub POS(2023)

StoreHub POS is a cloud-based point-of-sale system commonly used by small and medium enterprises. It follows a cloud-based client-server architecture, where data is stored and processed on remote servers and accessed through internet-enabled devices such as tablets or computers. The system integrates multiple modules including sales tracking, inventory management, and reporting, aligning with the concept of an integrated information system. It also utilizes database management system (DBMS) principles to ensure centralized and accurate data storage. While StoreHub improves efficiency and supports decision-making, it is designed for general retail operations and may include features that are not specifically tailored to bakery businesses.

Toast(2024)

Toast POS is a restaurant-focused digital system that demonstrates the use of advanced integrated platforms in business operations. It operates on a cloud-based architecture using a software-as-a-service (SaaS) model, allowing businesses to access the system through subscription-based services. Toast follows a modular architecture, where different components such as order processing, payment handling, and inventory management are combined into a unified platform. It also uses relational database management systems (RDBMS) to manage large volumes of transactional data and implements real-time processing to monitor operations instantly. Despite its efficiency, the system can be complex and may require advanced setup, making it less suitable for small-scale businesses.

Domino's Pizza Online Ordering System (2024)

Domino's Pizza operates one of the most advanced food ordering systems globally, integrating web-based ordering, mobile ordering, and real-time delivery tracking into one platform. The system follows e-commerce architecture combined with real-time processing systems, allowing customers to place orders, customize products, and monitor order preparation and delivery status instantly. It also applies database management system (DBMS) principles to store customer data, order history, and payment transactions. One of its strengths is its ability to provide real-time visibility in the order lifecycle, improving customer trust and satisfaction. However, its advanced infrastructure and enterprise-level resources may not be suitable for small bakery operations. This system is relevant to PECTRACK because it demonstrates how digital ordering and order tracking improve customer experience, which is one of the main objectives of the proposed study.

Uber Eats Merchant System (2024)

Uber Eats provides merchants with a digital food-ordering system that connects customers, restaurants, and delivery personnel through a centralized cloud-based platform. It follows a platform ecosystem architecture, where different users interact

within one system environment. It also applies distributed systems theory, allowing the system to handle multiple simultaneous orders and deliveries efficiently across different locations. The system integrates real-time order processing, payment validation, and delivery tracking to improve customer convenience. Its advantage is its scalability and ability to manage high transaction volumes. However, because it acts as a third-party intermediary, businesses have limited control over customer data and pricing. This system is related to PECTRACK because it demonstrates effective order and delivery coordination, while PECTRACK offers direct control over business data and operations.

Synthesis

The reviewed local literature shows that predictive analytics can help businesses understand past sales, identify patterns, and estimate future product demand. The studies of Divina and colleagues (2025), Lo and colleagues (2025), Orquesta (2026), Culili (2026), and Custodio and Lucero and colleagues (2026) discuss how data analysis can support business planning and decision-making. These findings are relevant to PECTRACK because Pecto's Bakery can use its historical order and sales records to estimate which products may have higher demand. This can help the bakery plan its inventory and production based on available data rather than relying only on manual observation.

Foreign literature also highlights the value of predictive analytics in improving business operations. The works of Krishnamurthy and Nadukuru and colleagues (2024), van der Haar and Wellens and colleagues (2024), Pattnaik and Padhi and colleagues (2025), the AutoML study (2025), and Mansur and Sattar and colleagues (2025) discuss approaches to analyzing data and supporting predictions. These studies provide a basis for incorporating predictive analytics into PECTRACK. However, the system should use an approach that fits the bakery's available data and operational needs. The predictions will serve as a guide for planning and will not guarantee actual sales or demand.

The reviewed local studies on web-based sales, inventory, and order management systems show the importance of organizing business transactions through a centralized system. The studies of Cruz, Mendoza, and Dela Cruz (2022), Lopez, Garcia, and Reyes (2023), Santos and Villanueva (2023), Taqueban and Villamor (2024), and Salazar (2024) support the use of digital tools to manage orders, monitor product availability, and organize business records. These findings are relevant to PECTRACK because Pecto's Bakery currently handles orders and records through different methods, including Facebook messages, walk-in transactions, and handwritten documents. A web-based Order Management System can help organize these activities and make order information easier to access and monitor.

Foreign studies, including the work of Alzoubi and Snider (2022), further support the use of order management systems to improve coordination and organize order-related activities. These systems show the importance of managing order information throughout the transaction. For PECTRACK, this means providing a more organized way to record customer orders, monitor their status, and coordinate order preparation and fulfillment. By keeping order details in one system, the bakery may reduce problems caused by duplicate entries, unclear handwritten records, and missed order updates.

The reviewed local systems, such as Foodpanda, GrabFood, Jollibee, Red Ribbon, and Goldilocks, demonstrate how digital platforms can help customers access products and place orders. However, some of these platforms mainly focus on food delivery or are designed for larger businesses. Their features and operating processes may not fully match the specific needs of Pecto's Bakery. This supports the need for a system designed around the bakery's actual order management practices, including customer orders, product availability, order monitoring, and preparation for pickup or delivery.

Foreign systems such as Shopify, Square for Restaurants, StoreHub POS, Toast, Domino's, and Uber Eats also demonstrate different ways of managing orders, sales, and inventory through digital tools. Although these systems provide useful examples of digital business management, their features, costs, and requirements may not be suitable

for every small business. PECTRACK is therefore intended to provide features that are relevant to Pecto's Bakery and its existing operations, instead of simply copying the functions of larger commercial platforms.

Overall, the reviewed literature, studies, and existing systems highlight the importance of digital order management, inventory monitoring, and data-based decision-making in business operations. However, Pecto's Bakery needs a system that combines these functions based on its specific needs. PECTRACK addresses this need by providing a web-based Order Management System with inventory management and predictive analytics. It aims to organize customer orders, provide clearer order status information, monitor product availability, and demonstrate how predictive analytics can be used to estimate future product demand.

The reviewed sources also provide a foundation for the Purpose and Description and Scope and Limitations of the study. PECTRACK will focus on order management, inventory management, and reporting and analysis, including predictive analytics using assumed or simulated historical sales and order data. Since the actual historical data of Pecto's Bakery is not available for the study, the assumed data will be used to demonstrate the system's predictive analytics functionality. Therefore, the generated estimates will not represent the bakery's actual sales patterns or guarantee future demand. Despite this limitation, the system aims to demonstrate how order management, inventory monitoring, and predictive analytics can be integrated into one platform to support more organized operations and informed decision-making.

METHODOLOGY

Technical Background

• Methodology

The study will use the Agile Methodology, a software development approach that emphasizes iterative development, flexibility, and continuous collaboration with the client. Agile allows the proponents to develop the system in small and manageable increments to ensure that each feature is continuously evaluated and improved based on feedback.

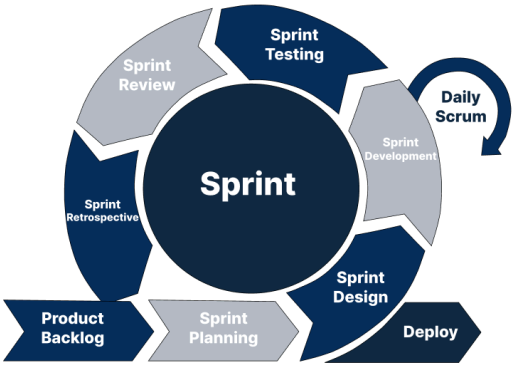


Figure 19: *Scrum Methodology*

Among the commonly used Agile frameworks, the proponents selected the Scrum Framework because it provides a structured process through Product Backlog, Sprint Planning, Sprint Design, Sprint Development, Sprint Testing, Sprint Review, and Sprint Retrospective. This framework is applied in the study since it can be used to facilitate the development in modules, specific sprint objectives, and frequent consultation with clients.

The development of the proposed system was categorized into three major parts which are the foundational modules, core operational modules, and decision support modules with deployment. This ensures the development process and each important feature of the system was implemented systematically and efficiently.

The foundational modules focused on establishing the core structure and security of the system, including user authentication, dashboard, and product management. These modules served as the foundation of the entire system since they manage user access, navigation, and product information. The core operational modules focused on the daily operational processes of the bakery such as order processing, inventory monitoring, and payment recording. These modules were developed to improve transaction handling, automate inventory updates, and support the

day-to-day activities of the bakery. Lastly, the decision support modules and deployment focused on sales reporting, customer order history, final testing, and system deployment. These modules were intended to support business monitoring and decision-making while also ensuring that the system was fully tested. After gathering feedback from the client and confirming that the system was already functional and met the operational requirements of the bakery, the system will prepare for actual deployment.

The circular flow of the Scrum Framework is the iterative aspect of Agile development where the cycle repeats over and over again with a series of sprints until all the requirements of the proposed system are fulfilled. After completing one sprint cycle, the proponents proceed to the next sprint by reviewing the completed tasks, gathering client feedback, identifying areas for improvement, and prioritizing the remaining system requirements. This continuous cycle allows the system to be gradually improved throughout the development process. It also ensures that errors, missing features, and client suggestions are addressed immediately before moving to the next phase of development. This looping process ensures that all modules of PECTRACK: A Web-Based Order Management System with Predictive Analytics and Inventory System for Pecto's Bakery at Lucban Quezon are continuously developed, tested, evaluated, and improved until all system objectives and project scope have been met.

Phase of Scrum Model

• **Phase1: Product Backlog -** The Scrum process was adopted when the proponents identified a potential client whose business operations could benefit from a computing solution. After selecting PECTO's Bakery as the client, the proponents conducted interviews and consultations with the owner to understand the current workflow, existing issues, and desired improvements in their order processing operations. The information gathered was adapted into system requirements such as order processing, inventory monitoring, payment recording,

and sales reporting, which were then documented in the Product Backlog. These requirements were then organized into three major parts which are foundational modules, core operational modules, and decision support modules with deployment, which will serve as the basis for the succeeding sprint phases.

• **Phase2: Sprint Planning -** The Sprint Planning phase will be adopted to organize and prioritize the items in the Product Backlog based on their importance to the bakery's operations. The proponents will adapt the gathered requirements into structured sprint goals by organizing them into three major parts, foundational modules, core operational modules, and decision support modules with deployment.

• **Phase3: Sprint Design -** The Sprint Design phase will be adopted to establish the system architecture and overall design of the proposed system. The proponents will adapt the client's requirements into system designs, including user interface layouts, database structure, and system flow diagrams. Wireframes will be created to visualize the system's functionality before the development of each sprint cycle begins.

• **Phase4: Sprint Development-** The Sprint Development phase will be adopted to implement the actual coding of the system. The proponents will adapt the approved design specifications into functional modules across three sprint cycles, foundational modules, core operational modules, and decision support modules with deployment. Within each sprint cycle, the proponents will conduct a Daily Scrum to monitor the day-to-day progress of each module being developed, ensuring that any issues will be immediately addressed and resolved. Each module will be developed iteratively, meaning that if a module does not meet the required functionality, the development cycle will repeat and go back through coding and adjustments, until the module is fully functional. The proponents will only proceed to the Sprint Testing phase once all modules within the current sprint cycle have been completed and are functioning as intended.

• **Phase5: Sprint Testing-** The Sprint Testing phase will be adopted to evaluate the functionality and performance of the developed modules. After completing all modules within a sprint cycle, the proponents will adapt testing procedures such as functional testing and usability testing to ensure that each module will perform according to the system requirements. Any errors and bugs identified during this phase will be fixed before proceeding to the next sprint cycle.

• **Phase6: Sprint Review -** The Sprint Review phase will be adopted to present the developed features to the client for evaluation. The proponents will adapt the feedback gathered from the bakery owner and staff regarding usability, functionality, and suggested improvements to ensure that the system will align with the actual operational needs of the bakery.

• **Phase7: Sprint Retrospective -** The Sprint Retrospective phase will be adopted to evaluate the overall sprint process and identify areas for improvement. The proponents will adapt the insights gathered from the client's feedback, testing results, and development challenges to improve the approach of the next sprint cycle. This continuous evaluation will ensure that each sprint cycle becomes more efficient and aligned with the project goals until all system requirements are fully met and the system is ready for deployment.

• Technologies to be Used

This section presents the technologies and development tools that will be used for developing PECTRACK: A Web-Based Order Management System with Predictive Analytics and Inventory System for Pecto’s Bakery at Lucban Quezon. These technologies will be the tech stack for the proposed system, which refers to the set of programming languages, design tools, and database management solutions used to build the web application.

REACT (Version 19.2)

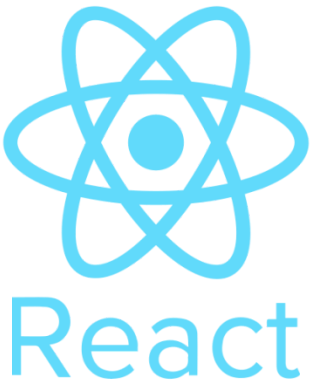


Figure 20. *Logo React*

React is a JavaScript library used for building dynamic and interactive user interfaces. This allows faster response time and smoother user experience for the bakery owner and staff while managing orders, inventory, and sales records. React enables the proponents to create reusable components such as forms, dashboards, navigation menus, and inventory tables, making the development process more organized and efficient. It also improves system performance by updating only the necessary parts of the user interface instead of reloading the entire webpage.

Tailwind CSS (Version 4.3)



Figure 21. *Logo Tailwind CSS*

Tailwind CSS provides utility-based classes that allow the researchers to create a modern, clean, and responsive layout that can adapt to different screen sizes, including desktop and mobile devices. This technology enhances the visual appearance and usability of the system by making it easier for the bakery owner

and cashier to navigate the different modules.

Node.js (Version 22)



Figure 22. *Logo Node.js*

Node.js is a JavaScript runtime environment used for developing the back-end functionalities of web applications. In the proposed system, Node.js will be used to handle server-side operations such as processing requests, managing system logic, and communicating with the database. This enables the system to process multiple requests efficiently through its event-driven and non-blocking architecture, making it suitable for real-time web applications. It will support the core functionalities of PECTRACK, including order processing, inventory monitoring, user authentication, and alert notifications.

Chart.js(Version 4.5)



Figure 23. *Logo Chart.js*

Chart.js is a JavaScript charting library used to display graphical representations of system data. In the proposed system, Chart.js will be used to generate visual reports such as daily sales, revenue summaries, inventory statistics, and product performance charts. This allows the bakery owner to easily monitor business performance and analyze operational data through interactive and understandable visualizations.

PostgreSQL (Version 18)



Figure 24. *Logo PostgreSQL*

PostgreSQL will serve as the Relational Database Management System (RDBMS) of the proposed system. It will manage and store essential data such as customer orders, product details, inventory records, payment transactions, user accounts, and sales reports. The use of an RDBMS enables the system to organize and maintain data efficiently through the establishment of relationships between database tables. This also ensures data integrity, consistency, reliability, and security, which are essential for supporting the daily operations and transaction processing of Pecto's Bakery.

LOGIN MODULE FLOW CHART

The Login Module ensures secure access to the system by verifying user identity and assigning role-based access permissions.

The process begins when the user enters their username and password. The system first validates the credentials against the database. If the credentials are invalid, access is immediately denied to prevent unauthorized entry. If the credentials are valid, the system proceeds to identify the user's assigned role, which determines the appropriate access level and dashboard.

For the Administrator, after successful credential validation, the system triggers a One-Time Password (OTP) verification as an additional security layer. The OTP is sent to the administrator's email. If the OTP is incorrect, access is denied. Only after successful OTP verification is the Administrator granted access to the Admin Dashboard, which provides full control over system modules including user management, orders, inventory, delivery assignment, and reports.

For the Cashier, once the credentials are validated, the system checks whether the account role is assigned as cashier. If the role does not match, access is denied even if the credentials are correct. If the role is valid, the cashier is redirected to the cashier dashboard, where they can manage orders, update order status, process payments, and handle customer-related transactions.

For the Customer, after successful credential validation, the system checks whether the account is active. If the account is inactive or suspended, access is blocked. Only active customers are granted access; they are taken to the Menu, where they can browse products, place orders, and track order status.

This module ensures secure authentication and proper role-based access control across all user types within the system.

USER MANAGEMENT MODULE FLOW CHART

The Administrator accesses the User Management Module through the Admin Dashboard. The system displays the complete list of registered customers stored in the database. The Administrator can view customer details, update existing customer information, or deactivate customer accounts. Customers create their own accounts through registration. When modifications are made, the system validates the entered data. If the information is invalid, an error message is displayed and the changes are not saved. If the data is valid, the system saves the updated information successfully.

The Cashier uses this module primarily for order processing. The Cashier begins by searching for an existing customer using identifiers such as name or contact number. The system checks whether the customer exists in the database. If no customer is found, the system notifies the Cashier that no record exists. If a customer is found, the Cashier can view the customer's details and attach the customer to the order. Corrections to a customer's record are made by the Administrator.

The Customer has limited access within this module. The process begins when the Customer opens their profile page. The system displays their personal information stored in the database. If the customer chooses to edit their profile, they may update fields such as their name and contact number, add or edit delivery addresses, or change their email address, which requires their current password.The system validates the updated information before saving it. If the data is valid, the system updates the profile successfully.

INVENTORY MANAGEMENT MODULE FLOW CHART

The Administrator oversees the inventory system through the Inventory Management Module. The process begins when the Administrator accesses the inventory dashboard. The system displays all products along with their current stock levels. The Administrator can add new products, update product details, and monitor inventory availability. If stock levels fall below the minimum threshold, the system can generate stock alerts to notify the Administrator.

The Cashier may view inventory information when processing orders. Before confirming an order, the system checks whether sufficient stock is available for the selected products. If stock is insufficient, the system notifies the Cashier and prevents the order from proceeding until inventory is updated.

Customers do not have direct access to the inventory system. However, the system automatically reflects product availability in the product listing page. If a product is out of stock, it will either be hidden or marked as unavailable.

ORDER MANAGEMENT MODULE FLOW CHART

The Administrator monitors all orders within the system through the Order Management Module. The process begins when the Administrator opens the order management interface from the dashboard. The system displays the complete list of orders including order details, customer information, order status, and assigned staff . The Administrator can review orders, update order status, or resolve order issues when necessary.

The Cashier manages most of the operational tasks in the Order Management Module. The process begins when the Cashier creates a new order for a customer. The Cashier selects the customer record and adds products to the order. The system calculates the total amount based on the selected products and quantities. Once the order is confirmed, the system records the order and assigns an initial order status. The Cashier can update the order status during preparation or processing stages.

The Customer creates orders through the website interface. The process begins when the customer browses available bakery products and adds selected items to the cart. After reviewing the cart, the customer confirms the order by selecting the preferred delivery option and payment method. The system records the order and displays the order status, allowing the customer to track the progress of their purchase.

PAYMENT AND BILLING MODULE FLOW CHART

The Admin does not really manage the order process. It can only view payment transactions and generate financial reports to pdf.

The Cashier primarily manages the Payment and Billing Module. The process begins after an order is created and ready for payment. The Cashier selects the order and records the payment method chosen by the customer. The system calculates the final amount and records the payment details in the database.

If the customer selects an online payment method, the system processes the transaction through the integrated payment gateway. If the payment is successful,

the system updates the payment status and confirms the order.

If the payment fails or is cancelled, the system notifies the user and allows another payment attempt.

Customers can also select their preferred payment method when placing an order. After completing the payment process, the system records the transaction and updates the order status accordingly.

REPORTING AND ANALYTICS MODULE FLOW CHART

The Administrator uses the Reporting and Analytics Module to monitor system performance and business activities. The process begins when the Administrator selects a report type from the reporting dashboard. The system retrieves relevant data from the database, including order transactions, sales records, inventory levels, and customer activity.

Once the data is collected, the system generates a report that can be viewed on the dashboard or exported for documentation purposes. The generated reports help the Administrator analyze sales performance, monitor inventory usage, and make informed business decisions.

The Administrator has full access to this module. Cashiers see the same reports except Export History, and reports are view-only for everyone, to ensure the integrity of business data.

The Cashier uses the Reporting and Analytics Module to review transaction records and daily sales performance. The process begins when the Cashier opens the reports section of the system and selects the type of report to view, such as the transaction log or the daily sales summary. Once the report type is selected, the system retrieves the relevant transaction data from the database and generates a report displaying completed sales and recorded transactions. The Cashier can review the generated report directly within the system interface. If the Cashier needs a copy of the report, the system allows the report to be exported as a PDF file for documentation and record-keeping purposes. This functionality helps the Cashier monitor completed transactions and maintain accurate sales records for daily operations.

The Customer uses the Reporting and Analytics Module to review their personal transaction history and order records. The process begins when the Customer opens the order history section of their account within the system.

My Orders lists the customer's orders, newest first, with their status. My Payments lists the customer's payments and can be filtered by date, payment method, and status. The Customer can open the receipt of any order and print it or save it as PDF for personal documentation.

This feature allows Customers to keep track of their previous purchases and verify their completed orders within the system.

• **Calendar of Activities**

Product Backlog

• Course Orientation and Discussion- The developers were informed about the requirements needed on the development of the projects.

• Group Formation - In this activity, a group had to go through a series of stages to innovate projects.

Sprint Planning

• Brainstorming. The developers were able to share ideas and information to create innovative topics.

• Title proposal and consultation. The researcher presented three (3) thesis titles until the last title was approved through consultation.

Sprint Design

• Introduction - The researchers developed the introduction of the study to provide an overview of the project and explain its significance.

• Background of the Problem - The group identified and discussed the existing problems encountered in the current process or system. This section justified the need for the proposed solution.

• Purpose and Description - The developers described the purpose of the system and explained how the proposed project would function and benefit the users. Objectives of the Study The researchers formulated the general and specific objectives that the project aimed to achieve.

• Scope and Limitation - The boundaries, coverage, and limitations of the system were identified to clearly define the extent of the study.

• Review of Related Literature - The group gathered related studies, articles, and references to support the proposed project and strengthen its credibility. Synthesis The researchers summarized and analyzed the

gathered related literature to identify connections and gaps relevant to the study.

Sprint Development

• Methodology - The developers selected and explained the methodology used in developing the system, including the processes and procedures followed during the project.

• Hardware and Software - The researchers identified the hardware and software tools required for the development and implementation of the system.

Sprint Resting

• Appendices - The developers compiled supporting documents such as questionnaires, interview forms, letters, screenshots, and other necessary materials for validation and documentation purposes.

• Gantt Chart - The researchers prepared the Gantt chart to monitor and track the schedule, progress, and completion of each project activity.

Sprint Review

• During this phase, the researchers reviewed the completed tasks, documents, and outputs to ensure that the project requirements and objectives were properly achieved. Feedback and suggestions from the adviser were considered for further improvements.

Sprint Retrospective

• The researchers evaluated the overall workflow, collaboration, and development process of the project. The team identified strengths, challenges encountered, and possible improvements that could help enhance future project activities and teamwork.

- **Resources**

  Hardware

  - **Computer**



Figure 25. *Computer*

The main hardware resource in the study will be the computer as the main development tool will be used to code, design interfaces, create databases, test and document. The minimum specifications suggested are an Intel Core i3 or Ryzen 3 processor, 8GB RAM, 256GB SSD drive, and Windows 10 or higher OS. The device is utilized to execute the development environment, Web-based system, PostgreSQL database, and prepare project documentation including the manuscript, diagrams, and Gantt chart.

Specification:

RAM - 16GB

CPU - Ryzen 5 5600g

GPU - RTX 5060

SATA- 256GB

- **Mobile Phone**



Figure 26. *Mobile Phone*

The device was used to record the interview sessions with the client so that all the requirements, features that were requested and operational issues of PECTO Bakery were properly recorded to be analyzed.

Specification:

Chipset -Mediatek Helio G100 Ultimate (6 nm)

CPU - Octa-core (2x2.2 GHz Cortex-A76 & 6x2.0 GHz Cortex-A55)

Internal Memory - 8GB RAM + 256 ROM

Software

- **Figma(Version 125)**



Figure 27. *Figma*

It will be used in drawing the wireframes, storyboard, user interface layouts and interactive prototypes of the system before the real coding process. Using Figma, the proponents will be able to visually show the proposed design to the client to approve it and receive their initial feedback on the layout, navigation, and usability of the system modules.

- **Visual Studio Code**



Figure 28. *Visual Studio Code*

Visual Studio Code will be the main code editor used by the proponents

for writing, editing, and organizing the React (JavaScript) front end, the Node.js and Express back end, and the Tailwind CSS styles during development. Its syntax highlighting, extension support, debugging options and project file management features make it more efficient and organized in the development process.

- **MS Word**



Figure 29. *Microsoft Word*

The documentation and word-processing software that will be used in the study is Microsoft Word. It will be applied in the writing and formatting of the research manuscript, development of the progress report, development of interview questionnaire, documentation of sprint output, and completion of the capstone paper to submit.

Requirements Analysis

This section presents the analysis of the client's existing business operations and the computing solutions that the proponents can offer to improve the current workflow of PECTO'S Bakery in Lucban, Quezon. The proponents first identified potential local businesses that could benefit from a computing solution and selected PECTO'S Bakery as the client due to its existing manual order processing, inventory monitoring, and sales recording procedures.

The proponents became aware of the bakery's operational challenges through Cristirene,

the sister of one of the proponents who worked part-time at PECTO'S Bakery. Through consultation and observation, the proponents identified issues related to manual transaction recording, stock monitoring, and report management. These findings led the proponents to propose a web-based order management system aimed at improving transaction efficiency, record accuracy, inventory monitoring, and sales reporting within the bakery's operations.

Product Backlog

This contains the complete list of the proposed system modules and features that directly respond to the bakery's identified operational needs. These include user authentication, dashboard, product management, order processing, inventory monitoring, payment recording, sales reporting and proof-of-delivery image capturing.

These modules were selected based on the identified challenges experienced by the bakery in handling transactions, monitoring stock levels, and organizing sales records manually. The user authentication module was included to ensure that only authorized users can access the system and manage bakery operations securely. The dashboard module provides an overview of daily transactions, inventory status, low-stock notifications, and quick access to important system functions. Product management allows the administrator to manage product information, prices, and stock quantities efficiently. The order management module was identified as one of the core features of the system since it directly supports the bakery's daily transactions. This module enables the recording and processing of customer orders while automatically updating inventory records through REST API integration. The inventory monitoring module helps track stock availability and provides alert notifications whenever products reach the minimum stock limit. The payment recording module was included to ensure that all customer payments are properly documented and reflected in the bakery's sales records. The proponents also included QR Ph integration to support digital

payment processing methods such as GCash. Additionally, the proof-of-delivery image capturing feature was included to allow delivery personnel to upload delivery confirmation photos taken with the phone's camera or chosen from its gallery. Lastly, the sales reporting module was added to generate organized reports and visual summaries of transactions, payments, and inventory movement. This feature will assist the bakery owner in monitoring daily operations and making informed business decisions.

Sprint Planning

In the Sprint Planning process, the backlog items identified were prioritized and grouped into sprints according to project scope. This enables the proponents to decide on the modules to be developed initially, and define the anticipated deliverables of every sprint. The first sprint will be the user authentication module because it is a core feature to ensure secure access and role-based usage for authorized bakery personnel. The dashboard module was added to give an overview of the daily transaction, inventory status and easy access to the important system functions. Lastly, the product management serves as the main data source for products, prices, and stock information.

The inventory monitoring will be included for the second sprint because stock deduction and product availability are directly dependent on processed orders. In addition, the payment recording module was included to ensure that all completed transactions are properly documented and reflected in the daily sales records.

The final sprint will be given to the sales reporting module including the deployment preparation The sales reporting module consolidates data from orders, payments, and inventory movement to generate accurate reports that can assist the bakery owner in monitoring performance and making business decisions

The requirements analysis process helped the proponents to discover and justify the operational issues of the client. This will serve as the final basis before proceeding to system documentation and development.

Requirements Documentation

This section presents the formal documentation of the agreed system requirements between the proponents and the owner of PECTO'S Bakery regarding the development of PECTRACK: A Web-Based Order Management System with Predictive Analytics and Inventory System for Pecto's Bakery at Lucban Quezon. The requirements documentation serves as the official reference for the design, development, testing, and implementation of the proposed system. It enumerates all software features in detail by providing a storyboard showing how the system would look once it has been fully designed and coded. This shows a complete and prioritized list of all system features and functionalities agreed upon by the client and the developers, and the Sprint Planning, which breaks down these agreed features into structured and time-bound development cycles.

Product Backlog

The Product Backlog contains the complete and prioritized list of all modules, system features, and functionalities agreed upon by the client and the proponents. It serves as the primary reference for all system requirements that will be developed throughout the project. The modules included in the Product Backlog are based on the operational needs identified during the requirements analysis phase. These modules include user authentication, dashboard, product management, order processing, inventory monitoring, payment recording, sales reporting.

Each module is documented according to its purpose, functionality, user accessibility, and expected system output. The user authentication module ensures that only authorized personnel can access the system. The dashboard module

provides an overview of daily transactions, inventory status, and system notifications. Product management allows the administrator to manage product information, pricing, and stock quantity.

The order management module enables the recording and management of customer orders, while the inventory monitoring module automatically updates stock levels based on completed transactions. The payment recording module documents payment details and transaction status, including digital payments processed through QR Ph.

The sales reporting module generates organized sales summaries and transaction reports that assist the bakery owner in monitoring business performance. **Sprint Planning**

The Sprint Planning phase records the way the approved modules and features are to be allocated into sprint cycles to be actually designed and developed. This will make sure that the agreed system requirements are systematically put into place as per the project timeline. According to the requirements approved by the client, the basic modules like user authentication, dashboard, and product management were attributed to the first sprint since they provide a basic framework and data foundation of the system.

According to the requirements approved by the client, the basic modules like user authentication, dashboard, and product management were attributed to the first sprint since they provide a basic framework and data foundation of the system.

The second sprint was assigned to the order management module, inventory control, and recording of the payment as these modules are the direct representatives of the daily operational activities of the bakery in the client interview. They were given priority following the foundational modules since they require the initial product and user data.

The last sprint would be sales reporting module, client revisions, and deployment preparation. This will help to make sure that the reports are prepared based on the validated transaction data and that all client requested improvements are implemented before implementation.

The requirements documentation formalized the agreed appearance of the system, functions, user roles and module features of the proposed study. Upon agreement with the client, the proponents used the Product Backlog and Sprint Planning phases to make sure that all the requirements approved were well documented and ready to be used in the next design and development phases. According to the requirements approved by the client, the basic modules like user authentication, dashboard, and product management were attributed to the first sprint since they provide a basic framework and data foundation of the system.

Design of Software, System, Product and /or Processes

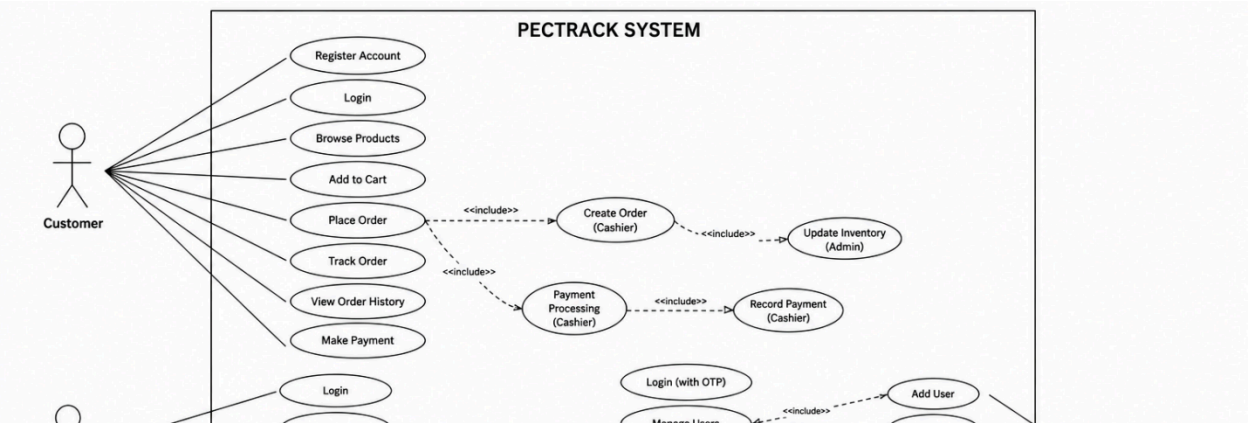


Figure 30. *Use Case Diagram of the Bakery Order Management System*

The Use Case Diagram illustrates the interactions between the users and the PECTRACK System. It is classified as a UML Behavioral Diagram specifically under the Use Case Diagram category because it represents the behaviors, functions, and interactions of the users within the system. The diagram identifies the four main roles namely the Admin, Cashier, and Customer, together with their corresponding tasks and system functionalities.

The Customer can register an account, log in, browse products, place orders, track orders, and make payments. The Cashier is responsible for processing customer orders, recording payments, updating order status, and managing customer information. Meanwhile, the Admin manages users, products, inventory, orders, reports, and overall system transactions.

The diagram provides a visual representation of how each role interacts with the PECTRACK System and helps explain the major processes and functionalities of the proposed system.

The system involves three main Roles: **Admin, Cashier, and Customer**. Each of the four roles has different responsibilities and levels of access within the system.

The **Admin** can log in to the system and perform administrative tasks such as managing products, approving stock changes and bulk orders, generating reports, assigning deliveries, and managing user accounts, including cashier accounts. Through these functions, the admin ensures that the system operates efficiently and that all necessary information is properly maintained.

The **Cashier** plays an operational role in handling daily transactions within the bakery. Cashiers can create orders for walk-in customers, update order status,

record cash and GCash payments. These actions help ensure that orders are processed accurately and that product availability is properly monitored.

The **Customer** interacts with the system primarily for purchasing bakery products. Customers will register an account first, can browse available products, place orders, track the status of their orders, and view their receipts after completing transactions. They can also cancel orders. These features allow customers to conveniently monitor and manage their purchases through the system.

The Use Case Diagram provides a high-level overview of the system's functionality and clearly defines how each user interacts with the Bakery Order Management System. It helps in understanding the scope of the system and the roles and responsibilities of each actor involved in the ordering and management process.



Figure 31. *Pectrack Logo.*

Logo

The Pectrack logo was designed to visually connect with the identity of our client, Pecto's Bakery in Lucban, Quezon. Since the system is specifically developed for their bakery, we made sure that our logo reflects their brand to maintain consistency and recognition.Pectrack highlights the purpose of the system. The word "TRACK" emphasizes order tracking and management, which is the main function of the web-based order management system. The clean and organized layout of the logo also symbolizes efficiency, accuracy, and digital service. The logo was created to show that Pectrack is not just a separate system, but a digital extension of Pecto's Bakery. It combines the bakery's traditional identity with digital technology to represent the goal of improving their operations through a web-based order management system.

System Security

Security is a fundamental aspect of PECTRACK to ensure that all data handled within the system, including customer information, order records, payment transactions, and inventory data remains protected from unauthorized access, data breaches, and malicious attacks. The following security measures are implemented across the system:

Authentication and Access Control: PECTRACK implements a role-based access control (RBAC) mechanism that restricts system features based on the user's assigned role: Admin, Cashier, or Customer. Each role is granted access only to the modules and functions relevant to their responsibilities, preventing unauthorized users from accessing sensitive areas of the system.

For the Administrator account, an additional layer of protection is provided through Two-Factor Authentication (2FA). After entering valid credentials, the administrator must confirm a one-time passcode (OTP) sent to their email before gaining access to the admin dashboard. This ensures that even if login credentials are compromised, unauthorized access to administrative functions is prevented.

Session management is also enforced throughout the system. User sessions are assigned a unique session token upon login, which expires after a defined period of inactivity. This prevents unauthorized users from hijacking active sessions and ensures that idle accounts are automatically logged out.

Password Security: All user passwords stored in the PECTRACK database are hashed using the bcrypt algorithm before storage. Bcrypt is a one-way hashing function: it turns each password into a hash that cannot be reversed, so even if the database is compromised, the actual passwords cannot be retrieved. Bcrypt also adds a random salt to every password, so two users with the same password are stored with different hashes. Plain-text passwords are never stored in the system at any point.

Additionally, the system enforces password complexity requirements during account registration and password changes. Users are required to create passwords that meet a minimum length and include a combination of uppercase and lowercase letters, numbers, and special characters to reduce the risk of brute-force attacks.

Input Validation and SQL Injection Prevention: PECTRACK uses server-side input validation on all form fields to ensure that data submitted by users conforms to expected formats before being processed or stored. This prevents malformed or malicious data from entering the system.

To protect the database from SQL injection attacks, a common method used by attackers to manipulate database queries through user input, the system uses prepared statements and parameterized queries in all database interactions. This approach separates SQL code from user-supplied data, making it impossible for injected code to alter the intended database query.

HTTPS and Data Encryption in Transit: All data transmitted between the user's browser and the PECTRACK server is encrypted using HTTPS (HyperText Transfer Protocol Secure) with a TLS (Transport Layer Security) certificate. This ensures that sensitive information such as login credentials, payment details, and personal customer data cannot be intercepted or read by third parties during transmission. Any attempt to access the system over an unencrypted HTTP connection is automatically redirected to the secure HTTPS version.

Payment Security: For online payment transactions, PECTRACK integrates with QR Ph, a Payment Card Industry Data Security Standard (PCI-DSS) compliant payment gateway. All payment processing — including GCash transactions — is handled directly by QR Ph secure infrastructure. PECTRACK does not store any raw payment credentials, card numbers, or sensitive financial data on its own servers. Upon transaction completion, QR Ph returns a payment confirmation

token that the system uses to verify and record the transaction status.

Data Privacy and Confidentiality: PECTRACK handles personal customer data, including names, contact numbers, addresses, and order histories, in accordance with the principles of the Republic Act 10173, also known as the Data Privacy Act of 2012. Access to customer records is restricted to authorized personnel only. Customers may only view and modify their own profile information and have no access to other users' data. The system does not share customer data with any third-party service beyond what is necessary for payment processing through QR Ph.

Error Handling and Logging: PECTRACK implements controlled error handling to prevent the system from displaying raw error messages, stack traces, or database query details to end users. Exposing such information could provide attackers with insights into the system's internal structure. Instead, user-friendly error messages are displayed, while detailed error logs are recorded on the server side and are accessible only to the system administrator for debugging and monitoring purposes.

System activity logs are also maintained to record significant events such as login attempts, failed authentications, order modifications, and payment transactions. These logs support accountability, allow administrators to detect suspicious behavior, and provide an audit trail for reviewing system activity.

Protection Against Brute-Force Attacks: To prevent automated brute-force login attempts, PECTRACK enforces an account lockout policy. After a defined number of consecutive failed login attempts, the account is temporarily locked and the user is required to wait before attempting again or to request a password reset. This significantly reduces the risk of unauthorized access through repeated password guessing.

Database Design

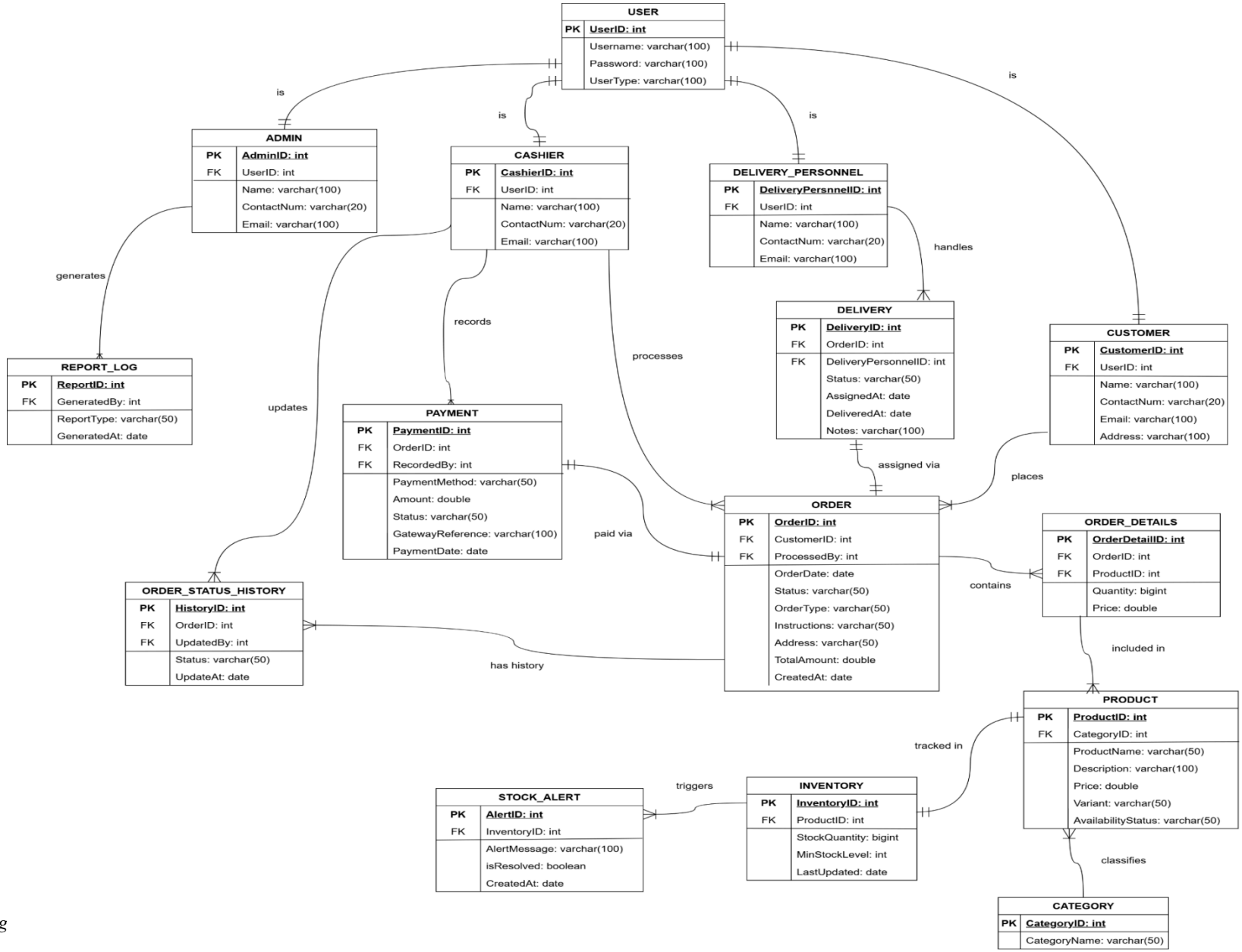


Figure 32. *Relational Database Diagram*

The Relational Database Diagram ensures a structured organization of data supporting the system's six primary modules: **Login, User Management, Order Management, Payment and Billing, Inventory Management, Notifications, and Reporting and Analytics**. By implementing normalized tables and clearly defined relationships, the system maintains data integrity, reduces redundancy, and ensures efficient data retrieval for operational and analytical purposes.

USER TABLE

The **USER** table stores the authentication credentials used to access the system. It contains the login information of all users regardless of role. Each user account is uniquely identified by a UserID and can be associated with one of the system roles such as Admin, Cashier, Customer.

Primary Key

- UserID

Attributes

- UserID(PK)
- Username
- Password
- UserType

Relationships

Referenced by:

- ADMIN
- CASHIER
- CUSTOMER

ADMIN TABLE

The **ADMIN** table stores the profile information of administrative users responsible for managing system operations such as generating reports, monitoring transactions, and supervising system activity.

Primary Key

- AdminID

Foreign Key

- UserID - USER(UserID)

Attributes

- AdminID(PK)
- UserID(FK)
- Name
- ContactNum
- Email

CASHIER TABLE

The **CASHIER** table contains the information of staff members responsible for processing orders, recording payments, and managing daily transactions within the system.

Primary Key

- CashierID

Foreign Key

- UserID - USER(UserID)

Attributes

- CashierID(PK)
- UserID(FK)
- Name
- ContactNum
- Email

CUSTOMER TABLE

The CUSTOMER table stores the profile of each registered customer — name, contact number, and email address — and links it to the customer's login account in the USER table. A customer's delivery addresses are stored separately in the CUSTOMER_ADDRESSES table, so that one customer can save several addresses.

Primary Key

- CustomerID

Foreign Key

- UserID - USER(UserID)

Attributes

- CustomerID(PK)
- UserID(FK)
- Name
- ContactNum
- Email
- Address

DELIVERY_PERSONNEL TABLE

The **DELIVERY_PERSONNEL** table stores information about delivery staff responsible for delivering customer orders.

Primary Key

- DeliveryPersonnelID

Foreign Key

- UserID - USER(UserID)

Attributes

- DeliveryPersonnelID(PK)
- UserID(FK)
- Name
- ContactNum
- Email

ORDER TABLE

The **ORDER** table records all customer orders placed within the system. It includes order status, order type (pickup or delivery), and the total amount of the transaction.

Primary Key

- OrderID

Foreign Key

- CustomerID - CUSTOMER(CustomerID)
- ProcessedBy - CASHIER(CashierID)

Attributes

- OrderID(PK)
- CustomerID(FK)
- ProcessedBy(FK)
- OrderDate
- Status
- OrderType
- Instructions
- Address
- TotalAmount

ORDER_DETAILS TABLE

The **ORDER_DETAILS** table stores the list of products included in each order. This table resolves the many-to-many relationship between orders and products.

Primary Key

- OrderDetailID

Foreign Key

- OrderID - ORDER(OrderID)
- ProductID - PRODUCT(ProductID)

Attributes

- OrderDetailID(PK)
- OrderID(FK)
- ProductID(FK)
- Quantity

- Status.

PRODUCT TABLE

The **PRODUCT** table contains information about bakery products available for sale such as cakes, bread, and pastries.

Primary Key

- ProductID

Foreign Key

- CategoryID - CATEGORY(CategoryID)

Attributes

- ProductID(PK)
- CategoryID(FK)
- ProductName
- Description
- Price
- Variant
- AvailabilityStatus

CATEGORY TABLE

The **CATEGORY** table organizes bakery products into groups such as cakes, breads, and pastries to simplify product management.

Primary Key

- CategoryID

Attributes

- CategoryID (PK)
- CategoryName

PAYMENT TABLE

The **PAYMENT** table records all payment transactions made for customer orders. It supports multiple payment methods such as cash for walk-in orders and QR Ph for online transactions like GCash payments.

Primary Key

- PaymentID

Foreign Key

- OrderID - ORDER(OrderID)
- RecordedBy - CASHIER(CashierID)

Attributes

- PaymentID(PK)
- OrderID(FK)
- RecordedBy(FK)
- PaymentMethod
- Amount
- Status
- GatewayReference
- PaymentDate

INVENTORY TABLE

The **INVENTORY** table monitors the stock levels of bakery products. It helps the system track available quantities and determine when restocking is necessary.

Primary Key

- InventoryID

Foreign Key

- ProductID - PRODUCT(ProductID)

Attributes

- InventoryID(PK)
- ProductID(FK)
- StockQuantity
- MinStockLevel
- LastUpdated

STOCK_ALERT TABLE

The **STOCK_ALERT** table records alerts generated when product inventory falls below the minimum stock level.

Primary Key

- AlertID

Foreign Key

- InventoryID - INVENTORY(InventoryID)

Attributes

- AlertID(PK)
- InventoryID(FK)
- AlertMessage
- IsResolved
- CreatedAt

REPORT_LOG TABLE

The **REPORT_LOG** table records system-generated reports for monitoring sales performance and operational activities.

Primary Key

- ReportID

Foreign Key

- GeneratedBy - ADMIN(AdminID)

Attributes

- ReportID(PK)
- GeneratedBy(FK)
- ReportType
- GeneratedAt

ORDER_STATUS_HISTORY TABLE

The **ORDER_STATUS_HISTORY** table stores the history of status changes for each order. This allows the system to track updates such as order confirmation, preparation, delivery assignment, and completion.

Primary Key

- HistoryID

Foreign Key

- OrderID - ORDER(OrderID)
- UpdatedBy - USER(UserID)

Attributes

- HistoryID(PK)
- OrderID(FK)
- UpdatedBy(FK)
- Status
- UpdatedAt

DATABASE NORMALIZATION

The database design follows the **Third Normal Form (3NF)** to ensure data integrity and eliminate redundancy.

First Normal Form (1NF)

All tables contain atomic values and each field stores only a single value. Repeating groups have been removed.

Second Normal Form (2NF)

All non-key attributes are fully dependent on the entire primary key.

Third Normal Form (3NF)

All tables eliminate transitive dependencies, ensuring that non-key

attributes depend only on the primary key.

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APPENDICES

APPENDIX A. RESOURCE PERSON

Mrs. Katherine R. Abillar

General Manager of Pecto's Bakery

Lucban Quezon

Facebook Account : Kat Rada Abillar

Alina Shane Oblea

Capstone Project Adviser

STI College Lucena

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Figure 33. Client Signing the request letter



Figure 34. Interview with the General Manager of Pecto's Bakery.



Figure 35. Second Interview with the General Manager of Pecto's Bakery.

APPENDIX B. INTERVIEW TRANSCRIPT

PROJECT TITLE: PECTRACK: A Web-Based Order Management System with Predictive Analytics and Inventory System for Pecto's Bakery at Lucban Quezon

DATE: 2026-04-11

LOCATION: Lucban Quezon

INTERVIEWER: Christian V. Luza
Christian C. Valencia
Yancy L. Ymata
Carla Mhaey G. Zoleta



INTERVIEWEE: Katherine R. Abillar

[00:00:01]

Zoleta: *"How do you receive customers order po"*

[00:00:03]

General Manager: *"Dito ba sa ano…tindahan"*

[00:00:03]

Zoleta: *"Opo, opo"*

[00:00:10]

General Manager: *"Ahhh, pa'nong ano? To receive customers order"*

[00:00:21]

Valencia: *"Op- para pong ano siguro, Halimbawa po 'yung may, halimbawa mag message po sainyo para pong gusto po nila ng certain tinapay po mga ganon. Paano n'yo po sila nare-receive"*

[00:00:27]

General Manager: *"Ahh okay so hindi yung everyday na ano dito. Tama ba?"*

[00:00:31]

Valencia: *"pwede, ano po siguro po, pareho po sila"*

[00:00:56]

General Manager: *"kasi dito kung ano yung inano nila edi yun yung i-aano namin. Pero yung mga ano, pre-orders sa page sila sa facebook page namin como-contact o tumatawag sila, pero pag yung everyday na mga walk in na customer edi ganyan pagka tinuro nila so yun yung i-aano namin sakanila, isusulat lang"*

[00:01:02]

Zoleta: *"Ano po, How do you process an order from start to finish? Like ano"*

[00:01:15]

Valencia: *"yung…halimbawa po yung, may nag, yun nga po pag may nag chat po sa page nyo ano po pano po yung mismong process po na para pong ito na po yung mismong order po nila na, ano po pano po i-ano yun"*

[00:01:44]

General Manager: *"Kung paano yung ano. So una i c-confirm namin kay customer, kung ito ba minsan nag sesend kami ng picture kasi minsan sasbihin nila pa order ng ensaimada, marami kaming klaseng ensaymada so… ang gagawin namin mag sesend kami ng picture to confirm tapos pag nag confirm na edi yon, isusulat namin tapos ibibigay namin don sa ano sa mag rereceive ng order sila mag p-pack. Yun."*

[00:01:50]

Zoleta*: "Ano po, Who is responsible for handling and managing customer order?*

[00:02:19]

General Manager: *"Pag sa page nag message, kami. Kasi kami nasa admin, pero pagka nag walk in dyan tas may order ng marami kung sino yung sales lady naka assign yun yung mag a-ano ng order tapos ang responsible dyan syempre susulatin lang naman namin yan e so ang gagawa talaga nyan yung nasa packaging, sila yung mag p-pack, sila yung ano mag bibigay don sa ay hindi ang mag bibigay din ay yung tindera din."*

[00:02:24]

Zoleta*: "Where do you record or list customer orders?"*

[00:02:26]

General Manager: *"uhhmm, Sa papel lang"*

[00:02:28]

Valencia: *"Para pong log book po ganon"*

[00:02:45]

General Manager: *"Oo, papel lang sya naka sulat."*

[00:02:52]

Zoleta: *"Ano po, When do errors usually occur in your order process po?"*

[00:02:53]

General Manager: *"When do errors?"*

[00:02:54]

Valencia: *"opo"*

[00:03:21]

General Manager*: "ahh, pag hindi nabasa, pag hindi nabasa nung pinag, kasi dinidikit lang namin yun sa labas e, meron kaming dikitan ng mga oders so pag pag hindi nila nabasa ayun don sya nag kakaron ng error or nabasa nila pero mali yung pagkakaintindi, halimbawa sa oras, or yung sa date, yun*

[00:03:26]

Zoleta: *"Why do you think this problem happens during order handling*

[00:03:46]

General Manager: *"hmm, baka ano, siguro miscommunication . So siguro dapat pag may order instead na idikit mo dyan i-ano mo muna don sa mismong tao "uy may order na ganto ganyan" saka mo idikit so para nagkakaintindihan na kayo hindi yung ididikit lang"*

[00:03:52]

Zeleta: *"What common problems do you encounter when managing orders?"*

[00:04:04]

General Manager: *"uhhm, what common ano, errors?, ano problems? Ang na e-encounter pag… ulit utli"*

[00:04:10]

Zoleta: *"When managing orders po, What common problems do you encounter when managing orders"*

[00:04:18] - [00:04:34]

General Manager: *"Ayun, minsan kasi, actually madalas na… halimbawa ako nabasa ko "uy ano may order na ganto ganyan" yung bang verbal lang, hindi ko sinulat"*

[00:04:35]

Zoleta: "*Ahh opo…"*

[00:04:50]

General Manager*: "So itong sinabihan ko, "ay sige-sige" edi akala-akala ko okay na, naisulat na yun pala hindi so yun yung kadalasan ng yayare dito kaya dapat once na ma receive mo sulat mona ka-agad tas ibigay mo"*

[00:04:55]

Zoleta: *"How do this problem affect your daily operations?"*

[00:05:27]

General Manager*: "Syempre ano, uhhm stress. Syempre mag papaliwanag ka sa customer e, lalo na kapag halimbawang marami syang, dibale kung mga 10 pcs 20 kayang i provide pero pag mga hundreds na tapos hindi mo na meet syempre pano mo e-explain sa customer. Merong iba na nag a-ano ni o-offer namin yung same price pero different na tinapay pero minsan hindi sila pumapayag, so yun talaga."*

[00:05:35]

Zoleta: *"Ano po, When do delays usually happen po, kunwari po sa sa mga pag busy po talaga."*

[00:06:05]

General Manager: "*ummh, pag ano, pag ka na late yung luto ng tinapay, halimbawa naluto yung tinapay ng mga six p-pick upin yung tinapay ng mga seven so hindi sya kaya*

kasi papalamigin pa, so dapat yun ay pag ka ganyang maagap ang order sasabihan na si baker na uunahin yun minsan nga miscommunication akala walang order ganyan so ano yung process nila yun ang sinusunod."

[00:06:10]

Zoleta: "*Why is it difficult to track order status manually po?"*

[00:06:12]

General Manager: *"Manually?"*

[00:06:13]

Zoleta: *"opo"*

[00:06:24]

Valencia: *"uhm, for example po is yung pong… halimbawa po i d-deliver nyo na po mismo, ahm china-chat nyo po ba yun sa mga customer nyo po, halimbawa po on the way na po yung inorder nila ganun po"*

[00:06:34]

General Manager: *"ahh pagka ano, ah minsan hindi na kasi may oras naman akong nakalagay. halimbawa , i d-deliver ba?"*

[00:06:35]

Valencia: *"opo"*

[00:06:51]

General Manager: "*halimbawa, nag order si ano, nag order ang school, nakalagay don 9am, so hindi na namin i m-message ang school darating nalang sya ng 9 so minsan pag na d-delay sila yung nag me-message sa amin kung bat wala pa. Tama ba? Yan ba yung…"*

[00:06:52]

Zoleta: *"opo opo"*

[00:06:52]

Valencia: *"opo."*

[00:06:53]

General Manager: "*parang ganon"*

[00:06:59]

Valencia: *"oh, may ano, may tanong pa kayo?"*

[00:07:06]

Luza: *"yung sa ano, yung sa ano lang po sino po ba yung gagamit nung sa system."*

[00:07:08]

General Manager: *"Baka kami, admin"*

[00:07:09]

Valencia: *"ahh admin"*

[00:07:10]

Luza: *"ilan po"*

[00:07:11]

General Manager: "*tatlo"*

[00:07:11]

Luza*: "tatlo lang po"*

[00:07:14]

Zoleta: *"admin lang ba"*

[00:07:16]

Valencia: *"Admin, then customer…"*

[00:07:17]

General Manager: *"Ano ba yan may app? Pano ba yan."*

[00:07:20]

Luza: *"Ahh bale sa website po sya"*

[00:07:32]

General Manager*: "ahh website, so parang ano sya- ahh hindi sya app oh edi yung parang ano yung meron dito yung para syang pasabuy"*

[00:07:34]

Luza: "*pasabuy po"*

[00:07:40]

General Manager: "*oo pero hindi sya app parang may pupuntahan kang website don ka oorder, parang ganon?"*

[00:07:40]

Luza*: "opo"*

[00:07:43]

General Manager: *"hindi sya yung d-download mo"*

[00:07:45]

Luza: "*oh hindi na po d-download"*

[00:07:48]

General Manager: *"o Yun parang yun nga yung nakikita ko"*

[00:07:58]

Luza*: "Bale yung sa ano po yung pag may ibang order po sa ibang bayan hanggang ilan po yung minimum na pwede?"*

[00:07:59]

General Manager*: "wala naman"*

[00:07:59]

Luza: *"Ahh wala naman"*

[00:08:36]

General Manager: "*kaya lang ang, ang ano don is yung depende sa o-orderin na tinapay kasi yung mga soft breads 3 days lang yun so kung dadalhin pa sya ng malayo halimbawa baguio, hindi sya p-pwede, pwede yung mga ano, mga broas, mga ibang ano na tumatagal ng months yun ang pwede. May mga omorder samin dito mga ganyan, Bicol, Baguio san pa nga ba 'to, tas yun Manila. Pinapa ano nalang namin, LBC or JNT"*

[00:08:48]

Zoleta: *"tanong mo kung ano pang problema na kailangan na itanong"*

[00:08:50]

Luza: "*ano?"*

[00:08:55]

General Manager: *"Tanong n'yo na"*

[00:08:56]

Valencia: *"wala ka naba tatanong?"(Valencia to Yancy)*

[00:08:57]

Ymata*: "Yan si Luza"*

[00:08:59]

General Manager: "*Tama ba yung mga sinagot ko"*

[00:09:00]

Zoleta: *"opo"*

[00:09:02]

General Manager: "*yun ba yung gusto nyong ano."*

[00:09:12]

Ymata: *"Ay yung sa user po ba, ahmm kasi po ano po kaso yun e meron po kasing bawal makita ng staff tapos nakikita po ng admin"*

[00:09:14]

General Manager: *"Ano-ano, pa'no?"*

[00:09:28]

Ymata: *"kunwari po uhhmm admin lang po yung nakakakita nung parang reporting tas sa staff, bawal po silang makita, so parang tinatanong ko lang po na parang like ilan po yung like gagamit nung parang sa Cashier"*

[00:09:49]

General Manager: *"yung, ah staff, siguro kahit samin nalang, kami nalang siguro ang ano tas i-ano nalang namin susulat tas sila nalang mag r-ready, para wala ding masyadong mag open ng website, baka kasi akala nung isa, pano ba yan pag nakita nung isa nag a-appear na sya as read?"*

[00:09:49]

Ymata*: "opo"*

[00:09:50]

General Manager: *"parang ganon?"*

[00:09:50]

Ymata*: "opo"*

[00:10:21]

General Manager: "*Ahh okayy, so dapat talaga pag nabukasan mo sya i-aano mona, kasi yung sa page namin. Tatlo kasi kaming admin din ay apat, halimbawa nabasa na nung kapatid ko mag a-appear parin sya sa'kin as unread, so minsan, 'yun nga nabasa na pala nya hindi kopa nakikita, so yan pala ay pag na open na ng isa consider ano na yun talaga."*

[00:10:21]

Luza*: "opo"*

[00:10:23]

General Manager*: "okay?"*

[00:10:37] - [00:10:38]

General Manager: *"Sinisi-sinisimulan nyo naba yan?"*

[00:10:41]

Luza: *"ano palang po, papel po"*

[00:10:44]

Zoleta: *"Proposal palang po"*

[00:10:47]

General Manager*: "Ahh Proposal~ so kakailanganin nyo ng pictures nyan ano?"*

[00:10:48]

Zoleta*: "opo"*

[00:10:58] - [00:11:00]

General Manager: "*Kailan deadline nyan?"*

[00:11:04]

Zoleta: "*ano po, mock defense na po namin sa 16 po."*

[00:11:10]

General Manager: *"16? Dedefense na kayo sa 16?"*

[00:11:11]

All*: "opo"*

[00:11:23] - [00:11:25]

Zoleta: *"Pagdating dito sa inventory?" (Zoleta to Luza)*

[00:11:25]

Luza*: "oo"*

[00:11:29]

Valencia*: "nag oo nalang e, gara"*

[00:12:34] - [00:12:41]

Zoleta: *"Ano po, why do you think a digital inventory records is better than current manual method po?"*

[00:12:47]

General Manager: *"Ahh pag sa websitve?"*

[00:12:48]

Zoleta*: "opo"*

[00:13:02]

General Manager: "*Ahhh syempre ma-iiba, parang lalawak lalo e unlike don sa nangyayri ngayon so kung sino lang nakakakilala sa pecto's, yun lang. 'Pag ba sa website ano yan, makikita ng kahit sino?"*

[00:13:03]

All: *"opo"*

[00:13:05]

General Manager*: "Ahhh okay~"*

[00:13:12]

Ymata*: "Pero bawal po sila omorder basta-basta pag di po sila parang like wala po silang account na gagawin"*

[00:13:13]

General Manager: "*ah kailangan"*

[00:13:20]

Ymata*: "gagawa po muna account, tsaka po sa admin, may account mo para sa admin"*

[00:13:26] - [00:13:39]

General Manager*: "para s'yang ano, shopee? Kasi may account ang shopee e, so may mga COD, COD din dyan"*

[00:13:43]

Ymata*: "opo uhmm sa payment po"*

[00:13:49]

General Manager: *"pero pano kung hindi pwede ang COD okay lang? payment agad"*

[00:13:52]

Ymata: "*Like sa ano po, gcash po?"*

[00:13:52]

General Manager: *"Kasi food yan e "*

[00:13:55]

Ymata: *"Online payment"*

[00:13:59]

Zoleta: "*nakalagay sa scope mo yan diba?" (Zoleta - Yancy)*

[00:14:00]

Ymata: "*Oo nga, tatagkalin kona."*

[00:14:21]

General Manager: "*Cod? Hirap kasi pagka ano kasi food yan e kasi na r-return nila diba, sa seller pag ayaw nilang kunin. Baka kasi mamaya ma lugi kami nyan kasi ipapadala namin e baka hindi bayaran o ibalik e durog-durog na haha."*

[00:14:31]

General Manager: *"anong course nyo nga?"*

[00:14:31]

All: *"IT po"*

[00:14:34]

General Manager: *"ohh it, graduating na kayo?"*

[00:14:35]

Luza: *"Hindi pa po 3rd year"*

[00:14:36]

General Manager: *"Ahh third year"*

[00:14:39]

Ymata: "*Hindi pa po pag na pasa po kami"*

[00:14:42]

General Manager: "*ahh kasi 4th year nyo ojt na?"*

[00:14:42]

Valencia*: "opo"*

[00:14:44]

General Manager: *"kaya n'yo 'yan"*

[00:16:34] - [00:16:40]

Zoleta: *"Ano po ma'am, What benefits can you get from having customer order list history po?"*

[00:16:58]

General Manager: "*What benefits? Ayun uhhm, makikita mo yung mga syllable items, tapos benefits uhmmm, ano ulit nga tama ba yung sabi ko?"*

[00:17:07]

Zoleta: "*What benefits can you get from having customer order list history?"*

[00:17:38]

General Manager: *"Uhhmm yun nga parqang makikita mo don kasi kung ano yung uhhmm, syllable na tinapay sa yung mga sa taga ibang bayan syempre kasi sila yung o-order dyan sa website e diba? Kasi kung yung mga taga dito lang e syempre mga simpleng mga tasty, ganyan pero pag sa ganyan don makikita na ah okay so gusto pala nila ng mga taga ganito yung broas namin so parang don, don sya papasok."*

[00:17:45]

Zoleta: *"Ano po, Why it is important to avoid duplicate customer record in the system?"*

[00:17:51]

General Manager*: "uhhmm, Duplicate records?"*

[00:17:59]

Luza: *"ah meron po bang ganon, parang pag omorder sila na d-duplicate po"*

[00:18:36]

General Manager*: "parang hindi pa nangyayare yun. Kasi parang ang ano kasi e, ito lang kasi yung contact nila e, unless alam nila yung landline tas tumawag or nag message pa so ang ano non, mahirap non ma dodoble yung magagayak naming order. Kunwari nag order ng pianono 200 tas iba naka receive tas ako na receive ko din so papagawa kami ng 400. E late na na realize na same person pala so sayang. Hindi mo naman pwedeng sabihin na na doble"*

[00:20:07] - [00:19:22]

Ymata: *"'yung sa delivery n'yo po ba, parang pano po pag ang nag order po ay ah kunwri uhm, hindi po sya, parnag hindi po sya yung sa deliver nyo po ba may ano po ba kayo, rider?"*

[00:19:50]

General Manager: "*rider? Ay sarili? Ano basta kasi diba may branches kami dito, sya yung nag de-deliver yung tinapay don, tapos yung mga orders pag within lucban sya din pero pag nag lucena na o tayabas kami na nag d-deliver non. Pag ka ano malayo edi yun mga courier na, LBC, JNT, yun."*

[00:21:05] - [00:21:13]

General Manager*: "Lalo na nung pandemic gamit na gamit yung pag d-deliver, kasi diba sarado lahat bawal lumabas."*

PROJECT TITLE: PECTRACK: A Web-Based Order Management System with Predictive Analytics and Inventory System for Pecto's Bakery at Lucban Quezon

DATE: 2026-04-22

LOCATION: Lucban Quezon

INTERVIEWER: Christian V. Luza
Christian C. Valencia
Yancy L. Ymata
Carla Mhaey G. Zoleta



INTERVIEWEE: Katherine R. Abillar

[00:00:012] - [00:00:23]

Zoleta: "*Ano po. Tatanong po sana namin kung ilan po yung nagiging customer like sa araw-araw po, estimated"*

[00:01:05]

General Manager: "*Nagiging customer? Madami, May count ka dyan? Ay, parang nakaano eh. Pero kasi nung nag bilang kami dati, ay parang nag ano more or less ay 700. sa isang araw, yun ay regular days. So yung mga ano, halimbawa ay nag-holiday or weekends, mas madami. Di na kasi namin, pero nung dati nag bilang kami. Yun nasa 700. Dito lang yun sa main. Hindi pa kasama yung outlet. Kasama ba yung outlet?"*

[00:01:15]

Luza: "*Ay, bali yun sa ano po ba? Di ba po ba yun sa Facebook po? Sino nga po ulit yung Naka kaano po? Yung sa admin?"*

[00:01:18]

General Manager: "*Ako, si ate, tsaka si ate Ren. Tatlo kami sa ano."*

[00:01:22]

Luza: "*Bawat sa ano po, Bawat. sa branch po?"*

[00:01:45]

General Manager: "*Ah, sa branch? Ahh kasama ba branch? Baka mahirapan kayo nyan, Dito na lang sa main. Kasi sa amin din naman nila ibabagsak yung orders. Kasi ang dami gagawa pa kayo nan. Kasi ang branch namin ay... apat. magiging lima ang gawin nyo. iba-iba"*

[00:01:46]

Luza: "*Sige po"*

[00:01:48]

General Manager: "*Yun ba gusto n'yo? Okay lang naman, kaya mahihirapan."*

[00:01:59] - [00:02:05]

Luza: "*Ay, tanong ko lang po. Sa nga po ba ulit naka locate po yung sa ano? Yung sa ibang branch po?"*

[00:02:05]

General Manager: "*Naka? Locate?"*

[00:02:06]

Luza: *"opo"*

[00:02:27]

General Manager: "*yung isa sa daan Tayabas sa may SLSU ay sa bago mag tulay, yung isa sa palengke, yung isa ay daan Mauban, tapos, ay yung isa kahit wag nyona isama kumisari kasi. Tatlo, apat."*

[00:03:17] - [00:03:29]

Luza: "*Ay, yung bali yung sa ano po, yung sa process po nung ano po. Paano po nila malalaman kung may order po? Yung sa baker po."*

[00:04:00]

General Manager: "*Sa baker? Nakasulat sa ano nila. Everyday kasi meron kaming parang order form. Tapos nakasulat dun yung ano. Tapos ay ano, naka-indicate lang doon kung may order. Halimbawa, Regular na ginagawa ay 100. So mag-a-add lang para sa order. Ganun lang. Manual lang, sulat."*

[00:04:14]

Ymata: "*Doon po ba sa part ninyong nagkakameron ng error? Parang nag d-doble?"*

[00:04:42]

General Manager: "*Na d-doble? May instance bang ganon… very minimal, siguro lang pag ano pagka nasabi na nung isa tas inulit, na d-doble. O kaya'y nag-order dun sa isa, tas umorder din dun nag-message na sa akin, tapos naisulat na, tas tumawag pa rin sa landline."*

[00:04:43]

Luza: "*ahhh"*

[00:05:07]

General Manager: "*Tapos ay ano halimbawa ay sa kamay ni hesus. Ang umorder ay halimbawa ay iba ang pangalan tas ang umorder sa page ay iba din ang pangalan. So yun yun, parang nadu-duplicate. Pero kapag naman parehong sinabing nawasal, kinoconfirm naman namin kung same din ba yun. Lalo kung same yung items."*

[00:05:14]

Ymata: "*Pag sa branch n'yo ba nag-order? Parang dito rin yung ano…"*

[00:05:14]

General Manager: "*Nanggagaling 'yung tinapay?*

[00:05:15]

Ymata: "*opo"*

[00:05:15]

General Manager: "*dito din"*

[00:05:19]

Ymata: "*dun lang po parang nag t-take po ng orders?"*

[00:05:37]

General Manager: " *halimbawa sa outlet? oo, Sa kanila lang nag p-place ng order pero dito pa rin ang bagsak non Kaya para din na kayo hindi na makomplikado gagawin niyo, Ito na lang. Kailangan ba marami?"*

[00:05:42]

Luza: "*Hindi po Mas ano lang po. Parang kinaklaro lang po kung paano."*

[00:05:47]

Ymata: "*Kasi po, akala po nila na parang yung ibang branch po is parang ganto rin po."*

[00:05:48]

General Manager: "*kalaki?"*

[00:05:50]

Ymata: "*Opo, tapos doon din po na order."*

[00:06:13]

General Manager: *"Actually walang gawaan doon. Ito lang talaga tsaka yung isang komisary. Tapos sila binabagsakan na lang ng tinapay. So parang ano lang sya, parang stall. Walang gawaan. dito talaga yung gawaan sa tindahan"*

[00:06:39] - [00:06:44]

Ymata: "*May mga question po kasi na parang unexpected po."*

[00:06:45]

General Manager: "*Katulad ng?"*

[00:06:51]

Valencia: "*Yung po, ano po, siguro po yung sa dapat, yung about sa ingredients po ng mga tinapay."*

[00:06:53]

General Manager: "*Ay, tinanong din?"*

[00:07:00]

Valencia: "*Diba po may inventory pong kasama yung namin? Parang mag-a-alert daw po sa kanila kung may stock po ng ingredients."*

[00:07:03]

General Manager: "*Ah, pati raw materials? Ah, okay."*

[00:07:15]

Ymata: "*Baka kasi po ano mag-indicate na available pa rin kahit, kunwari po maubusan po ng ingredients pala yung tinapay niyo. Like, wala na palang ingredients yung..."*

[00:07:15]

General Manager: "*Flour, ganyan."*

[00:07:30]

Luza: "*parang If may nawalan po ng ingredients, parang dito po dapat nalabas din. Dito po dapat nalabas din na parang available pa rin."*

[00:07:32]

General Manager: "*Ahh okayy"*

[00:07:36]

Ymata*: "May instance po ba na ganon na parang minsan po naubos po?"*

[00:07:55]

General Manager: " *uhmm Kasi may nagchecheck kasi sa amin, may nag-e-inventory kasi ng stock. So parang hindi naman, unless wala talagang ma-provide yung supplier, yun yung time. Ahh kung baga, ang ibig sabihin, dapat hindi siya lalabas sa system. Dapat siya i-sold out, ganyan? Ahhhh"*

[00:08:02]

Ymata: "*Kaya po pinapa consider na maglalagyan sa ingredients din."*

[00:08:31]

General Manager: "*Ah, ingredients din? Nako dami nyan. 'Di Actually, nauubos lang naman sa amin yung ube. Ano ate Rin? Yung munggo, munggo tsaka ube na filling sa tinapay. Yun lang yung usually na... na uubos na minsan ay ano pero kasi ang nangyayari pag wala kaming actually na nagagawa namin ng paraan kasi may nabili sa Lucena eh so hindi siya nang, parang hindi siya nangyayari"*

[00:08:32]

Ymata: "*na parang hindi biglaan"*

[00:08:35]

General Manager: " *oo, hindi siya biglaan na, "uy wala na pala"*

[00:08:50]

Zoleta: "*Kaya tinatanong din po ni sir sa amin kung gaano po kami sure na mag aano po yung mga customer niyo sa website."*

[00:08:52]

General Manager: "*yung mag-o-order?"*

[00:08:56]

Zoleta: "*Kaya po tinatanong po niya po ilan po ba yung mga nabili po."*

[00:09:05]

Ymata: "*Tapos ano rin po, ang isang difference din yung parang sa walk-in po kaysa sa pag-o-order po sa website."*

[00:09:59]

General Manager: "*Hindi, yun yung sabi ko sa inyo, yung mga ano, yung mga taga-ibang bayan. Pero actually kunti lang naman yun. Hindi pa siya ganun kadami. Mga walk-in talaga ang ano, everyday, ang madami. Sabihin nyo siguradong ano yun, kasi... Paano ba? Paano magagamit? Paano makakasigurang o-order sa online? Ang gusto ng teacher niyo ay talagang magagamit siya lagi."*

[00:10:05]

Ymata: "*Parang masasolve niya po yung problema niyo."*

[00:10:23]

General Manager: "*Mmm, ganun. Paano ba yun ate Ren? Tanong daw sa kanila ng sir nila pano daw makakasigurong ano maraming o-order sa online."*

[00:10:25]

Ate Ren: "*Edi I m-market, pwedeng i market"*

[00:10:40]

General Manager: "*Pero yung walk-in namin, naabot kami 700-800 per day."*

[00:10:45]

Zoleta: "*Sa ano po kaya? Sa page nyo po"*

[00:11:35]

General Manager: "*Sa page? Actually, maunti lang eh. Parang sa isang buwan, less than 100. Nang omo-order. Pero kung yan ay gagawin nating ano. halimbawa may mga customer na gusto naka ready na agad. So kahit sabihin mo taga dito sila, pwede silang mag-order sa website. Na para pagdating nila, ready to pick up na. Yun, pwede. Ba, yung mga nagmamadali, o sige, habang nasa biyahe ako, from Lucena, order na ako. Para pagdating ko ng Lucban, pick up na lang. Tapos ang bayad ay Gcash, kung ano man."*

[00:11:49]

Luza: "*Ay, pag may umorder po ba ng salang, sa ibang bayan po ba? Partial lang po ba yung bayad or fully paid na po?"*

[00:11:50]

General Manager: "*Full."*

[00:11:52]

Luza: "*Ah, full na po."*

[00:12:07]

General Manager: "*Kasi ano yun eh, 3 days lang yung aming pinapay. Baka madurog o ano. Para walang return. Nag-gisa pala kayo."*

[00:12:36]

Ymata: "*paano daw po niya magamit yun kong walking distance na daw po yun?"*

[00:12:44]

General Manager: "*Ayun nga, sabihin nyo. Pag gusto mo ay ano, ayaw mo magintay. Gusto mo pick up na lang."*

[00:12:51]

Ymata: "*Tsaka makita mo yung stock eh. Makakapag-browse ka hindi ka pupunta dito."*

[00:13:12]

General Manager: "*Yan. Kasi sa Amerika, ganyan na din eh. naka order ka na. Tapos parang ito drive-thru mo na lang. Paparating na ako. Baka yun lang ang gustong sagot ni Sir. Nakalagay ba? Sinagot nyo ba yun?"*

[00:13:15]

Ymata: "*Hindi po, na mental block"*

[00:13:17]

Luza: "*Nasa papel"*

[00:13:24]

Valencia: "*Alam daw po namin sabi nung advisor namin. Kaya nangyari po na parang nalito-lito na"*

[00:14:11]

General Manager: "*Baka yun lang yung gustong sagot. Diba minsan? Diba sa mga ano? Sa mga, di ba, dyan sa buddies? Pag ayaw mo magintay doon, tawag ka. Tama, taga dito lang kami. Tawag ako sa buddies. Tapos, pagka, ano, for pick-upna saka ako mapunta. Ay, ayaw mo magintay doon eh. Ay, kung meron namang, pwede namang online. O Jollibee, may ganun din. piliin mo lang yung for pickup. For convenience. baka nalito kayo hindi nyo naisip."*

[00:17:22] - [00:17:31]

Ymata: "*Parang pano natin ma s-solve yun kung sulat pa rin. Yung binibigay sa Baker."*

[00:17:35]

General Manager: "*ibig sabihin Gusto ni sir niyo ay lahat ay ano na."*

[00:17:38]

Ymata: "*Like, din po, yung pano po magagamit yung system din hanggang kahit sa loob."*

[00:17:45]

General Manager: "*Systemize na lahat. Hanggang sa dulo?. Medyo matagal na proseso rin yun."*

[00:17:50]

Zoleta: "*may lalagay na monitor para mag display yung-"*

[00:17:53]

Ymata: "*Oo, yung parang sa Jollibee."*

[00:18:54]

General Manager: "*Oo. Sa mga malalaking company ganyan. Sa Jollibee, pag umorder ka, diba may kanya-kanyang designated ano? Meron sa fries, meron sa burger. Pag may burger, may papasok na sa'yo. Pag fries, chicken, papasok. Sa kanila yun. Dito wala. Ano pa lang. Tsaka piling ko hindi siya ano e. Kasi may oras ang baker namin. Katulad dito, gabi ang pasok ng baker. Ano sila Night shift. Ang start nila 12am. So, bago palang, ah, kung o-order man si customer, talagang isusulat namin. Kasi walang re-receive. Kasi walang tao na dyan. Sarado na kami ng time na papasok yung mga baker. Panggabi kasi sila, gabi ang production namin para pagbukas ng bakery. Bago na ang tinapay. Yun ang aming style."*

[00:19:06]

Ymata: "*Ah! Alala ko. Ah, oo Gets ko na. Ibigsabihin "di kailangan yun? Alala ko po kasi parang, parang pag in-order po, ginagawa, tsaka nyo palang po ginagawa parang…"*

[00:19:07]

General Manager: "*Hindi!"*

[00:19:08]

Ymata: "*Naka ana pala."*

[00:19:13]

General Manager: "*Naka-ano yan? Naka…"*

[00:19:14]

Ymata: "*Handa na po?"*

[00:19:29]

General Manager: "*Ano na siya parang... Naka-standard na siya. Kung ilan. So, nagdadagdag lang kami pag weekends. Ganun. Tapos pag halimawa, maraming sabak ngayon. Magbabawas pa rin tomorrow."*

[00:19:31]

Ymata: "*Gets ko na."*

[00:19:32]

General Manager: "*Every day siya ginagawa."*

[00:19:40]

Ymata: "*Gets ko na kung bakit ganoon yung tanong ni sir. Mali pala sya. Akala niya po. Akala niya po yung sa fastfood po"*

[00:19:45]

General Manager: "'*di hindi, hindi pwede yun, edi ang baker namin nandyan magdamag."*

[00:19:48]

Ymata: "*Yun nga po"*

[00:19:55]

General Manager: "*akala ay made to order na as in pagka omorder. May mga stock na sya"*

[00:20:02]

Ymata: "*yun po, parang ini-expect niya po siguro na ano. Pwede. Parang ganoon sa Jillibee na parang nalabas po sa baker"*

[00:20:23]

General Manager: "*Ah, hindi. pag nagpapo-order kami ay at least a day before. Two or a day before. A day before nakailangan."*

[00:21:55] - [00:22:02]

Ymata: "*paano po ba yung parang buong process po? kunwari po, may order posainyo na"*

[00:22:22]

General Manager: "*Actually yung ano yung nadadagdag lang yun, Ang nadadagdag lang actually dun, pag may umorder ng bulk, yun lang yung parang nagiging additional sa kanilang gawa. Pero pag wala, naka standard na yun, everyday pare-pareho."*

[00:22:25]

Ymata: "*May ano po kayong parang estimate po na kung hanggang ilan lang po yung gagawin?"*

[00:22:54]

General Manager: "*oo, Ganun lang, naka estimate. Sa pag-weekend, sa holiday, naka-times two kami, times three, ganyan. Pero pag regular days, yun. Halimbawa, regular days namin, ang pandesal namin ay, halimbawa, isang sako, ganyan. Regular days. Pieces ba kailangan niyo? Pagkahalimbawa. Pieces na. Pwede naman, ma p-provide namin yan mga pieces-pieces."*

[00:27:30] - [00:27:36]

Ymata: "*Yung sa, paano po naman po malalaman ng baker? Yung susulat po nga? Susulat po ng stock?"*

[00:28:57]

General Manager: "*Isusulat? Oo, isusulat. Meron kaming tagagawa ng order. So ang gagawin ni tindera, isusulat dito sa, may susulat kami dyan ng mga order sa labas. Tapos ang edi eche-check nung gumagawa ng order, isasama dun sa, sa order form nung baker. So naka ganyan lang siya. Nakalagay yung pangalan ni Baker. Tapos ilan yung order ganyan lang, minsan naka-computerized. Pero pagka... Pag akong gumawa, naka-computerize. Tamad ako mag sulat eh. Pero pag sakanila…"*

[00:28:58]

Ymata: "*Pini-print nyo po?"*

[00:29:04]

General Manager: "*Pini-print. Pero ganun din, ganto din naman yung ano. Naka-per-baker siya."*

[00:29:19]

Ymata: "*Pano kung gawin nalang natin yun ano. Parang pag morder Para s'yang malalagay sa isang form. Tapos ready to print siya."*

[00:29:21]

Luza: "*Tapos ididikit nalang."*

[00:29:26]

Ymata: "*I-didikit nalang. Iwas redudancy siya."*

[00:29:13]

Valencia: "*Yan din yung isa pang pinunto sa atin eh. Gawa ng gawa ng. Isusulat-sulat."*

[00:29:34]

Luza: "*Para di na ma-sulat, edi gawa ng…"*

[00:29:41]

Valenncia: "*Ano? O ano?"*

[00:33:12] - [00:33:31]

Valencia: "*Ma'am, about doon po sa page niyo po. Di ba po, tatlo po kayong admin. Which is pag yung, halimbawa po na-view na po ng isang admin niyo nag-order po doon sa mismong page. Tapos nag-open po kayo. Ano pong mangyayari doon sa, parang... sa message po nung nag-order? Ano po, mag-mark as read na po ba yun or"*

[00:34:03]

General Manager: "*yun ang hindi ko sure. Kasi ang gamit namin ay yung sa... Pero feeling ko, kasi meron kaming automated na nag r-reply. Tulad yan, pag may nag-inquire. Ayan o, yung may ganyan. So, hindi pa namin yan nasi-seen. Tapos... Pero ang alam ko, kasi kaming magkakapatid dito ang admin. Pag na seen na ni ate, nagmamark pa rin siya sa akin na, ano…"*

[00:34:04]

Valencia: "*Unread?"*

[00:34:05]

General Manager: "*oo"*

[00:34:11]

Ymata: "*Pero pag na-take niya po kunwari po, na-take niya po yung…"*

[00:34:18]

General Manager: "*Yes Pero parang pag na-replyan niya na, nag-reread na siya sa akin. Hindi ko na din nakikita. Parang ganun."*

[00:34:22]

Luza: "*Ah, nag-reread na."*

[00:35:24]

General Manager: "*Dito yun sa page, sa business suite ng Facebook. Gusto niyo sa isa lang admin, mahirap ba pag tatlo? kaya lang naman kami nagpapatagdag ng admin in case hindi ako pwede o hindi pwede isa. May makakakita pa rin. Ayun. Tanong na mamaya b-byahe na naman kayo paparito?"*

[00:41:03] - [00:41:07]

Ymata: "*Ano'ng mali dyan? May mali ba dyan?Uy"*

[00:41:09]

General Manager: "*Alin?"*

[00:41:12]

Ymata: "*yung logo po namin sa website"*

[00:41:30]

General Manager: "*sa website namin? Ba't hindi na lang yung Pecto's din? Ayaw niyo?"*

[00:41:38]

Luza: "*Pwede naman po, ay parang ginanakod tayo. Ano sabi ni Ma'am?"*

[00:41:41]

General Manager: "*Para alam nila na ano"*

[00:41:44]

Valencia: "*Ah, alam kona. Babaguhin ko yan. Parang alam pala."*

[00:41:53]

General Manager: "*Baka kasi mamaya sabihin, baka hindi yan sa pectos ah. Kung yan din yung lalabas sa website niyo. Baka malito, "pektos ba ito?"*

[00:41:55]

Valencia: "*alam kona gagawin ko"*

[00:42:08]

Ymata: "*kayo lang po ba nag mamanage po ng mga orders po? Like online? Sa admin lang po kita yun?"*

[00:42:09]

General Manager: "*oo."*

[00:42:10]

Ymata: "*a hindi po nila kita yun?"*

[00:42:22]

General Manager: "*edi yung iba, yung walk-in sa tindera. Pero yung mga papasok sa page at yung mag-direct sa sa'min, edi kami. Kung sino mang kumuha ng order, edi ikaw ang responsible doon."*

[00:42:38] - [00:42:41]

General Manager: "*Saan nyo pala ilalagay yun? Kailangan nyo ng computer?"*

[00:42:44]

Ymata: "*Opo, website."*

[00:42:50]

General Manager: "*Ah, website. Kala ko nasa isang iPad."*

[00:42:57]

Ymata: *Hindi. Mahirap po kasi yun. Kailangan po naming gumastos. Pagka may hardware."*

[00:45:33] - [00:45:47]

Luza: "*yung sa ano po kaya pagdagdi-deliver po, paano po dito lang po yung nagsasabi? O kahit sa online na rin po? Yung dito lang po sa Lucban."*

[00:46:04]

General Manager: "*Pagdito sa lucban? Pag sinasabi nila kung pick-up or deliver. Pag-deliver, kami din na nangdadala. Pag sa online usually yan ay deliver talaga kasi mga taga-ibang bayan. Yeah."*

[00:46:15]

Ymata: "*Parang dito lang. May rider nga po ba kayo?"*

[00:46:16]

General Manager: "*Meron."*

[00:46:19]

Ymata: "*so makagamit din siya?"*

[00:46:22]

Luza: "*Parang ito yung…"*

[00:46:25]

General Manager: "*Parang kami lang ang nagsasabi."*

[00:46:42]

Ymata: "*Ah. halimbawa po kasi possible na malagyan din siya. Kailangan niya rin gumamit.*

[00:47:02]

General Manager: "*Pero yung rider namin, siya rin talaga yung ano eh... Siya kasi nagdi-deliver na mga tinapay sa outlet. Tapos siya din yung naghahakot ng mga ingredients. So kung may mga order, eh di parang siya na din yung ano, rider. Siya na lahat."*

[00:47:18]

Luza: "*Ang ibig sabihin ata ni sir ay yung ano. Pag may omorder, makikita agad."*

[00:47:31]

General Manager: "*Parang matrack. Yung omorder kung nasan na yung order niya? Kailangan mo ba yan? baka mas lalong magiging komplikado."*

[00:47:56]

Ymata: "*Eh, nasa limitation kasi natin yan, JNT po ba ang pinapapa deliveran nyo?"*

[00:48:00]

General Manager: "*pag malalayo? LBC, JNT. GRS."*

[00:48:01]

Ymata: *"may sarili naman yung ano, di ba?"*

[00:48:04]

General Manager: "*Oo, sila. Meron. Meron sila."*

PROJECT TITLE: PECTRACK: A Web-Based Order Management System with Predictive Analytics and Inventory System for Pecto's Bakery at Lucban Quezon

DATE: 2026-05-12

LOCATION: Messenger Call

INTERVIEWER: Christian V. Luza
Christian C. Valencia
Yancy L. Ymata
Carla Mhaey G. Zoleta



INTERVIEWEE: Katherine R. Abillar

[00:00:05] – [00:00:14]

Luza: *"Ma'am, ask lang po sana namin tungkol po sa delivery details ng Pecto's. Paano po usually yung process?"*

00:00:20]

General Manager: *"Depende kasi kung within Lucban lang o outside Lucban."*

[00:00:28]

Luza: *"Pag dito lang po sa Lucban?"*

00:00:38]

General Manager: "*Pag dito lang, kami lang din nagde-deliver. Meron kaming rider."*

[00:00:45]

Luza: *"May sarili po kayong rider?"*

[00:00:52]

General Manager: *"Oo. Siya rin naghahatid sa outlets minsan."*

[00:01:02]

Valencia: *"Pag outside Lucban naman po?"*

[00:01:12]

General Manager: *"Usually LBC, J&T, o GRS na."*

[00:01:18]

Valencia: *"sila na po bahala sa shipping?"*

[00:01:25]

General Manager: *"Oo. Pag malalayo sila na."*

[00:01:40] – [00:01:52]

Luza: *"Ma'am, tungkol naman po sa users ng system. Sino-sino po kaya yung gagamit?"*

[00:02:00]

General Manager: *"Syempre kami mga admin."*

[00:02:08]

Luza: *"Pwede po kaya makuha names niyo po para sa documentation?"*

[00:02:18]

General Manager: *"Oo pwede naman. Ise-send ko na lang sa message."*

[00:02:28]

Luza: *"Ah sige po ma'am."*

[00:02:40] – [00:02:55]

Valencia: *"Ma'am, dun naman po sa PayMongo. Need po kasi yata ng legal documents at business permit."*

[00:03:05]

General Manager: *"Ahh okay."*

[00:03:12]

Valencia: *"Okay lang po kaya na maasikaso niyo po? Need lang po kasi yung business permit niyo po."*

[00:03:24]

General Manager: *"Sige okay lang naman. Meron naman kami nun."*

[00:03:32]

Valencia: *"Need lang po talaga siya for integration."*

[00:03:40]

General Manager: *"Oo sige. Ipapasa namin."*

[00:03:55] – [00:04:08]

Valencia: *"Ma'am, ilan nga po ulit yung admin?"*

[00:04:12]

General Manager: *"Tatlo kami."*

[00:04:18]

Valencia: *"Hindi po kaya mahirapan pag tatlo pa rin po gagamit?"*

[00:04:28]

General Manager: *"Pwede naman siguro isa na lang."*

[00:04:35]

Valencia: *"Kayo na lang po?"*

[00:04:40]

General Manager: *"Oo, ako na lang."*

[00:04:48]

Valencia: *"Mas madali po kasi ma-manage."*

[00:04:55]

General Manager: *"Oo para hindi rin magulo."*

[00:05:10] – [00:05:18]

Luza: *"Ma'am, ano pong devices yung gamit niyo po?"*

[00:05:26]

General Manager: *"May company phone kami."*

[00:05:34]

Valencia: *"Ahh dun po usually chine-check yung orders?"*

[00:05:42]

General Manager: *"Oo. Dun lang din."*

[00:06:10] – [00:06:22]

Luza: *"Ma'am, dun naman po sa username format. Ano pong gusto niyo gamitin?"*

[00:06:30]

General Manager: *"Kahit ano kayo na bahala."*

[00:06:38]

Luza: *"Pwede pong pangalan o apelyido?"*

[00:06:45]

General Manager: *"Oo kahit alin."*

[00:06:52]

Luza: *"Ah sige po kami na bahala mag-format."*

[00:07:05] – [00:07:18]

Luza: *"Ma'am, pwede rin po kaya makahingi ng pictures, list, at prices ng products?"*

[00:07:28]

General Manager: *"Oo. Hindi ko pa nga pala nase-send."*

[00:07:35]

Luza: *"Okay lang po ba may kasama nang prices?"*

[00:07:42]

General Manager: *"Oo okay lang."*

[00:07:50]

Luza: *"Need po kasi namin ilagay yung mga products po"*

[00:07:58]

General Manager: *"Sige isesend ko na lang."*

[00:08:12] – [00:08:20]

Valencia: *"Ay ma'am, pati po pala signature niyo po sana para sa transcript."*

[00:08:30]

General Manager: *"Ah sige."*

[00:08:36]

Valencia: *"yung kahit digital signature po"*

[00:08:42]

General Manager: *"oo sige meron ako non dito i s-send ko nalang din"*

[00:08:55] – [00:09:10]

Luza: *"Yun lang naman po ma'am. Maraming salamat po sa time niyo."*

[00:09:18]

General Manager: *"Sige okay lang."*

[00:09:24]

Luza: *"Thank you po ulit ma'am."*

[00:09:30]

Valencia: *"thank you poo"*

[00:09:38]

General Manager: *"Goodluck sa inyo."*

[00:09:45]

Luza: "*Thank you po. Ingat po."*

APPENDIX C. Flow Chart

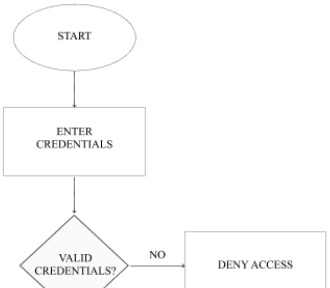


Figure 36. Login FLow Chart

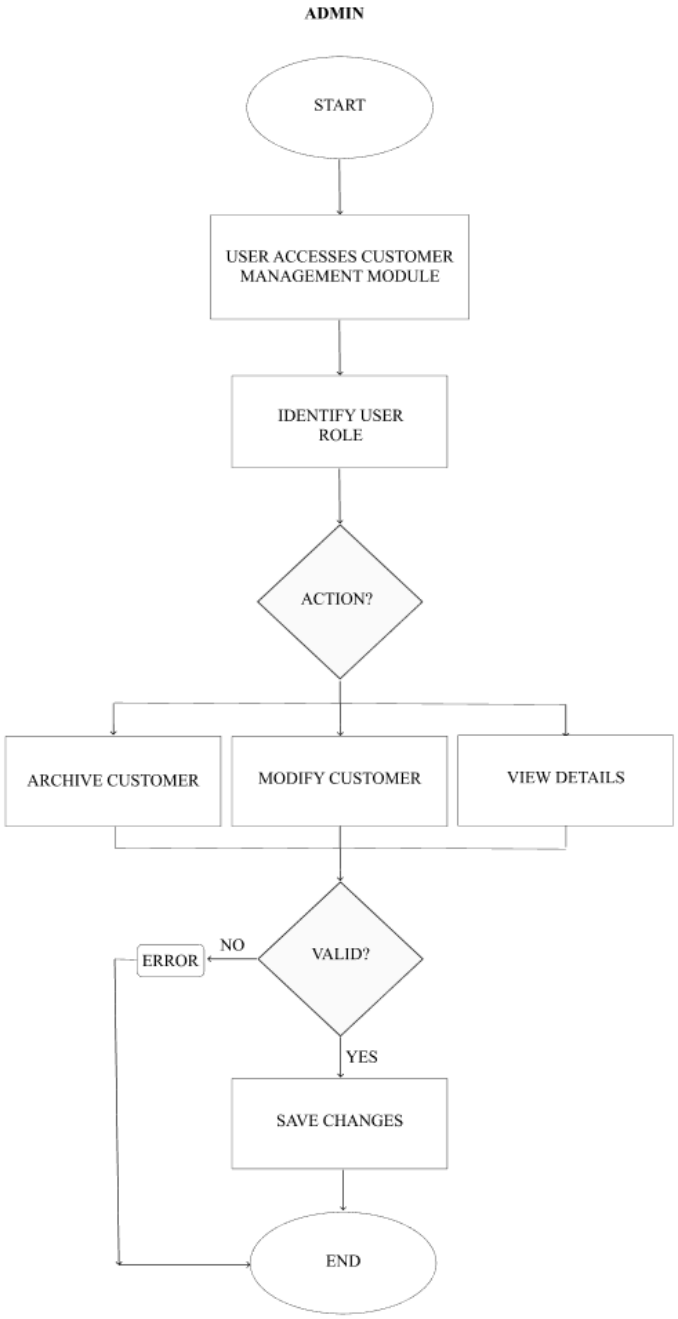


Figure 37. Admin Customer Management Flow Chart

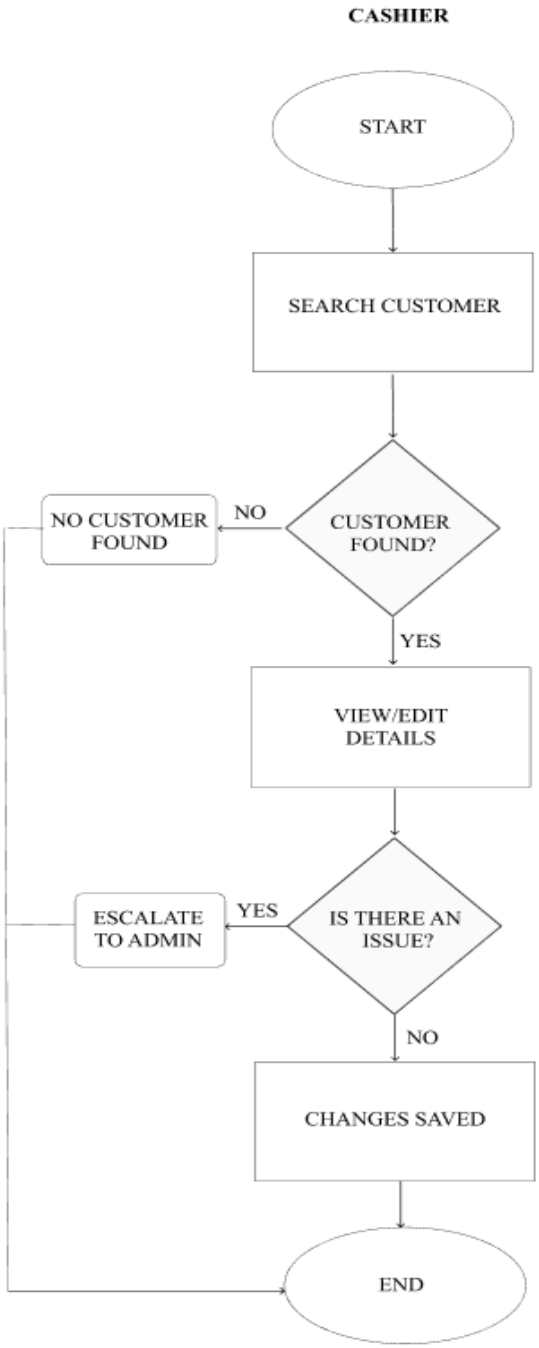


Figure 38.Cashier Customer Management Flow Chart

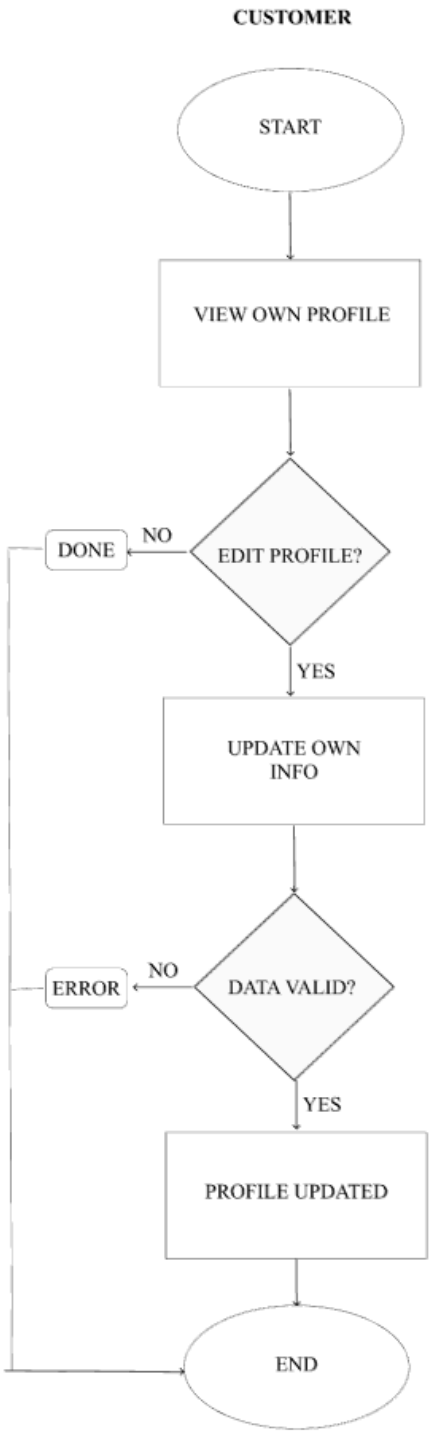


Figure 39. Customer Management Flow Chart

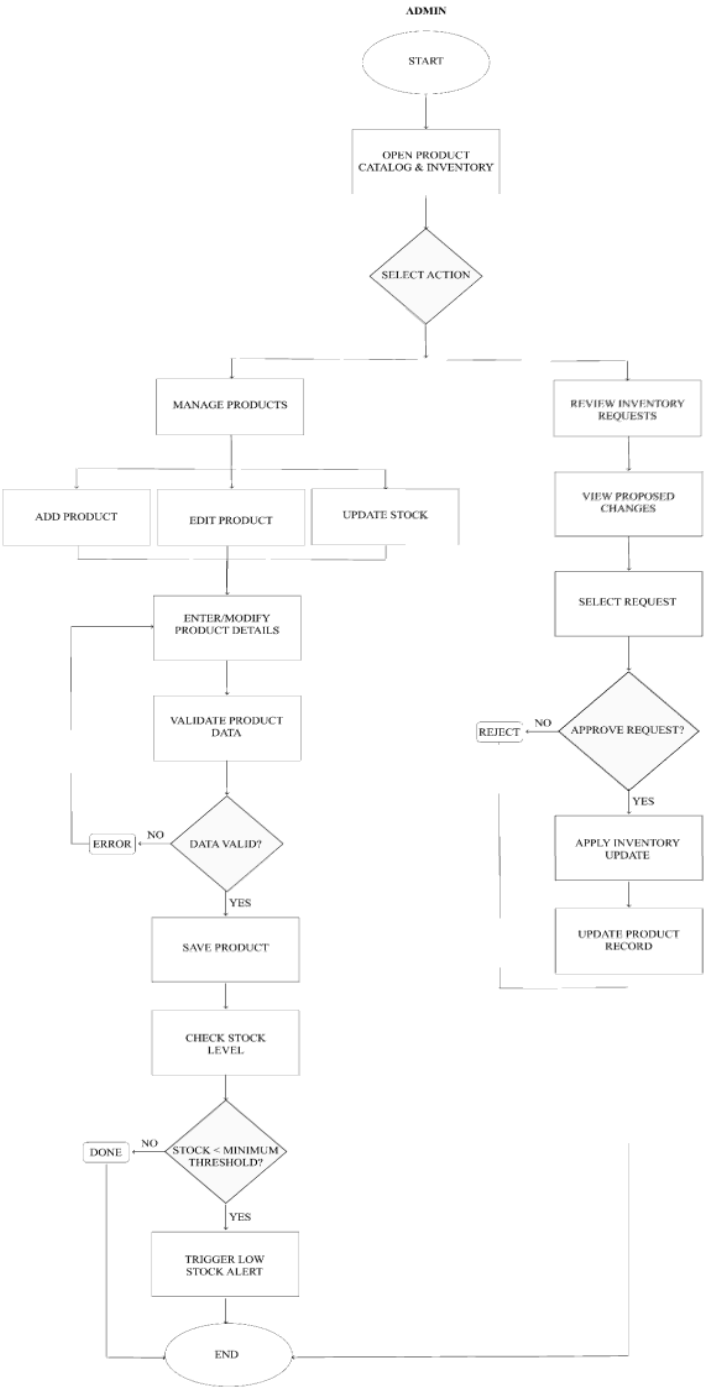


Figure 41. Admin Inventory Management Flow Chart



Figure 42.Cashier Inventory Management Flow Chart

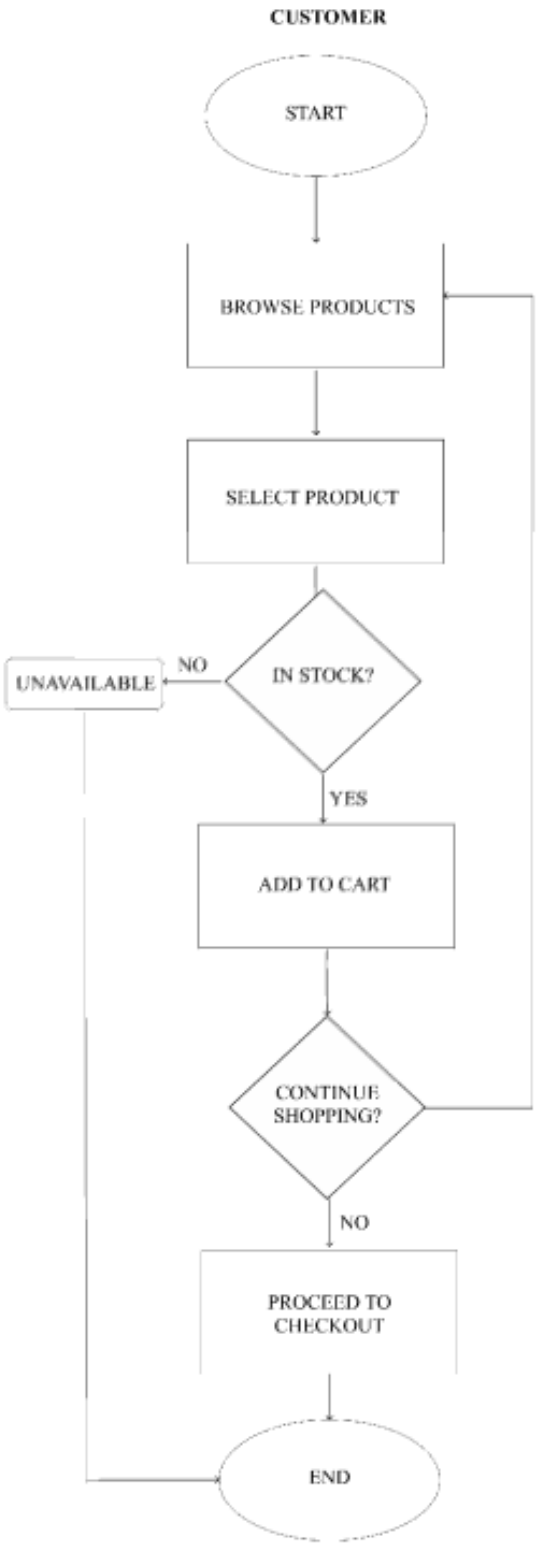


Figure 43.Customer Inventory Management Flow Chart

Figure 44. Delivery Person Inventory Management FLow Chart

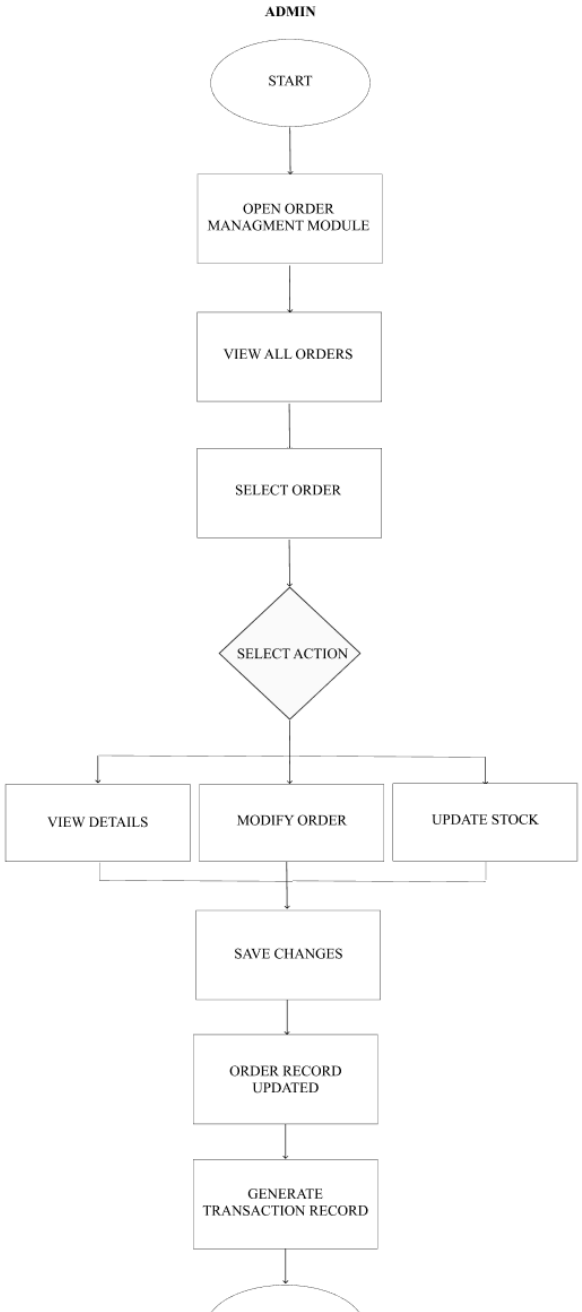


Figure 45. Admin Order Management Flow Chart

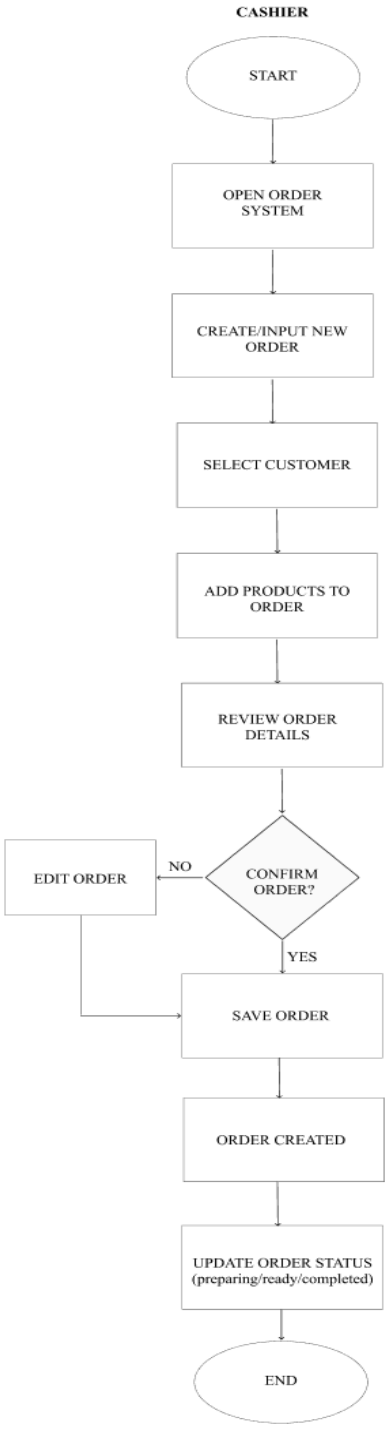


Figure 46. Cashier Order Management Flow Chart

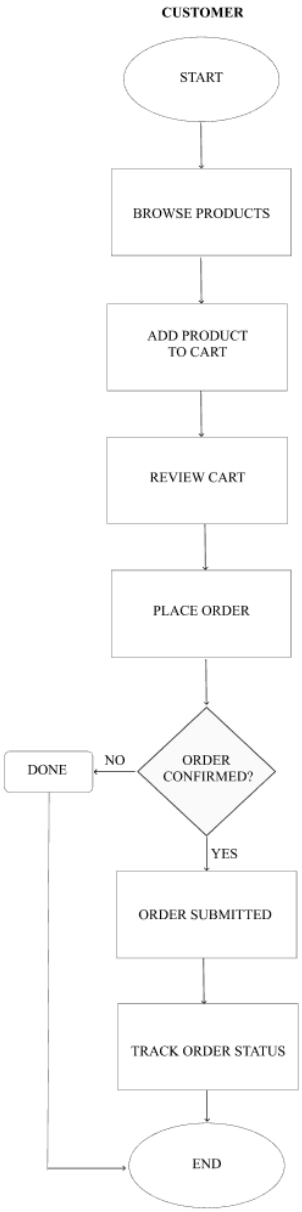


Figure 47. Customer Order Management Flow Chart

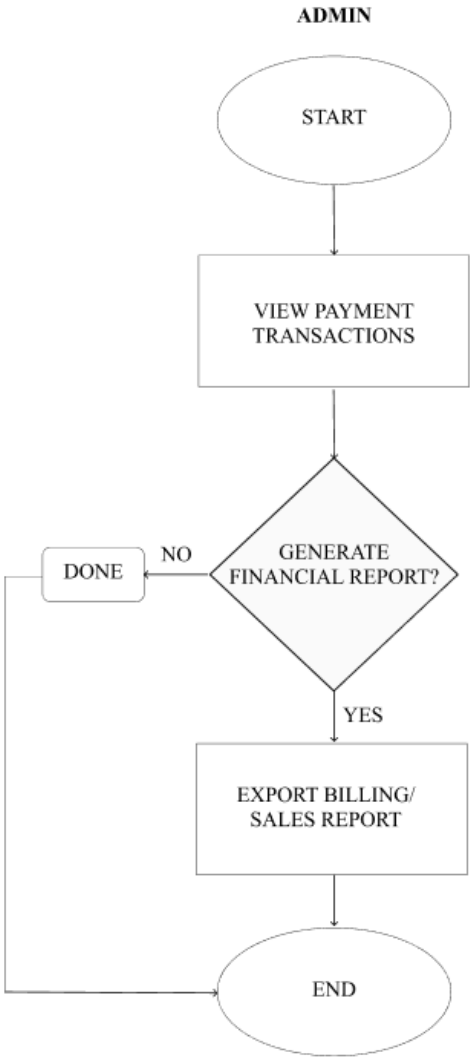


Figure 49. Admin Payment and Billing Flow Chart

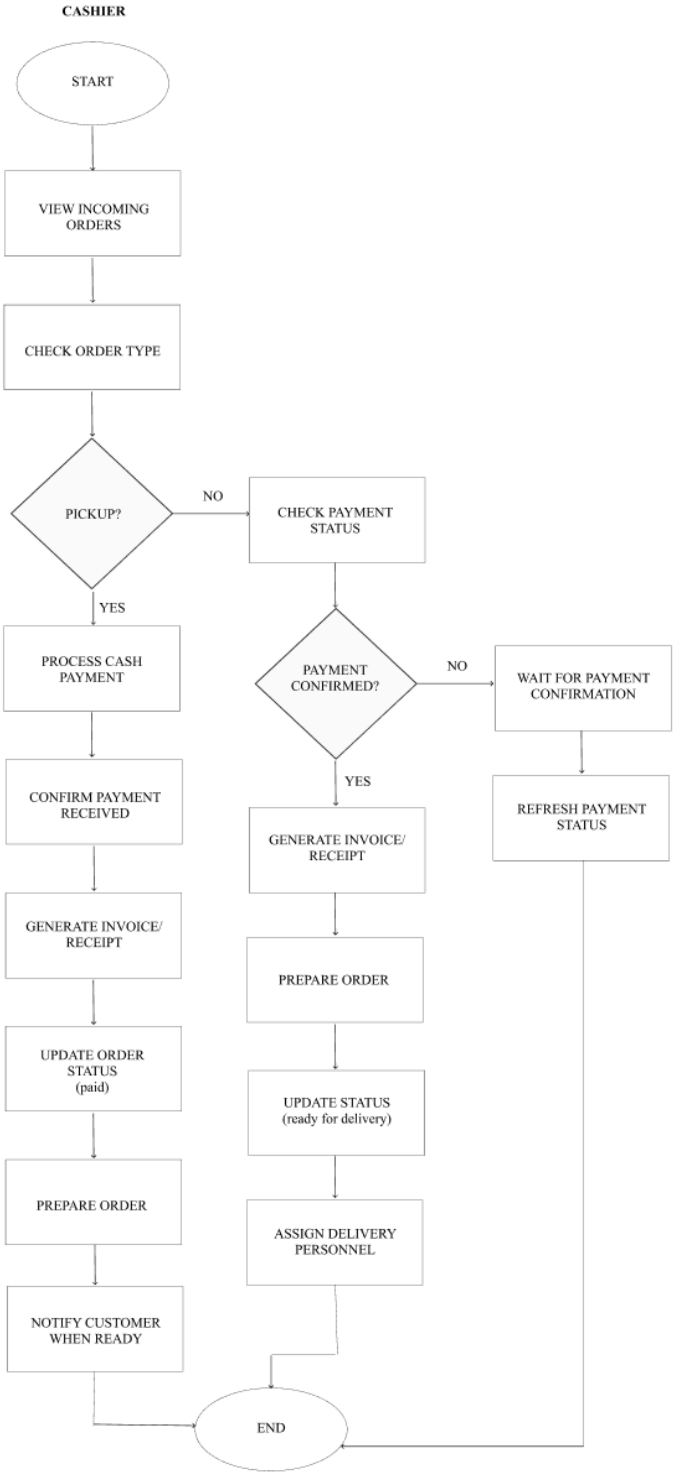


Figure 50. Cashier Payment and Billing Flow Chart

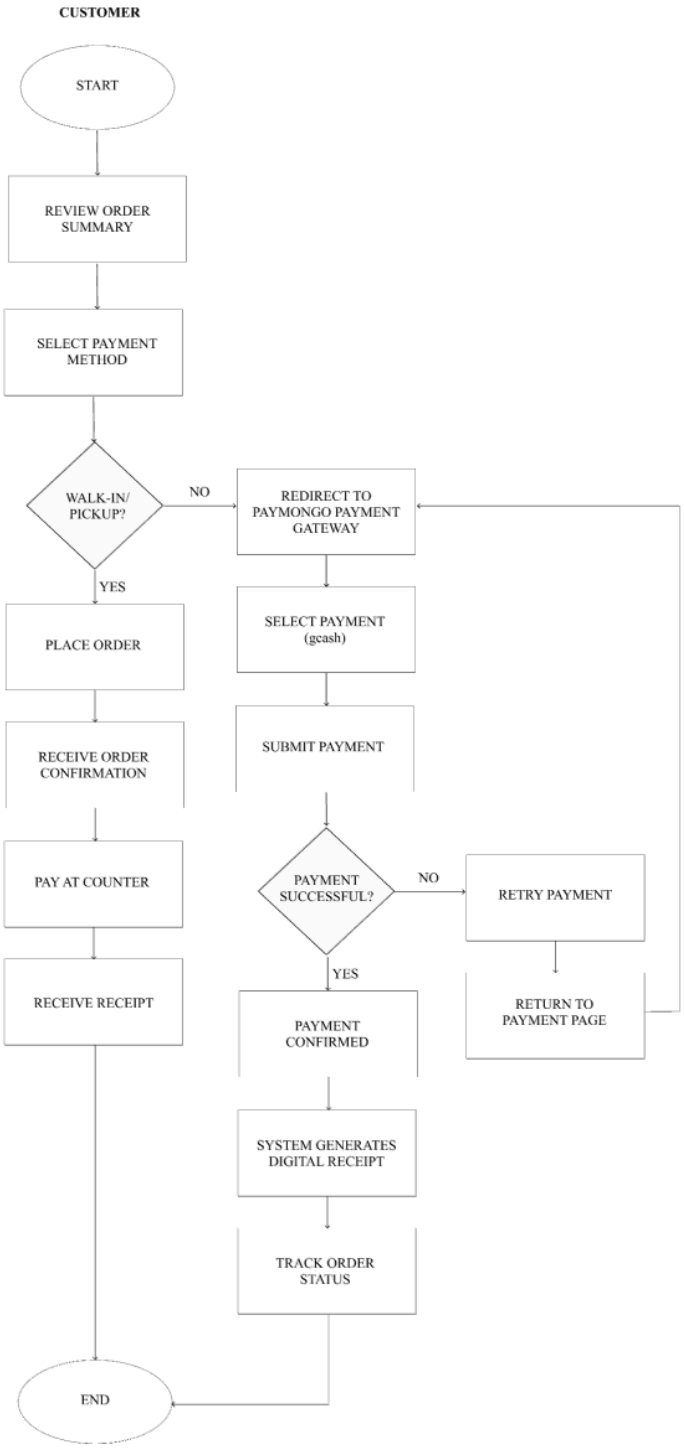


Figure 51. Customer Payment and Billing Flow Chart

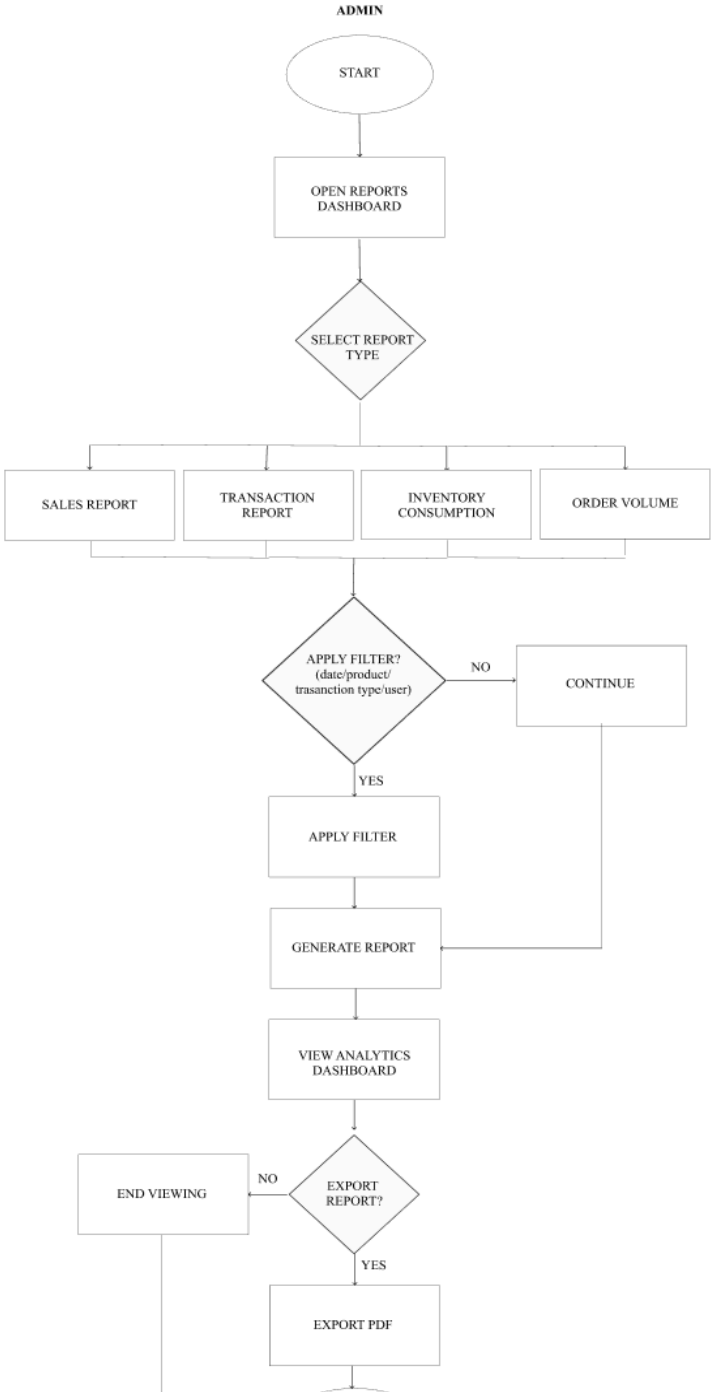

ADMIN
START
OPEN REPORTS DASHBOARD
SELECT REPORT TYPE
SALES REPORT
TRANSACTION REPORT
INVENTORY CONSUMPTION
ORDER VOLUME
APPLY FILTER? (date/product/ trasanction type/user)
NO
CONTINUE
YES
APPLY FILTER
GENERATE REPORT
VIEW ANALYTICS DASHBOARD
END VIEWING
NO
EXPORT REPORT?
YES
EXPORT PDF

Figure 53. Admin Reporting and Analytics Flow Chart

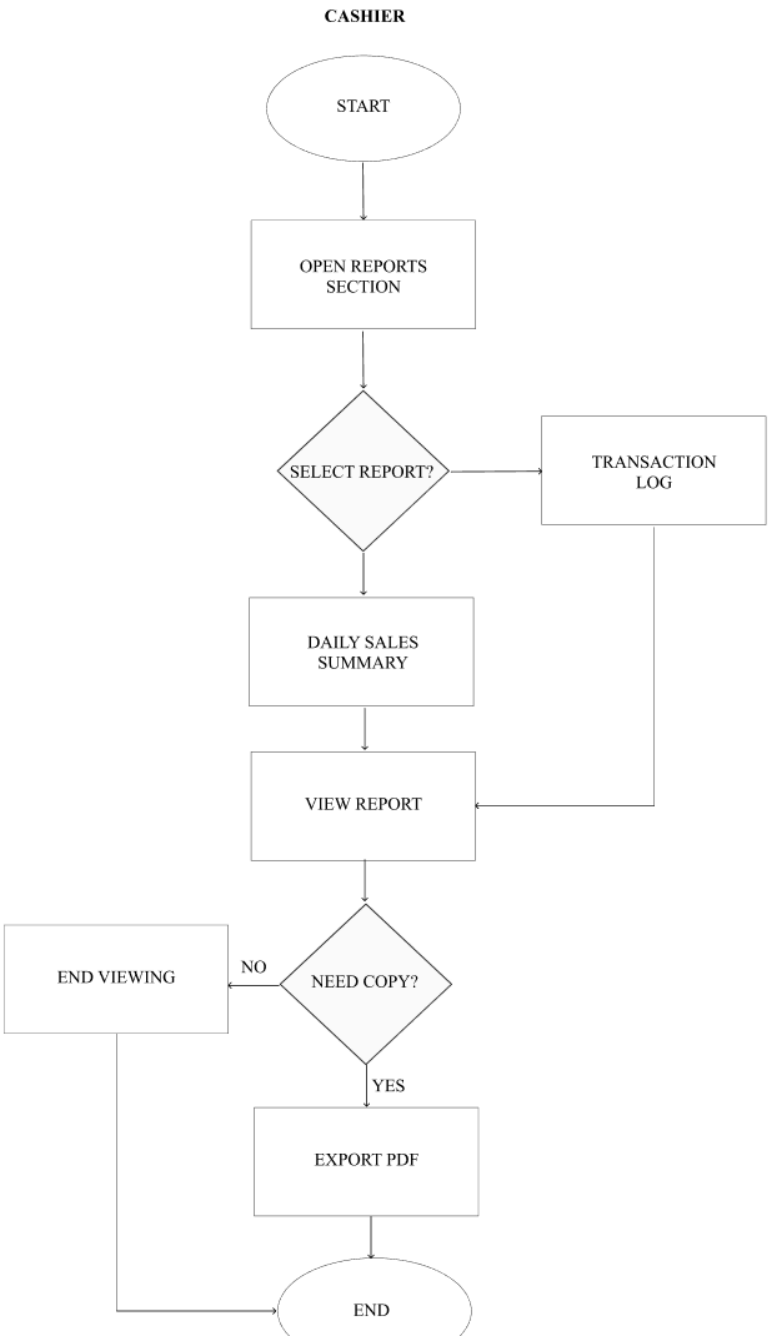


Figure 54. Cashier Reporting and Analytics Flow Chart

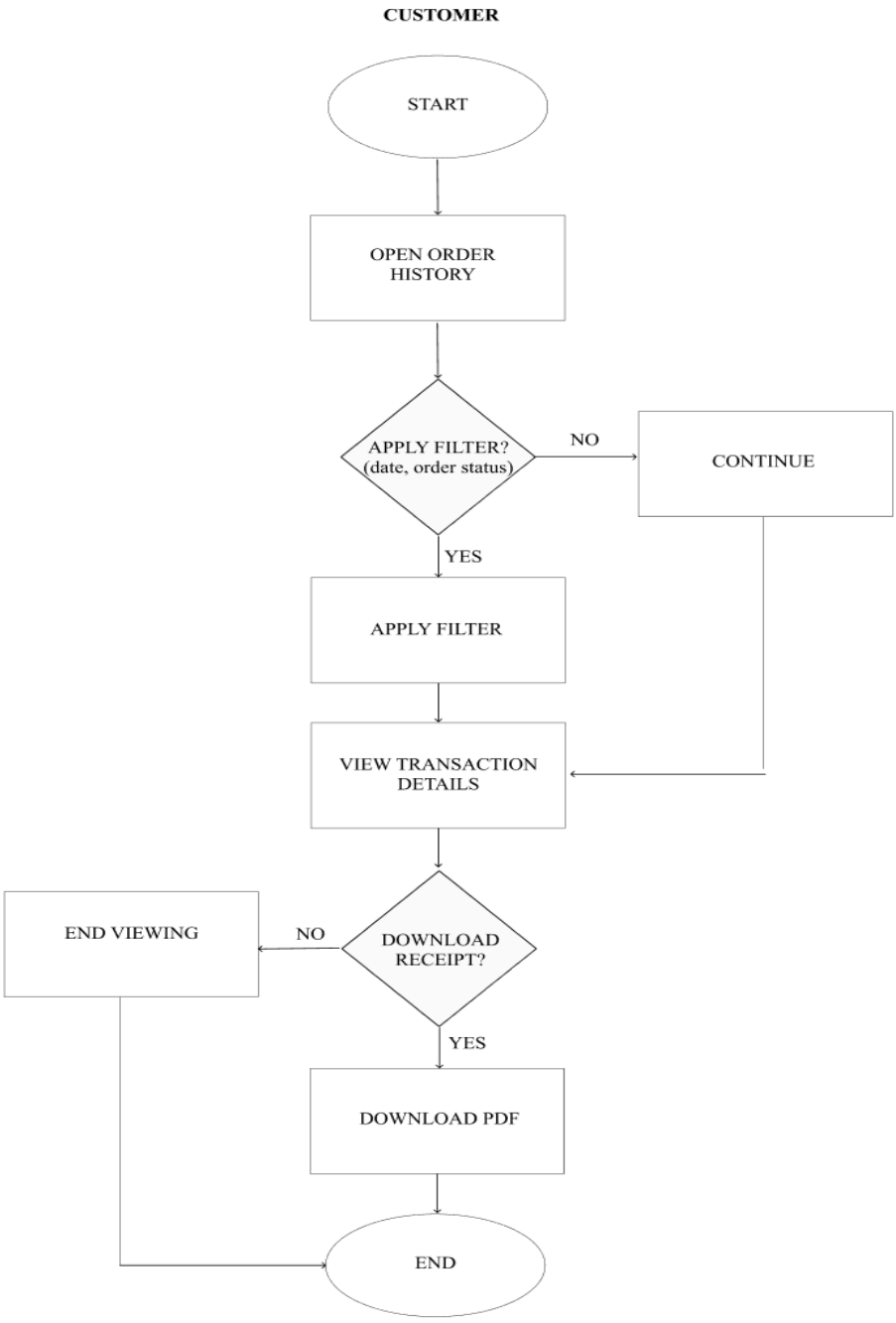


Figure 55. Customer Reporting and Analytics Flow Chart

APPENDIX D. Accomplishment Form

ACCOMPLISHMENT AND CONSULTATION FORM

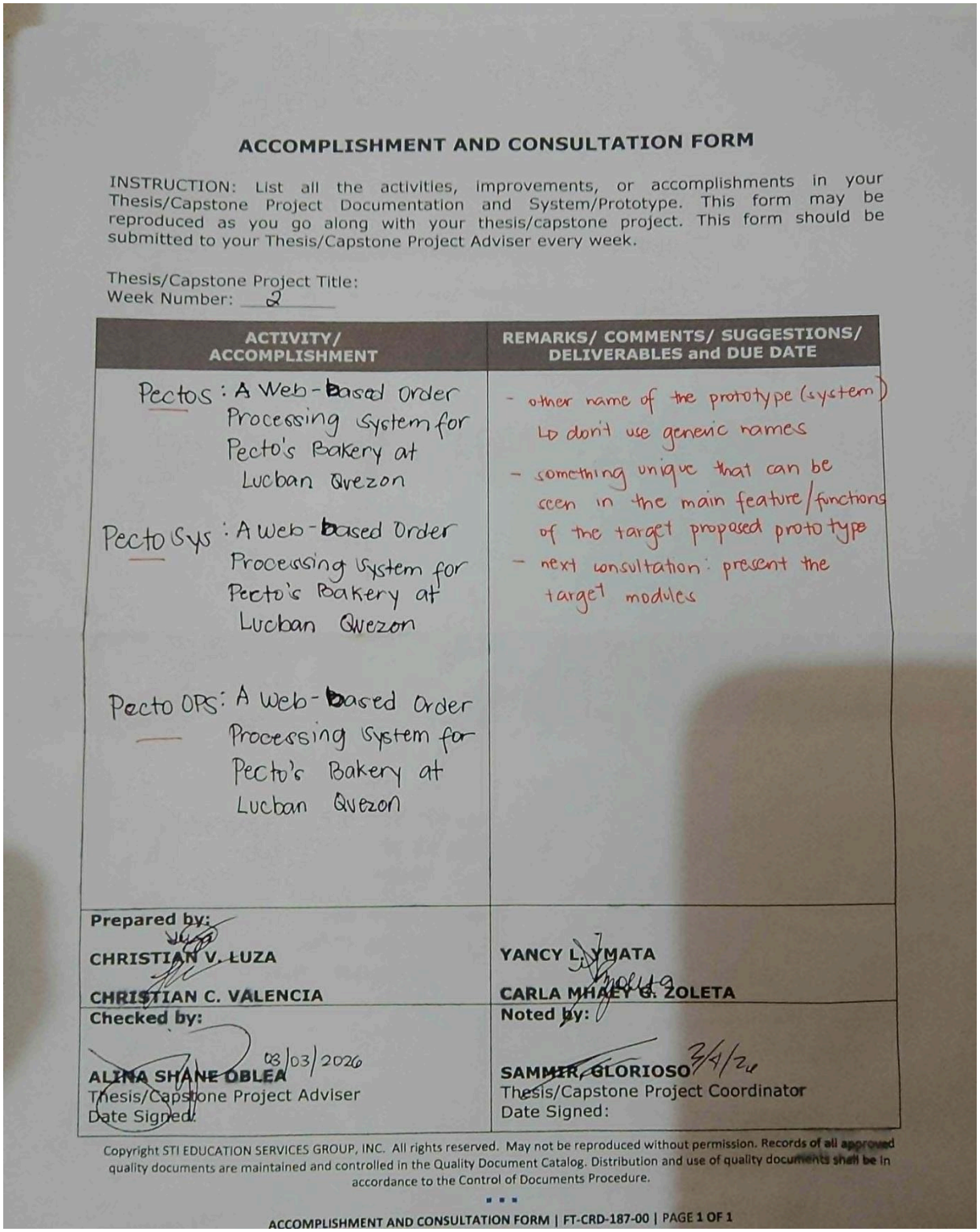
INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title:
Week Number: 1

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
1. PEC-WEB: A Web-based Order Processing system with Inventory Monitoring, Alert Notification, and Dashboard Reporting for Pecto's Bakery at Lucban, Quezon.	→ change title/name system → Web-based → Order Processing System (general) → remove redudancy
2. OPSIMAN: A web-Based Ordering Processing System with Inventory Monitoring and Alert Notification for Pecto's Bakery at Lucban Quezon.	→ change title/name system → remove redudancy → realtime API
3. Ponde Buns: A Web-Based Order Processing, Inventory Monitoring, and Sales Reporting System with Alert Notification for Pecto's Bakery at Lucban, Quezon	→ change title/name system → remove redudancy ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ~ ORDER PROCESSING SYSTEM *Analytical Dashboard → content/concept finalize
Prepared by: **CHRISTIAN V. LUZA** **CHRISTIAN C. VALENCIA**	**YANCY L. YMATA** **CARLA MHAE G. ZOLETA**
Checked by: 02/24/26 **ALINA SHANE OBLEA** Thesis/Capstone Project Adviser Date Signed:	**Noted by:** 3/4/24 **SAMMIR, GLORIOSO** Thesis/Capstone Project Coordinator Date Signed:

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Figure 57. Consultation week 1



ACCOMPLISHMENT AND CONSULTATION FORM

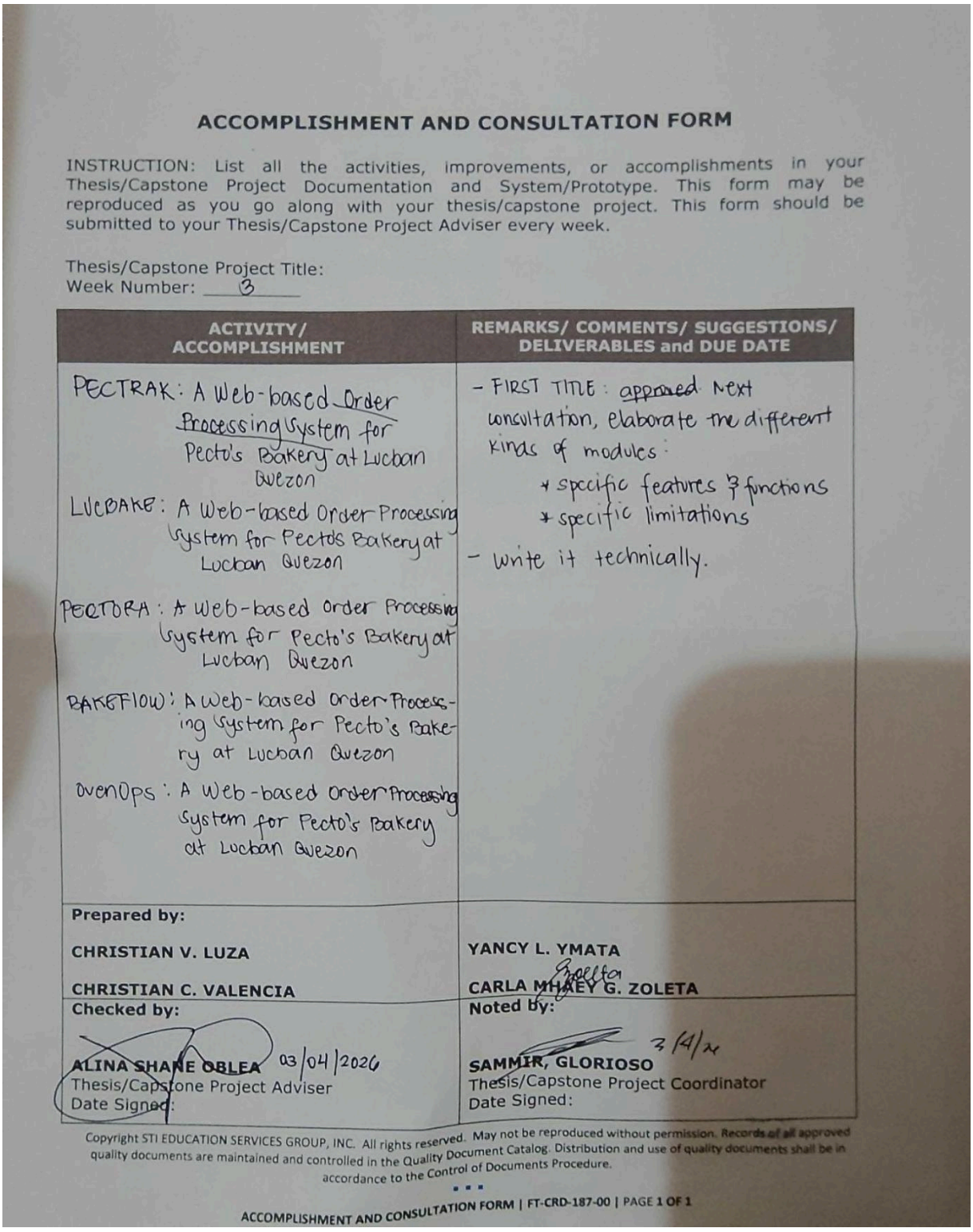
INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title:
Week Number: 2

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
Pectos : A Web-based Order Processing System for Pecto's Bakery at Lucban Quezon Pecto Sys : A Web-based Order Processing System for Pecto's Bakery at Lucban Quezon Pecto OPS: A Web-based Order Processing System for Pecto's Bakery at Lucban Quezon	- other name of the prototype (system) Lb don't use generic names - something unique that can be seen in the main feature/functions of the target proposed prototype - next consultation: present the target modules
Prepared by: **CHRISTIAN V. LUZA** **CHRISTIAN C. VALENCIA**	**YANCY L. YMATA** **CARLA MHAEY G. ZOLETA**
Checked by: 08/03/2026 **ALINA SHANE OBLEA** Thesis/Capstone Project Adviser Date Signed:	**Noted by:** 3/4/26 **SAMMIR GLORIOSO** Thesis/Capstone Project Coordinator Date Signed:

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Figure 58. Consultation week 2



ACCOMPLISHMENT AND CONSULTATION FORM

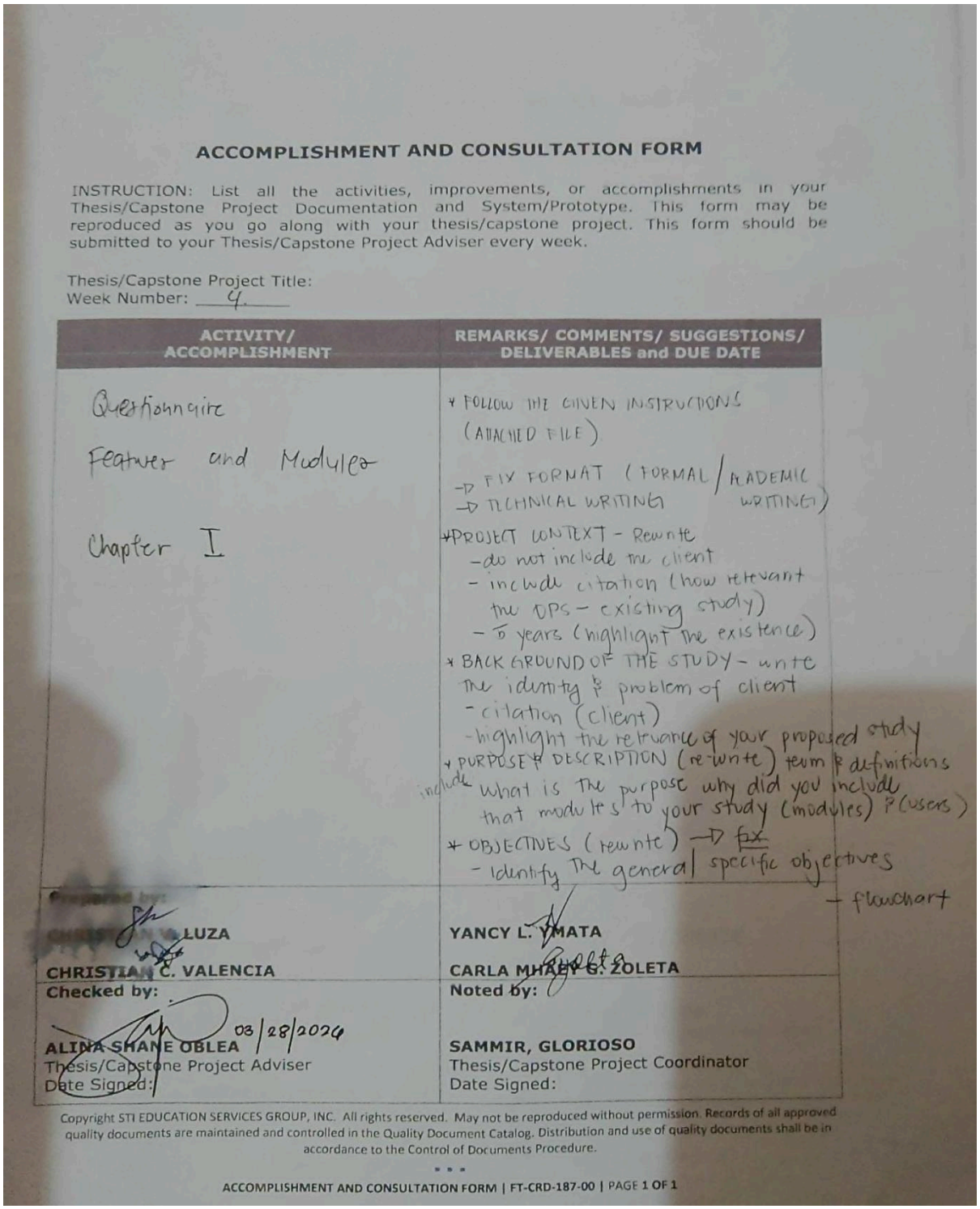
INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title:
Week Number: 3

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
PECTRAK: A Web-based Order Processing System for Pecto's Bakery at Lucban Quezon LUCBAKE: A Web-based Order Processing System for Pectos Bakery at Lucban Quezon PECTORA: A Web-based Order Processing System for Pecto's Bakery at Lucban Quezon BAKEFLOW: A Web-based Order Processing System for Pecto's Bakery at Lucban Quezon ovenOps: A Web-based Order Processing System for Pecto's Bakery at Lucban Quezon	- FIRST TITLE: approved Next consultation, elaborate the different kinds of modules: * specific features & functions * specific limitations - write it technically.
Prepared by: **CHRISTIAN V. LUZA** **CHRISTIAN C. VALENCIA**	**YANCY L. YMATA** **CARLA MHAEY G. ZOLETA**
Checked by: **ALINA SHANE OBLEA** 03/04/2026 Thesis/Capstone Project Adviser Date Signed:	**Noted by:** 3/4/26 **SAMMIR, GLORIOSO** Thesis/Capstone Project Coordinator Date Signed:

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Figure 59. Consultation week 3



ACCOMPLISHMENT AND CONSULTATION FORM

INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title:
Week Number: 4.

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
Questionnaire Feature and Modules Chapter I	* FOLLOW THE GIVEN INSTRUCTIONS (ATTACHED FILE) -> FIX FORMAT (FORMAL / ACADEMIC WRITING) -> TECHNICAL WRITING *PROJECT CONTEXT - Rewrite -do not include the client - include citation (how relevant the OPS - existing study) - 5 years (highlight the existence) * BACKGROUND OF THE STUDY - write the identity & problem of client - citation (client) -highlight the relevance of your proposed study * PURPOSE & DESCRIPTION (re-write) term & definitions include what is the purpose why did you include that modules to your study (modules) ? (users) * OBJECTIVES (rewrite) -> fix - identify the general/ specific objectives - flowchart

Prepared by:

CHRISTIAN C. VALENCIA

YANCY L. YMATA

CARLA MHAEY G. ZOLETA

Checked by:

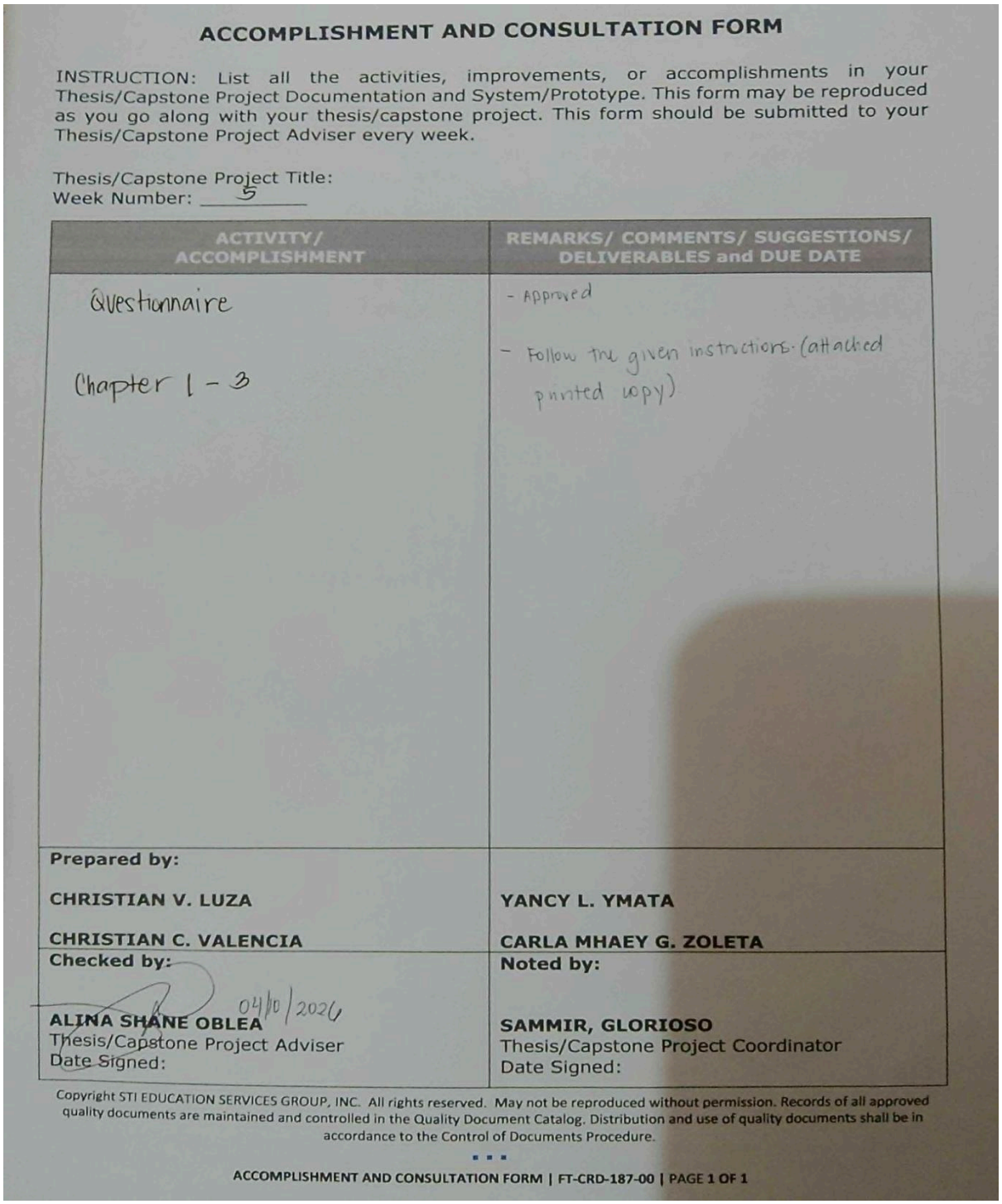
ALINA SHANE OBLEA
Thesis/Capstone Project Adviser
Date Signed: 03/28/2024

Noted by:

SAMMIR, GLORIOSO
Thesis/Capstone Project Coordinator
Date Signed:

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Figure 60. Consultation week 4



ACCOMPLISHMENT AND CONSULTATION FORM

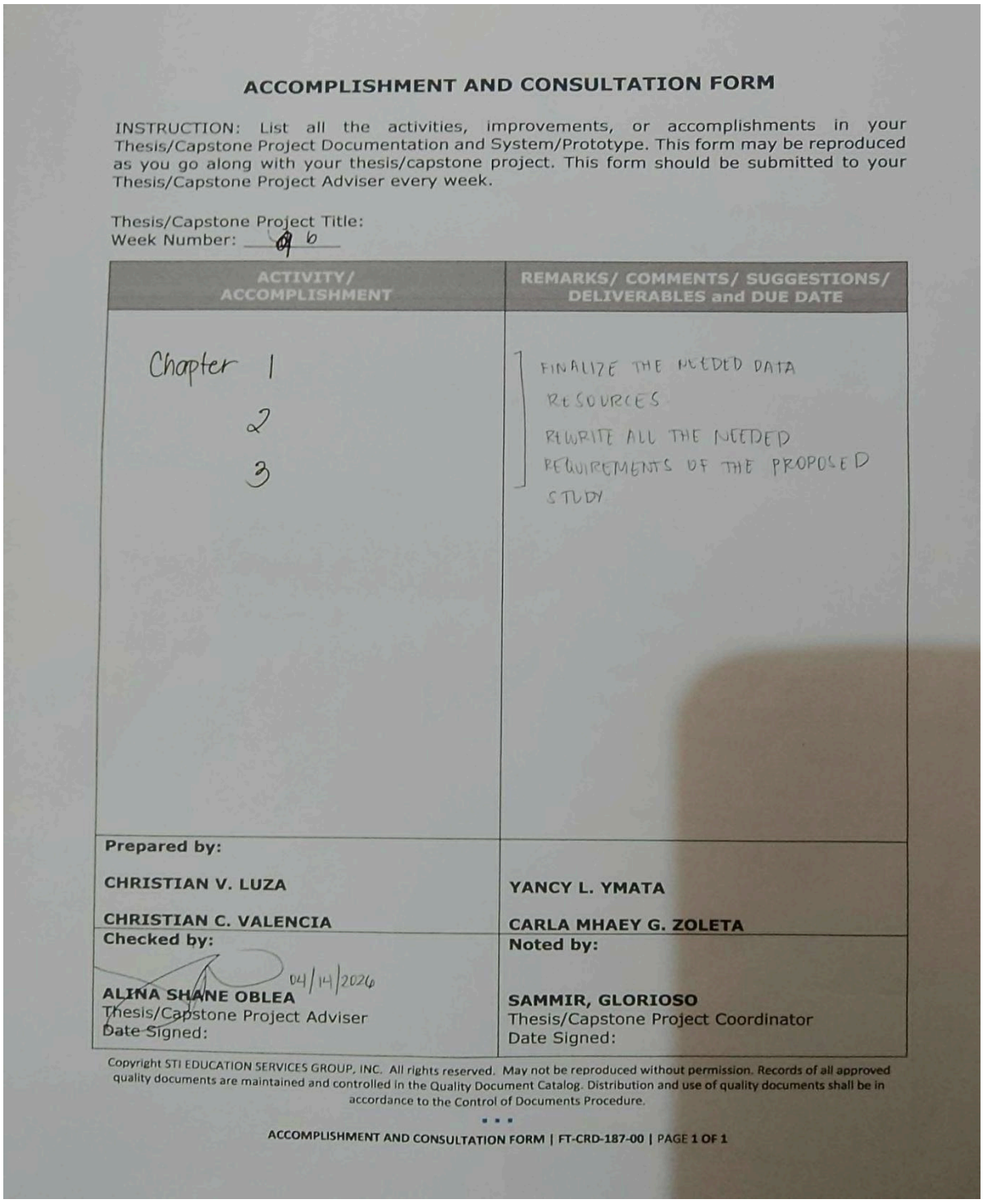
INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title:
Week Number: 5

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
Questionnaire Chapter 1 - 3	- Approved - Follow the given instructions. (attached printed copy)
Prepared by: **CHRISTIAN V. LUZA** **CHRISTIAN C. VALENCIA**	 **YANCY L. YMATA** **CARLA MHAEY G. ZOLETA**
Checked by: **ALINA SHANE OBLEA** 04/10/2026 Thesis/Capstone Project Adviser Date Signed:	**Noted by:** **SAMMIR, GLORIOSO** Thesis/Capstone Project Coordinator Date Signed:

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Figure 61. Consultation week 5



ACCOMPLISHMENT AND CONSULTATION FORM

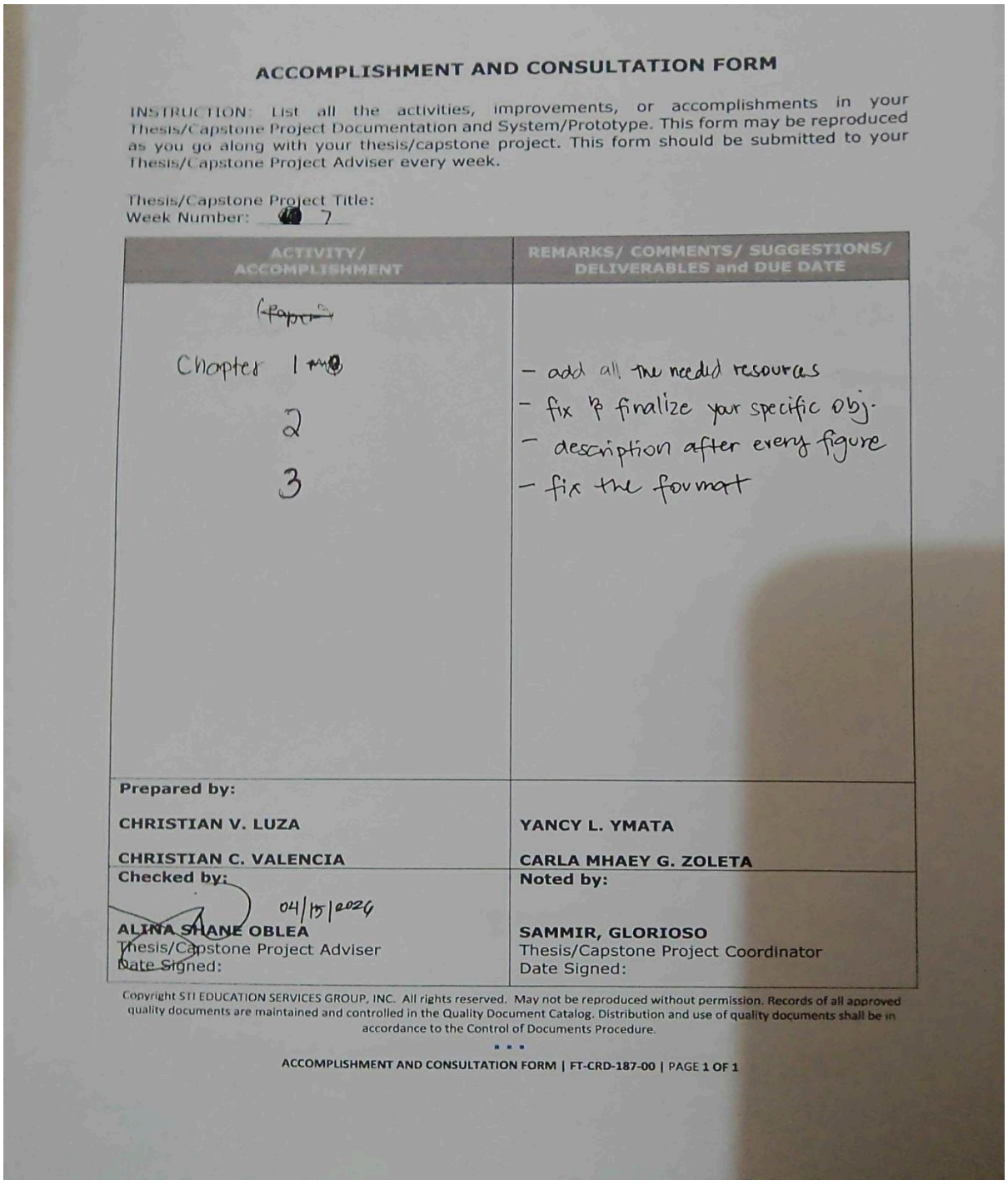
INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title:
Week Number: 6

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
Chapter 1 2 3	FINALIZE THE NEEDED DATA RESOURCES REWRITE ALL THE NEEDED REQUIREMENTS OF THE PROPOSED STUDY
Prepared by: **CHRISTIAN V. LUZA** **CHRISTIAN C. VALENCIA**	**YANCY L. YMATA** **CARLA MHAEY G. ZOLETA**
Checked by: 04/14/2026 **ALINA SHANE OBLEA** Thesis/Capstone Project Adviser Date Signed:	**Noted by:** **SAMMIR, GLORIOSO** Thesis/Capstone Project Coordinator Date Signed:

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Figure 62. Consultation week 6



ACCOMPLISHMENT AND CONSULTATION FORM

INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title:
Week Number: 7

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
~~Paper~~ Chapter 1 rev 2 3	- add all the needed resources - fix & finalize your specific obj. - description after every figure - fix the format
Prepared by: **CHRISTIAN V. LUZA** **CHRISTIAN C. VALENCIA**	**YANCY L. YMATA** **CARLA MHAEY G. ZOLETA**
Checked by: 04/15/2026 **ALINA SHANE OBLEA** Thesis/Capstone Project Adviser Date Signed:	**Noted by:** **SAMMIR, GLORIOSO** Thesis/Capstone Project Coordinator Date Signed:

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Figure 63. Consultation week 7

ACCOMPLISHMENT AND CONSULTATION FORM

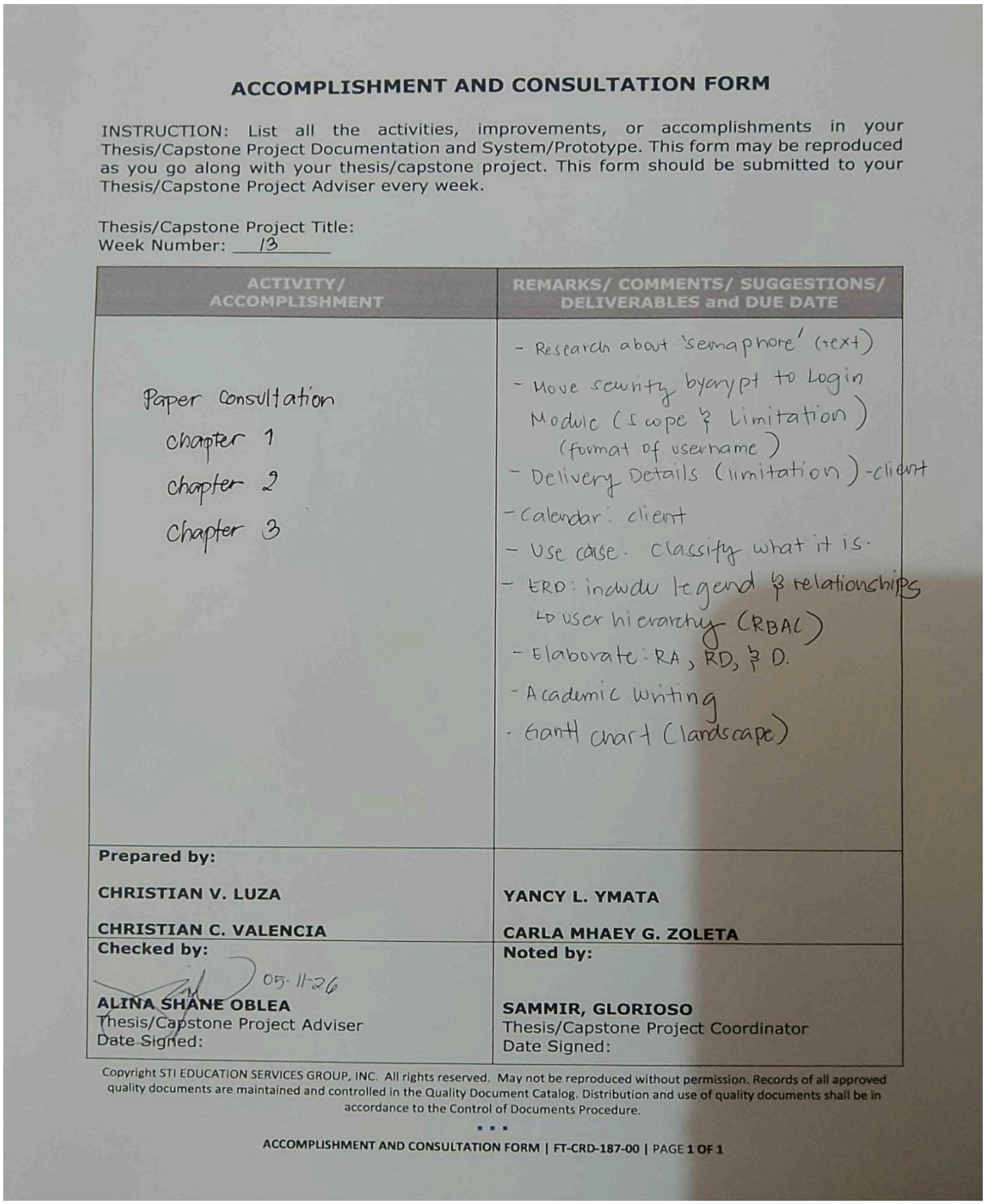
INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title:
Week Number: 12

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
Paper Consultation chapter 1 chapter 2 chapter 3	- Define what will be your tools. - Fix the format/margin. - fix the technicalities (specify the details) - Add additional local & foreign. - Rewrite the synthesis. - Specify the specific number of duration. - Remove the unnecessary details. - Methodology "adopt & adapt". when writing each description. - Add the details, based on your diagram. - Include the versions. - RA resources (find) - RD deliverables (how) - D template/format
Prepared by: **CHRISTIAN V. LUZA** **CHRISTIAN C. VALENCIA**	 **YANCY L. YMATA** **CARLA MHAEY G. ZOLETA**
Checked by: 05-04-2024 **ALINA SHANE OBLEA** Thesis/Capstone Project Adviser Date Signed:	**Noted by:** **SAMMIR, GLORIOSO** Thesis/Capstone Project Coordinator Date Signed:

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Figure 64. Consultation week 12



ACCOMPLISHMENT AND CONSULTATION FORM

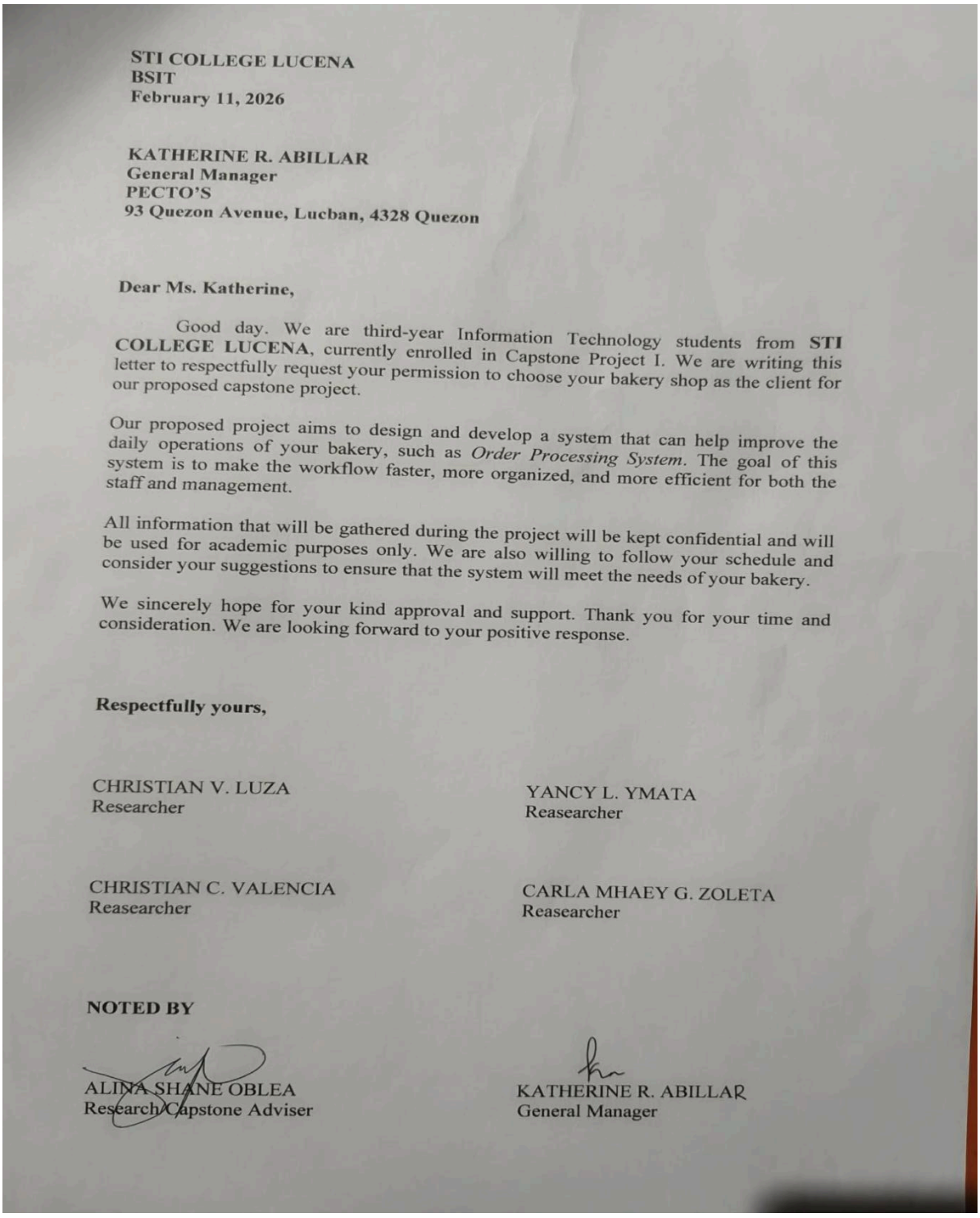
INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title:
Week Number: 13

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
Paper Consultation chapter 1 chapter 2 chapter 3	- Research about 'semaphore' (text) - Move security bycrypt to Login Module (Scope & Limitation) (format of username) - Delivery Details (limitation) -client - Calendar: client - Use case: classify what it is. - ERD: indvdu legend & relationships ↳ user hierarchy (RBAC) - Elaborate: RA, RD, & D. - Academic Writing - Gantt chart (landscape)
Prepared by: **CHRISTIAN V. LUZA** **CHRISTIAN C. VALENCIA**	**YANCY L. YMATA** **CARLA MHAEY G. ZOLETA**
Checked by: 05-11-26 **ALINA SHANE OBLEA** Thesis/Capstone Project Adviser Date Signed:	**Noted by:** **SAMMIR, GLORIOSO** Thesis/Capstone Project Coordinator Date Signed:

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Figure 65. Consultation week 13



STI COLLEGE LUCENA
BSIT
February 11, 2026

KATHERINE R. ABILLAR
General Manager
PECTO'S
93 Quezon Avenue, Lucban, 4328 Quezon

Dear Ms. Katherine,

Good day. We are third-year Information Technology students from **STI COLLEGE LUCENA**, currently enrolled in Capstone Project I. We are writing this letter to respectfully request your permission to choose your bakery shop as the client for our proposed capstone project.

Our proposed project aims to design and develop a system that can help improve the daily operations of your bakery, such as *Order Processing System*. The goal of this system is to make the workflow faster, more organized, and more efficient for both the staff and management.

All information that will be gathered during the project will be kept confidential and will be used for academic purposes only. We are also willing to follow your schedule and consider your suggestions to ensure that the system will meet the needs of your bakery.

We sincerely hope for your kind approval and support. Thank you for your time and consideration. We are looking forward to your positive response.

Respectfully yours,

CHRISTIAN V. LUZA
Researcher

YANCY L. YMATA
Reasearcher

CHRISTIAN C. VALENCIA
Reasearcher

CARLA MHAEY G. ZOLETA
Reasearcher

NOTED BY

ALINA SHANE OBLEA
Research Capstone Adviser

KATHERINE R. ABILLAR
General Manager

Figure 66. Request Letter for Client

March 27, 2026

Ms. Katherine R. Abillar
General Manager PECTO'S
93 Quezon Avenue , Lucban, 4328 Quezon
Subject: Request for Consent to Conduct an Interview or Written Questionnaire

Dear Ms. Katherine,
Greetings!

We are 3rd year BSIT students of STI College Lucena currently conducting a study on a Web-based order management system in Pecto's bakery for our Capstone Project 1.

In this regard, we respectfully request your consent to participate in our data-gathering procedure. You may choose to participate in one of two ways:

- Personal Interview – We can schedule a time at your convenience to discuss the current pre-assessment process, including the procedures followed, the challenges encountered, and your recommendations for improvement.

- Written Questionnaire – If more convenient, we can provide the questions in writing, and you may submit your responses at your preferred time.

All information collected will be used strictly for academic purposes only and will be treated with the utmost confidentiality and respect.

We are hoping for your kind approval and support for our study. Thank you very much for your time and consideration.

Respectfully yours,

The Researchers:

Christian V. Luza

Christian C. Valencia

Yancy L. Ymata

Carla Mhaey G. Zoleta

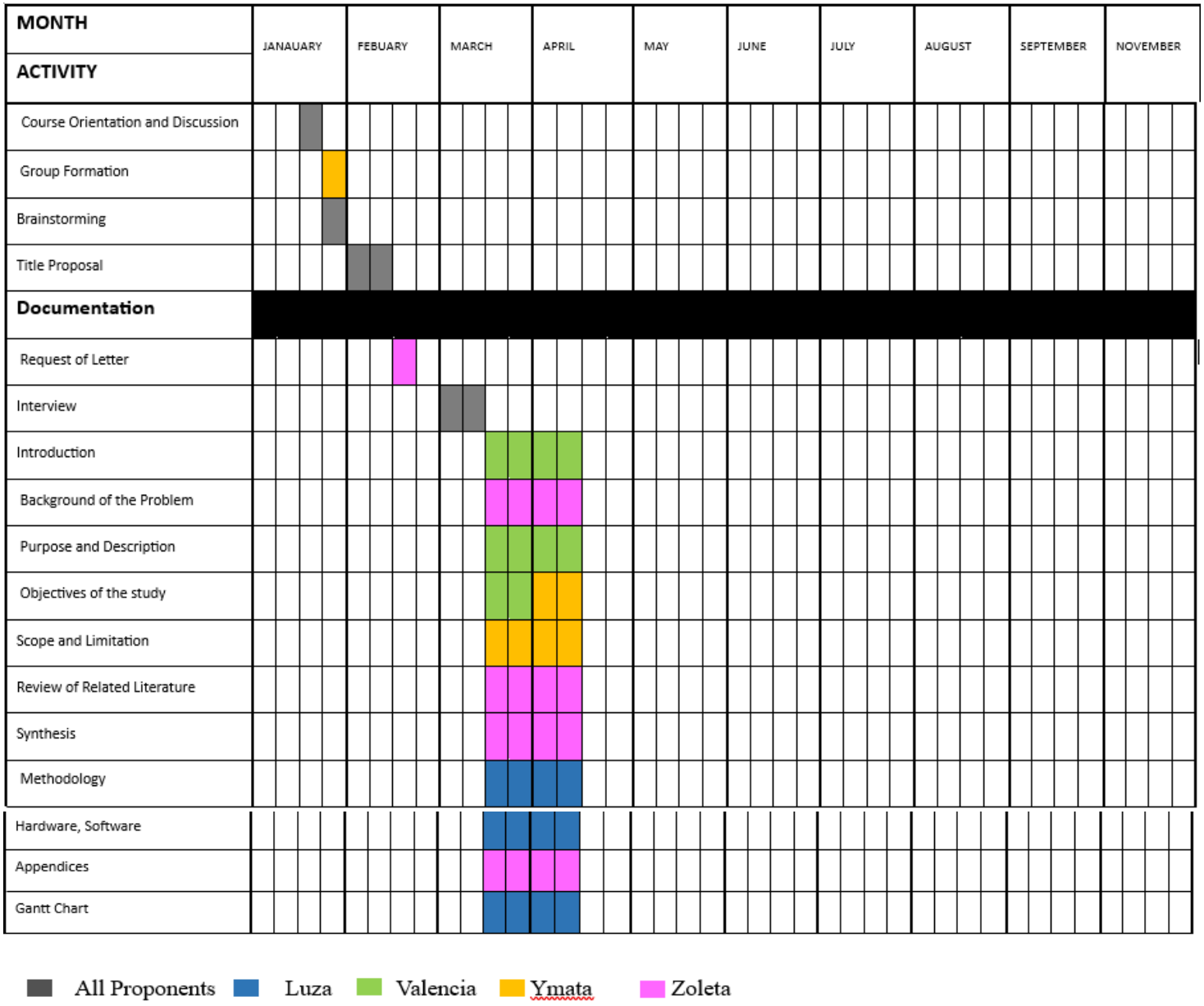


Noted by:

Ms. Alina Shane Oblea
Capstone Project Adviser

Katherine R. Abillar
General Manager

APPENDIX E. Gantt Chart


MONTH
ACTIVITY
JANAUARY
FEBUARY
MARCH
APRIL
MAY
JUNE
JULY
AUGUST
SEPTEMBER
NOVEMBER
Course Orientation and Discussion
Group Formation
Brainstorming
Title Proposal
Documentation
Request of Letter
Interview
Introduction
Background of the Problem
Purpose and Description
Objectives of the study
Scope and Limitation
Review of Related Literature
Synthesis
Methodology
Hardware, Software
Appendices
Gantt Chart
All Proponents
Luza
Valencia
Ymata
Zoleta

APPENDIX F. Curriculum Vitae



Curriculum Vitae of

YANCY L. YMATA

Purok Matahimik Looban, Brgy Cotta. Lucena City. Quezon Province
yancyymata0@gmail.com
09197814385

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of School/Institution
Tertiary	2023 - Present	STI College Lucena
Senior High School	June 2021- May 2023	Holy Rosary Catholic School
Junior High School	June 2017 - May 2021	Cotta National High School
Elementary	June 2011 - May 2017	South 1 Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/Job Title	Name and Address of Company or Organization
March 1, 2023 - March 30, 2023	Work Immersion	City Government of Lucena

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
N/A	N/A	N/A

SKILLS

- **Knowledgeable in using MS Office**
- **Knowledgeable in using HTML, CSS, JAVASCRIPT, Microsoft sql, and JAVA**
- **Knowledgeable in using figma**

TRAININGS, SEMINARS, OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
N/A	N/A



Curriculum Vitae of

Christian V. Luza
9 Rizal Avenue, Brgy 7 Lucban, Quezon Province
cluza540@gmail.com
09817318480

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of School/Institution
Tertiary	2023-Present	STI College Lucena
Senior High School	June 2021- May 2023	STI College Lucena
Junior High School	June 2017 - May 2021	Paaralang Sekundarya ng Lucban
Elementary	June 2011 - May 2017	Paaralang Elementary ng Lucban

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/Job Title	Name and Address of Company or Organization
March 1, 2023 - March 30, 2023	Work Immersion	Simon Tech Lucena

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
N/A	N/A	N/A

SKILLS

- **Knowledgeable in using MS Office**
- **Knowledgeable in using HTML, CSS, JAVASCRIPT, Microsoft sql, and JAVA**
- **Knowledgeable in using figma**

TRAININGS, SEMINARS, OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
N/A	N/A



Curriculum Vitae of

Christian C. Valencia

Brgy Potol. Tayabas City. Quezon Province
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09946954648

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of School/Institution
Tertiary	2023-Present	STI College Lucena
Senior High School	June 2021- May 2023	Luis Palad International High School
Junior High School	June 2017 - May 2021	Luis Palad International High School
Elementary	June 2011 - May 2017	Potol Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/Job Title	Name and Address of Company or Organization
March 1, 2023 - March 30, 2023	Work Immersion	Guinayan

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
N/A	N/A	N/A

SKILLS

- **Knowledgeable in using MS Office**
- **Knowledgeable in using HTML, CSS, JAVASCRIPT, Microsoft sql, and JAVA**
- **Knowledgeable in using figma**

TRAININGS, SEMINARS, OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
N/A	N/A



Curriculum Vitae of

Zoleta, Carla Mhaey G.

Brgy Poblacion. Guinayangan Quezon
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EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of School/Institution
Tertiary	2023-Present	STI College Lucena
Senior High School	June 2021- May 2023	Guinayangan Senior High School
Junior High School	June 2017 - May 2021	Guinayangan National High School
Elementary	June 2011 - May 2017	Guinayangan Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/Job Title	Name and Address of Company or Organization
March 1, 2023 - March 30, 2023	Work Immersion	Guinayangan Senior High School

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
N/A	N/A	N/A

SKILLS

- **Knowledgeable in using MS Office**
- **Knowledgeable in using HTML, CSS, JAVASCRIPT, Microsoft sql, and JAVA**
- **Knowledgeable in using figma**

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